INTRODUCTION TO GENERAL PHYSI
VOLUME I

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PREFACE

Our intention in preparing this two-volume work was to provide a uniform text on general physics for university undergraduates in the first year of their studies for a first degree in the physical sciences, the biological sciences or agricultural science. In the treatment, a basic level of knowledge of the subject is assumed. We have aimed at presenting the physical aspect of topics as much as possible, and then at providing the mathematical arguments and formulae necessary for a thorough understanding.

We have chosen the level to which the subject has been treated in a way which we hope will satisfy two groups. The first is those who need an introduction to a subject that they will study in greater detail and depth in their subsequent years using other, more advanced texts. The second group contains those who do not propose such studies but who need a complete and, as far as possible, up-to-date coverage suitable for a less specialized course. Historical details have also been given in some cases to provide a balanced perspective of the subject. To help the student, numerous worked examples have been included in the text.

Volume 1 of this textbook covers dimensions, vectors, dynamics, kinematics, work and energy, elasticity, fluids and thermodynamics. We are responsible for the final result and for any mistakes it may contain.

4th May, 2010

Cape Coast

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Chapter One

PHYSICAL QUANTITIES

Physical Quantities and Units

The field of physics is divided into the areas of mechanics, waves, optics, thermodynamics, and electricity which make up the classical physics developed between the sixteenth and nineteenth centuries by Galileo, Newton, Faraday, Maxwell, and others. Physics also includes modern topics developed in the twentieth century such as relativity, quantum physics and quantum field theory, as well as their appropriate applications in condensed matter, atomic and nuclear phenomena, elementary particles, and astrophysics. Physics is sometimes defined as the study of energy and matter and their relationships. Physics is also inherently a science of measurements and the ultimate test for any theory in physics is its agreement with observations and the measurements of the physical phenomenon. Thus, any number or set of numbers used for the quantitative description of a physical phenomenon is called a *physical quantity*. To define a physical quantity we must have a procedure for measuring such quantity or specify a way of calculating it. We measure a physical quantity in its own units and by comparison with a standard unit.

SI Standard Units

By international agreement a small number of physical quantities such as length, time and mass (called Fundamental quantities) have been assigned standard units. We can define all other physical quantities in terms of these fundamental ones. The units of these fundamental quantities are standard units (Table 1.1) and are accessible and invariable.

TABLE 1.1

SI Units for Fundamental Quantities

Quantity	*Unit Name*	*Unit Symbol*
Length	metre	m
Time	second	s
Mass	kilogram	kg
Temperature	kelvin	K
Electric Charge	Coulomb	C
Amount of a substance	mole	mol

Derived Units

From the fundamental quantities set up by the International System of Units (SI), all other units of measurement are formed. These other units are called SI derived units. Derived units are also considered part of the standard.

The names of the base SI units are always written in lowwercase. The unit symbols named after persons are, however, always spelled with an initial capital letter. For example the symbol of pascal is Pa; but metre which is not a person's name becomes m.

TABLE 1.2

SI Units for Derived Quantities

Quantity	*Unit Name*	*Unit Symbol*
Angle	radian	rad
Solid angle	steradian	sr
Area	square metre	m^2
Volume	cubic metre	m^3
Density	kilogram per cubic metre	kg/m^3
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s^2
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s^2
Force	Newton	N ($kg\ m/s^2$)
Torque	newton-metre	Nm
Work, Energy, heat	joule	J (N m)
Power, radiant flux	watt	W
Pressure, stress	pascal	Pa (N/m^2)
Frequency	hertz	Hz (1/s)
Electric current	ampere	A
Voltage	Volt	V (W/A)
Electric field	volt per meter	V/m
Electrical resistance	ohm	Ù (V/A)
Capacitance	farad	F (A s/V)
Magnetic flux	wèber	Wb (V s)
Inductance	henry	H (V s/A)
Magnetic Flux density	tesla	T (Wb/m^2)
Luminous flux	lumen	Lm
Luminance	lux	lx
Activity (radioactive)	becquerel	Bq (s^{-1})
Absorbed dose	gray	Gy (J/kg)
Dose equivalent	sievert	Sv (J/kg)

Standard Prefixes

In physics we sometimes deal with numbers that are very small and other times we deal with numbers which are very large. It gets cumbersome to write numbers in the conventional decimal notation. Once the fundamental units are defined, it now becomes easy to introduce larger and smaller units for the same physical quantities. In the metric system measurements additional units always relate to the fundamental units by multiples of 10 or or $\frac{1}{10}$. The multiplicative factors are most conveniently expressed in exponential notation; thus, $10000 = 10^4$, $\frac{1}{100} = 10^{-2}$ etc. The names of the additional units are always derived by adding a prefix to the name of the fundamental units.

TABLE 1.3

Prefixes for the Power of Ten

Power of ten	*Prefix*	*Abbreviation*
10^{-24}	yocto-	y
10^{-21}	zepto-	z
10^{-18}	atto-	a
10^{-15}	fermto-	f
10^{-12}	pico-	p
10^{-9}	nano-	n
10^{-6}	micro-	μ
10^{-3}	milli-	m
10^{-2}	centi-	c
10^{+3}	kilo-	k
10^{+6}	mega-	M
10^{+9}	giga-	G
10^{+12}	tera-	T
10^{+15}	peta-	P
10^{+18}	exa-	E
10^{+21}	zetta-	Z
10^{+24}	yotta-	Y

The following are a few examples of the use of the multiples of ten and their prefixes.

1 nanometre = 1nm = 10^{-9} m
1 micrometre = 1μm = 10^{-6} m

1 millimetre	= 1mm	=	10^{-3} m
1 centimetre	= 1cm	=	10^{-2} m
1 kilometre	= 1km	=	10^{3} m
1 milligram	= 1mg	=	10^{-6}kg
1 gram	= 1g	=	10^{-3} kg
1 nanosecond	= 1ns	=	10^{-9} s
1 microsecond	= 1ms	=	10^{-6}s

Unit Conversions

Relations among physical quantities are often expressed by equations in which physical quantities are represented by algebraic symbols. An algebraic symbol always denotes both a number and a unit. For example, x might represent distance of 5m, t a time of 30s and m a mass of 10kg. We often need to convert the units in which a physical quantity is expressed. We do so by multiplying the original measurement by a conversion factor which is a ratio of the units in question that equals unity. For example:

$$\frac{1\min}{60s}=1 \text{ and } \frac{60s}{1\min}=1$$

This is so because multiplying any physical quantity (or algebraic symbol) by unity leaves the quantity unchanged. For example we can convert 5min into seconds by multiplying it by the conversion factor of $\frac{60s}{1\min}$ to have

$$5\min=(5\min)(1)=(5\min)(\frac{60s}{1\min})=300s$$

Dimensions

The fundamental quantities used in physical descriptions are called dimensions. For example length, time and mass are dimensions. You could measure the distance between two points and express it in units of metres, centimetres, or feet, but the quantity would still have the dimension of length. It is common to express dimensional quantities by bracketed symbols as [L], [M] and [T] for length, mass and time respectively. Derived quantities are a combination of dimensions. For example, velocity v has dimensions of $[L]/[T]$ (think of it as metres/second = length/time) and volume, V has dimensions $[L] \times [L] \times [L]$, or $[L^3]$.

Example 1.1

What are dimensions (or the dimension of a physical quantity)? What is the dimension of acceleration?

Solution

The dimensions of a physical quantity are the powers (or exponents) to which the base or the fundamental quantities are raised to represent that quantity. For example, acceleration can be expressed as

$$acceleration=\frac{length}{(time)^2}=\frac{[length]}{[time]^2}=\frac{L}{T^2}=LT^{-2}$$

The dimensions of acceleration are then given by $[acceleration] = [LT^{-2}]$

Thus, we conclude that, acceleration has zero or no dimension in mass, one dimension in length, and –2 dimensions in time. Any quantity raised to zero is equal to 1. So, all those fundamental quantities that do not appear in the dimensional formula of acceleration have dimension 0. For example, here, the dimension of mass is 0.

Dimensions and Unit Analysis

Any equation must always be dimensionally consistent or homogenous. This means that any two terms added or subtracted in an equation must have the same units. For example, given the equation; 3.0m + 2.0m = 5.0m (similar to $a+b=c$) we see that both sides we have $[L]+[L]=[L]$ or in words length plus length gives you another length not $2[L]$. The above procedure for checking units in equations is called *dimensional analysis*. It is formally defined as a procedure by which the dimensional consistency of any equation may be checked. One major advantage of dimensional analysis is its usefulness for checking whether an equation that has been derived or being used to solve a problem has the correct form. The dimensions and units of some common quantities are given in Table 1.3.

Example 1.2

What are dimensional equations?

Solution

Dimensional equations are equations which represent the dimensions of a physical

quantity in terms of the fundamental quantities. When one equates a physical quantity with its dimensional formula then we have a dimensional equation of the physical quantity. The dimensional equations can be obtained from the equations representing the relations between the physical quantities. This is something that we have been doing, writing the quantity on the left hand side of the equation and its dimensions on the right hand side.

Example 1.3

In the gas equation $(p + \frac{a}{V^2})(V - b) = RT$, what are the dimensions of the constants a and b?

Solution

The dimensions of the terms in the equation are as follows:

$$[p] = [\text{Pressure}] = \frac{[\text{Force}]}{[\text{Area}]} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

$$[V] = [volume] = L^3$$

Since $\frac{a}{V^2}$ is added to pressure and we assume the given equation to be homogenous it must be pressure.

$$\Rightarrow \left[\frac{a}{V^2}\right] = \frac{[a]}{[V^2]} = \frac{[a]}{L^6} = [p] = ML^{-1}T^{-2}$$

$$\therefore [a] = ML^{-1}T^{-2} \times L^6 = ML^5T^{-2}$$

The constant b must represent a volume since it is being subtracted from a volume
$\Rightarrow [b] = L^3$

Example 1.4

By using dimensional analysis, derive an expression for the volume V of water passing any section of a pipe in time (i.e. the rate of volume flow V/t) if it depends on (a) the coefficient of viscosity η of the liquid (*Note:* $[\eta] = ML^{-1}T^{-1}$) (b) the radius r of the pipe (c) the pressure gradient $\Delta p/l$, where Δp is the pressure difference between the ends of the pipe and l its length.

Solution

We express the relations of the flow rate to the viscosity, radius and pressure gradient as:

$$\frac{V}{t} = k\eta^x r^y \left(\frac{\Delta p}{l}\right)^z$$

Where k is a dimensionless constant and x, y and z are unknown indices. Since each side of the equation must have the same dimensions we can write

$$\left[\frac{V}{t}\right] = [\eta^x][r^y]\left[\left(\frac{\Delta p}{l}\right)^z\right]$$

$$\therefore L^3T^{-1} = (ML^{-1}T^{-1})^x (L)^y (ML^{-2}T^{-2})^z$$

$$\Rightarrow L^3T^{-1} = M^x L^{-x} T^{-x} \cdot L^y \cdot M^z L^{-2z} T^{-2z}$$

$$\Rightarrow M^0 L^3 T^{-1} = M^{x+z} L^{y-x-2z} T^{-x-2z}$$

Equating the indices of M, L and T on both sides

For M: $0 = x + z$ (1)
For L: $3 = y - x - 2z$ (2)
For T: $-1 = -x - 2z$ (3)

Solving equations 1, 2 and 3 simultaneously gives $x = -1$, $z = 1$ and $y = 4$. The relationship is therefore

$$\frac{V}{t} = k\frac{r^4}{\eta}\left(\frac{\Delta p}{l}\right)$$

This relation is called Poiseuille's Law or relation (note that $k = \pi/8$)

Example 1.5

Consider a sphere of radius r moving with velocity v through a fluid whose coefficient of viscosity is η. The sphere experiences a viscous force F which acts in the opposite direction to that in which the sphere is moving. Use dimensional analysis to obtain an expression for $F \propto r\eta v$ (where $\propto$ denotes proportionality).

Solution

$F = kr^x \eta^y v^z$

$[F] = [r^x][\eta^y][v^z]$

$MLT^{-2} = L^x (ML^{-1}T^{-1})^y (LT^{-1})^z$

$M^1 L^1 T^{-2} = M^y L^{x-y+z} T^{-y-z}$

For M: $1 = y$ (1)

For L: $1 = x - y + z$ (2)

For T: $-2 = -y - z$ (3)

Solving equations 1, 2 and 3 simultaneously gives $y = 1$, $z = 1$ and $x = 1$. Thus

$F = kr\eta v$

Note that $k = 6\pi$.

Example 1.6

Check if the following equations are dimensionally correct:

(a) $x = at$ (b) $x = at^2$, where x is distance, a is acceleration and t is time.

Solution

(a) Equation can be expressed dimensionally as

$[L] = \frac{[L]}{[T^2]} \times [T] = \frac{[L]}{[T]}$ which is not true. Thus cannot be a correct equation.

(b) Equation can be expressed dimensionally as $[L] = \frac{[L]}{[T^2]} \times [T^2] = [L]$. This equation is dimensionally correct but as you will see later it is not always physically correct. The correct form of $x = at^2$ both dimensionally and physically is $x = \frac{1}{2}at^2$. The fraction $\frac{1}{2}$ has no dimensions i.e. it is a dimensionless number.

Doing dimensional analysis with the symbols is fine but in practice it is often more convenient to use actual unit abbreviations such as m, s, and kg. Units can also be treated as symbols and canceled. Using units instead of symbols in dimensional analysis is called units analysis.

Exercise 1.7

Is the equation $s = ut + 1/2\,at^2$ dimensionally consistent?

Solution

We are given $s = ut + 1/2\,at^2$ or
s(displacement) = u(initial velocity) X t(time) + 1/2 X a(acceleration)X t^2(time)

We solve by going through the following steps:

Step 1: Identify the terms in the equation and identify which symbol stands for which physical quantity. There are three terms here: s, ut, at^2

Step 2: Write down the dimensional formula of each symbol used in the equation
[s] = [L], [u] = [LT^{-1}], [t] = [T], [a] = [LT^{-2}]

Step 3: Calculate the dimension of each term in the equation.
Term 1: [s] = [L]
Term 2: [ut] = [LT^{-1}.T] = [L]
Term 3: [at^2] = [LT^{-2}.T^2] = [L]

Note that, as stated above, we have canceled dimensions from numerator and denominator like [T^{-2}.T^2].

Conclusion

The equation is dimensionally consistent since all the terms have the same dimensions.

Note: The following kinds of terms are dimensionless in an equation.

- A pure number or the ratio of similar physical quantities, such as angle as the ratio (length/length), refractive index as the ratio (speed of light in vacuum/speed of light in medium) etc are dimensionless.
- The arguments of special functions, such as the trigonometric, logarithmic and exponential functions are also dimensionless.

Example 1.8

Check the dimensional consistency of the following equations:

(a) $K = (1/2)mv^2 + ma$ (b) $K = (1/2)mv^2$ (c) $K = (3/16)mv^2$

Solution

(a) The first term (on the LHS),

K = *Kinetic Energy* = *Work* = Force × Displacement

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$$[(1/2)mv^2] = [1/2][m]\cdot[v^2] = [m]\cdot[v]^2 = [M]\cdot[LT^{-1}]^2 = [M]\cdot[L^2T^{-2}] = [ML^2T^{-2}]$$

The third term involves, m = Mass and a = acceleration

$$[ma] = [m][a] = [M][LT^{-2}] = [MLT^{-2}]$$

It can be recognized that the dimensions of the first two terms are the same but that of the third term differs. Therefore, the equation $K = (1/2)mv^2 + ma$ is dimensionally inconsistent.

(b) The first term (on the LHS),

K = *Kinetic Energy* = *Work* = Force × Displacement

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$$[(1/2)mv^2] = [1/2][m]\cdot[v^2] = [m]\cdot[v]^2 = [M]\cdot[LT^{-1}]^2 = [M]\cdot[L^2T^{-2}] = [ML^2T^{-2}]$$

Since the dimensions of the two terms of the equation are dimensionally similar then the equation $K = (1/2)mv^2$ is dimensionally consistent.

(c) Just like in (a) and (b), the first term (on the LHS),

K = *Kinetic Energy* = *Work* = Force × Distance

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$$[(3/16)mv^2] = [3/16][m]\cdot[v^2] = [m]\cdot[v]^2 = [M]\cdot[LT^{-1}]^2 = [M]\cdot[L^2T^{-2}] = [ML^2T^{-2}]$$

Since the dimensions of the two terms of the equation are dimensionally similar then the equation $K = (3/16)mv^2$ is dimensionally consistent. Note that the dimension of a number, an argument or a ratio is unity.

TABLE 1.4

Dimensions and Units

Quantity	*Dimension*	*Unit*
mass	$[M]$	kg
time	$[T]$	s
length	$[L]$	m
area	$[L^2]$	m²
volume	$[L^3]$	m³
velocity	$[L]/[T]$	m/s
Acceleration (a or g)	$[L]/[T^2]$	m/s²

Precision and Accuracy

Accuracy is a measure of how close a result of an experimental observation comes to the "true" value. Precision on the other hand is a measure of how carefully a result is determined without reference to any "true" value. Measurements can never be made with absolute precision in that physical quantities obtained from experimental observations always have some uncertainty. A distance measured with an ordinary ruler or metre rule is usually precise only to the nearest millimetre, whilst a precision micrometre caliper can measure a distance to the nearest 0.01mm or even less. The precision of a measurement or number is often indicated by following the number with the symbol ± and a second number indicating the maximum likely error. Errors are defined as the difference between measured experimental values and the "true" value. It is generally applied to the uncertainty in measurements but not blunders or mistakes. If for example, the diameter of a metal rod is given as 48.64±0.01mm, then it means that the true value is likely to be less than 48.65 or greater than 48.63. Precision can also be expressed in terms of the maximum likely fractional error or percentage error. A resistor labeled 47Ω, 10 percent probably has true resistance differing from 47Ω by no more than 10 percent of 47Ω or about 5Ω i.e. the true resistance is between 42Ω and 52Ω.

Significant Figures

If a student is asked to find the numerical value of π (*pi*) by using a calculator, after measuring the diameter and circumference of a circle as 135mm and 424mm respectively, *pi* is the ratio of the circumference to the diameter of a circle. They get an answer like 3.140740741.... Depending on the type of calculator, they could get decimal places more than the number stated. Whether this number or any other agrees with the true value of ð (correct value to ten digits is 3.141592654) or not is the question. To answer this question, we must recognize that at least the last six digits in the student calculation are meaningless unless they imply a greater precision in the result than possible with their measurement of π. The number of meaningful digits in a number is called the number of significant figures. In general, no numerical result can have more significant figures than do numbers from which it was computed. Thus the student value of π has only three significant figures and should be stated as 3.14 or (3.141 when rounded of to four figures). The rules for significant figures and rounding offs are as follow.

Rules for Significant Figures

1. The leftmost non-zero digit is the most significant digit/figure.
2. If there is no decimal point, the rightmost nonzero digit is the least significant digit/figure.
3. If there is a decimal point, the rightmost digit is the least significant digit/figure, even if it is zero.
4. All digits between the least and the most significant digits are counted as the significant figures.

Round off

Rounding off is simply to truncate a number to the specified number of significant figures and treat the excess (remainder) digits as a decimal fraction.

1. If the fraction is equal to or greater than ½ (0.5) then increase the new least significant figure by one.
2. If the fraction is less than ½ (0.5) then leave the new least significant figure unchanged.

Rules for Reporting with the proper Significant Figures

Addition/Subtraction

When adding or subtracting numbers, leave the same number of decimal places in the answer as there is in the number with the least number of significant figures.

For example, suppose we need to add 701.3, 104.0, 41.216 and 0.5462. The result from the calculator is 847.0622. Now, look at the data and observe that in the two of the numbers being added it is not clear what the figure in the second decimal place is. The result must be rounded off to one decimal place by dropping 622. Hence, the answer with proper significant numbers is 847.0.

Multiplication/Division

When multiplying and dividing numbers, leave as many significant figures in the answer as there are in the number with the fewest number of significant figures. Usually, multiplication and division produce more figures that are significant than the result must be rounded off.

For example, suppose the measured length and width of a plate is 2.64cm and 1.18cm respectively. The area, therefore, equals $2.64cm \times 1.16cm = 3.1152cm^2$. The result appears to have five significant figures, although the original quantities have three each. Therefore, the result can only have three significant figures and is $3.12cm^2$ (rounded off to three significant figures i.e. rounded to the second decimal place).

Example 1.9

Express the following numbers in scientific notation with three significant figures: (a) 0.00241 (b) 79,241 (c) 0.000452.

Solution

The numbers are expressed in scientific notation with three significant as:

(a) $0.00241 = 2.41 \times 10^{-3}$
(b) $79{,}241 = 7.92 \times 10^{4}$ [Note: 7.9241 has been rounded off to 7.92]
(c) $0.0000452 = 4.52 \times 10^{-5}$

Example 1.10

Express the following numbers as their decimal equivalents:

(a) 49.6×10^{-2} (b) -6.50×10^{-4} (c) 0.4291×10^{3} (d) 1.21×10^{0}

Solution

The decimal equivalents are as follows.

(a) $= 0.496 \times 10^{-2} = 0.496$ [by moving the decimal point two places left]
(b) $-6.50 \times 10^{-4} = -0.000650$ [by moving the decimal point four places left]
(c) $0.4291 \times 10^{3} = 429.1$ [by moving the decimal point three places right]
(d) $= 12.1$ [by moving the decimal point three places right]

Example 1.11

Express the final result of the following expressions in scientific notation:

(a) 51.5×0.0000521 (b) $\dfrac{2348}{83.1 - 91.8}$

Solution

(a) 51.5×0.0000521

First, express each term in the expression in scientific notation before final computation.

51.5×0.0000521
$= (5.15 \times 10^{1}) \times (5.21 \times 10^{-5})$
$= (5.15 \times 5.21) \times 10^{-4} = 26.8315 \times 10^{-4}$

The result is not correctly written in scientific notation. To get the decimal place in the right place, we move one decimal place to the left, increasing the exponent by +1. Thus, the final answer in scientific notation is 2.68315×10^{-3}. Keeping the results up to three significant figures, we round off the result and rewrite it as 2.68×10^{-3}.

(b) We first express each number in the fraction in scientific notation.

$$\frac{2348}{83.1 - 91.8} = \frac{2.348 \times 10^{3}}{8.31 \times 10^{1} - 9.18 \times 10^{1}} = \frac{2.348 \times 10^{3}}{-0.87 \times 10^{1}} = -2.70 \times 10^{2}$$

Here the result is shown in scientific notation up to three significant figures.

Equations and Greek Symbols

Different symbols and letters are used when stating and solving equations in physics. Table 1.5 shows symbols used in writing physical equations and Table 1.6 shows Greek letters are often used to represent physical quantities in equations.

TABLE 1.5

Symbols and letters in physical equations

Symbol	*Meaning*	*Symbol*	*Meaning*
=	Equal to	$\therefore$	Therefore
$\approx$	Approximately equal to	Σ	Sum
$\sim$	Same order as	Π	Product
>	Greater than	$\sqrt{\ }$	Square root
$\geq$	Greater than or equal to	$x!$	Factorial of x
<	Less than	$\neq$	Not equal to
$\leq$	Less than or equal to	∞	Infinity
>>	Much greater than	$\propto$	Proportional to
<<	Much less than	$\pm$	Plus or minus
$\equiv$	Equivalent to	$\mp$	Minus or plus
$\Rightarrow$	Implies	Δx	Change in x
/	Divided by		

TABLE 1.6

Greek Letters

Letter	*Uppercase*	*Lowercase*	*Letter*	*Uppercase*	*Lowercase*
Alpha	A	α	Nu	N	ν
Beta	B	β	Xi	Ξ	ξ
Gamma	Γ	γ	Omicron	O	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	e	Rho	P	ρ
Zeta	Z	ζ	Sigma	Σ	σ
Eta	H	η	Tau	T	τ
Theta	Θ	θ	Upsilon	Υ	υ
Iota	I	ι	Phi	Φ	ϕ
Kappa	K	κ	Chi	X	χ
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	M	μ	Omega	Ω	ϖ

Exercise 1

1.1 Show that the expression $V=V_o + at$ is dimensionally correct, where V and V_o represent velocities, a is acceleration, and t is a time interval

1.2 Suppose we are told that the acceleration a of a particle moving with uniform speed V in a circle of radius r is proportional to some power of r, and some power of v. Use dimensional analysis to predict the form of the relation. (**Ans:** $a = Kr^{-1}V^2 = KV^2/r$)

1.3 If $x = a + bt + ct^2$ where x is in metres (m) and t in seconds (s), then what is the unit of c? (**Ans:** ms^{-2})

1.4 The frequency of vibration, f, of a mass m, at the end of a spring that has stiffness constant k is related to m and k by the relation of the form $F = (\text{constant}) m^a k^b$. Use dimensional analysis to find a and b. It is known that $[F] = [T]^{-1}$ and $k = [M][T]^{-2}$ (**Ans:** $a = -1/2$; $b = 1/2$)

1.5 Use dimensional analysis to determine which of the following equations is certainly wrong: $s = vt$, $F = m/a$, $F = mv/t$, $h = v^2/2g$, $V = (2gh)^{1/2}$ (**Ans: F=m/a**)

1.6 The micrometre (1lm) is often called the micron. How many microns make up 1.0km? (**Ans: 10^9 lm**)

1.7 Express the speed of light $3.0 \times 10^8 ms^{-1}$ in millimeters per picosecond (mm/ps). (**Ans: 0.3mm/ps**)

1.8 How many significant figures are there in the figure, $0.007m^2$? (Ans: 1)

1.9 How many significant figures do we have in the following? (i) $2.642 \times 10^{24} \times 10^{-24}$ kg (ii) 0.0006032. (**Ans: [i] 3, [ii] 4**)

1.10 The significant figures in $0.2370 gcm^{-3}$ is.....(Ans: 4)

Chapter Two

VECTORS

Vectors and Scalars

Some physical quantities are scalar quantities whereas others are vector quantities. When a particle is confined to a straight line, it moves in only two directions. It is positive in one direction and negative in the other. In order to know the temperature outside so that you will know how to dress, the only information you need to know is a number and the unit. Not all physical quantities involve a direction. A quantity like temperature is an example of a scalar quantity.

A scalar quantity is specified by a single value with an appropriate unit and has no direction. That is, a scalar quantity has only magnitude but no direction. Other examples of scalar quantities are volume, mass, time intervals, pressure, and energy.

If a particle is not confined to a straight line but moves in three dimensions, however, a plus sign or minus sign is no longer enough to indicate the direction of the motion. To describe such a particle a vector is needed.

A vector quantity has magnitude as well as direction. Examples of vector quantities are displacement and velocity.

Components of a Vector

Sometimes graphical method of adding vectors is not recommended whenever greater accuracy is required or in three-dimensional problems as it can be very tedious. An easier technique involves algebra but with the requirement that the vector be placed on a rectangular coordinate system so that it is possible to make projections of the vector along coordinate axes. These projections are called the components of the vector.

Consider a vector A lying in the *xy* plane and making an arbitrary angle è with the positive *x* axis as shown in Figure 2.1.

This vector can be expressed as the sum of two other vectors $\mathbf{A}_x$ and $\mathbf{A}_y$. From Figure 2.1, the three vectors form a right-angled triangle and that $\mathbf{A} = \mathbf{A}_x + \mathbf{A}_y$. Often refered to as "the components of a vector A," written A_x and A_y (with no boldface). The component A_x represents the projection of **A** along the *x* axis, and the component A_y represents the projection of **A** along the *y* axis. If $\mathbf{A}_x$ points in the positive *x* direction the component A_x is positive and *vice versa*. The same is true for the component A_y. From Figure 2.1and the definition of sine and cosine, then

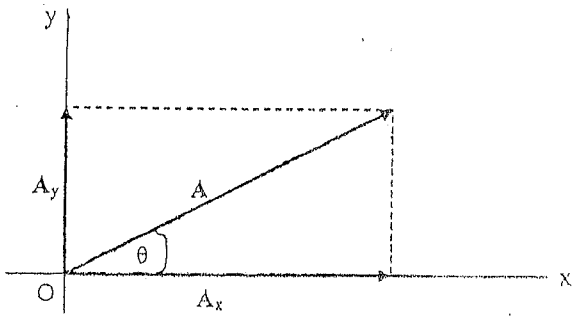


Fig. 2.1: Any vector A lying in the xy plane can be represented by a vector A_x lying along the x axis and by a vector A_y lying along the y axis, where $A = A_x + A_y$.

$\cos\theta = A_x/A$ and $\sin\theta = A_y/A$. Hence the components of A are

$$A_x = A\cos\theta \qquad 2.1$$

$$A_y = A\sin\theta \qquad 2.2$$

It follows from Figure 2.1 that the magnitude and direction of A are related to its components through the expressions

$$A = \sqrt{A_x^2 + A_y^2} \qquad 2.3$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right) \qquad 2.4$$

Notice that the signs of the components A_x and A_y depend on the angle θ. As an example, if θ = 120°, then A_x is negative and A_y is positive. If θ = 225°, then both and are negative.

Unit Vectors

A unit vector is a vector that has a magnitude of 1 and points in a particular direction. Unit vectors are used to specify a given direction and have no other physical significance so that it lacks both dimension and unit. The unit vectors in the positive directions of the *x, y, z* axes are represented by and as shown in Figure 2.2. The unit

vectors form a set of mutually perpendicular vectors in a right-handed coordinate system as shown in Figure 2.2. The magnitude of each unit vector equals 1; that is $|\mathbf{i}| = |\mathbf{j}| = |\mathbf{k}| = 1$.

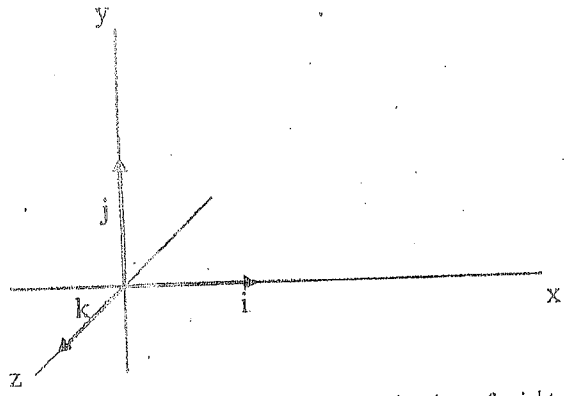


Fig. 2.2: Unit vectors i, j and k define the directions of a right-handed rectangular coordinate system.

Consider a vector A lying in the xy plane as shown in Figure 2.3 The product of the component A_x and the unit vector i is the vector $A_x\mathbf{i}$, which lies on the x axis and has magnitude $|A_x|$. $A_y\mathbf{j}$ is a vector of magnitude $|A_y|$ lying on the y axis.

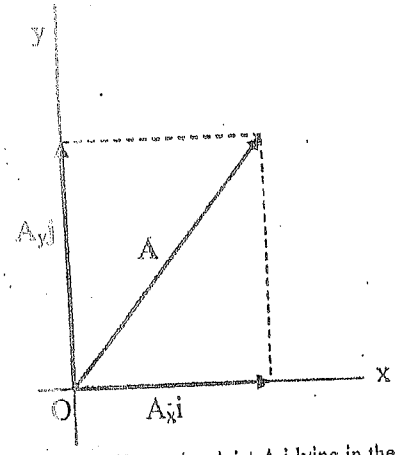


Fig. 2.3: Vector $A = A_x\mathbf{i} + A_y\mathbf{j}$ lying in the xy plane has components A_x and A_y

The unit vector notation for the vector A then is

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j}$$

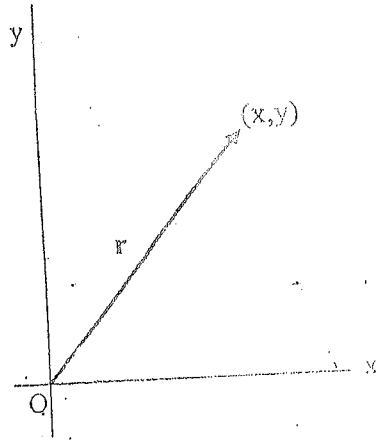


Fig. 2.4: The point whose Cartesian coordinates are (x, y) can be represented by the position vector r = xi + yj

If a point lies in the plane with cartesian coordinates (x, y) as shown in Figure 2.4, then its **position vector** in unit-vector form is given by

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$$

Equation 2.6 tells us that the components of r are the lengths x and y. If we want to add vectors A and B in component form so that the resultant is R, then,

$$\mathbf{R} = (A_x\mathbf{i} + A_y\mathbf{j}) + (B_x\mathbf{i} + B_y\mathbf{j})$$

or

$$\mathbf{R} = (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j}$$

But $\mathbf{R} = R_x\mathbf{i} + R_y\mathbf{j}$, so that the components of the resultant vector are

$$R_x = A_x + B_x$$
$$R_y = A_y + B_y \qquad 2.9$$

The magnitude of $\mathbf{R}$ then becomes

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(A_x + B_x)^2 + (A_y + B_y)^2} \qquad 2.10$$

and the angle that $\mathbf{R}$ makes with the x axis is

$$\tan\theta = \frac{R_y}{R_x} = \frac{A_y + B_y}{A_x + B_x} \qquad 2.11$$

If $\mathbf{A}$ and $\mathbf{B}$ both have x, y and z components, they can be expressed in the form

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$$
$$\mathbf{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k} \qquad 2.12$$

The sum of $\mathbf{A}$ and $\mathbf{B}$ is

$$\mathbf{R} = (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j} + (A_z + B_z)\mathbf{k} \qquad 2.13$$

Example 2.1

An expeditionary team in search of antiques in a cave travels a net 2.6km westward, 3.9km southward, and 25m upward. What was their displacement vector?

Solution

Let their horizontal displacement be d_h. Then by the Pythagoras theorem the magnitude of d_h is

$$d_h = \sqrt{(2.6km)^2 + (3.9km)^2} = 4.69km$$

The angle θ of their displacement vector relative to the west is given by

$$\tan\theta = \frac{3.9km}{2.6km} = 1.5$$

or

$$\theta = \tan^{-1} 1.5 = 56^0$$

We now find the magnitude of the overall displacement d,

$d = \sqrt{(4.69km)^2 + (0.025km)^2} = 4.69km$ and the angle ϕ,

$$\phi = \tan^{-1}\frac{0.025km}{4.69km} = 0.3^0$$

Example 2.2

A car travels 20km due north and then 35km in a direction west of north, as shown in Figure 2.4b. Find the magnitude and direction of the car's resultant displacement.

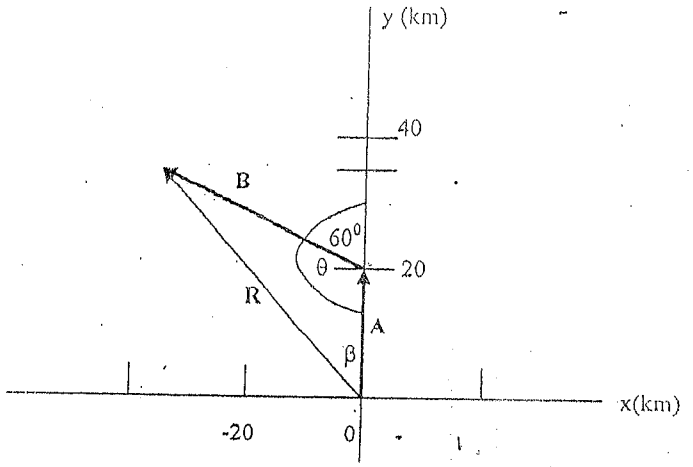


Fig. 2.4b: Graphical method for finding the resultant displacement vector R = A + B

Exercise 2

2.1 A force of 3N acts at 90° to a force of 4N. Find the magnitude and direction of their resultant, R. (Ans: R=5N, θ=37°)

2.2 A particle which is moving due east at $4ms^{-1}$ changes direction and starts to move due south at $3ms^{-1}$. Find the change in velocity. (Ans: velocity = $5ms^{-1}$ at 37° S of W)

2.3 Calculate the horizontal and vertical components of a force of 50N which is acting at 40° to the horizontal. (Ans: Horizontal = 38N; Vertical = 32N)

Use the information below to answer questions 2.4–2.6. Consider two vectors $\mathbf{A} = 3\underline{i} - 2\underline{j}$ and $\mathbf{B} = -\underline{i} - 4\underline{j}$. Calculate:

2.4 (i) 2.1 $\mathbf{A} + \mathbf{B}$; (ii) $\mathbf{A} - \mathbf{B}$. (Ans: [i] $2\underline{i} - 6\underline{j}$, [ii] 4+2)

2.5 (i) 2.1 $|\mathbf{A} + \mathbf{B}|$; (ii) $|\mathbf{A} - \mathbf{B}|$. (Ans: [i] $\sqrt{40}$ units, [ii] $\sqrt{20}$ units)

2.6 The direction of (i) $|\mathbf{A} + \mathbf{B}|$; (ii) $|\mathbf{A} - \mathbf{B}|$ (Ans: [i] θ=288°, [ii] θ=26.57°)

2.7 The resultant force on a body of mass 4.0kg is 20N. What is the acceleration of the body? (Ans: $5.0ms^{-2}$)

2.8 A body of mass 3.0kg slides down a plane which is inclined at 30° to the horizontal. Find the acceleration of the body: ($g=10ms^{-2}$) (a) if the plane is smooth (b) if there is a frictional resistance of 9.0N. (Ans: [a] $5.0ms^{-2}$, [b] $2.0ms^{-2}$)

2.9 A body of mass 6.0kg moves under the influence of two oppositely directed forces whose magnitudes are 60N and 18N. Find the magnitude and direction of the acceleration of the body. (Ans: $7.0ms^{-2}$ in the direction of 60N force)

2.10 A body of mass 5.0kg is pulled along smooth horizontal ground by means of a force of 40N acting at 60° above the horizontal. Find: (a) the acceleration of the body (b) the force the body exerts on the ground. ($g = 10ms^{-2}$) (Ans: [a] $4ms^{-2}$, [b] 15.4N)

Chapter Three

DYNAMICS

Mechanics

Mechanics is the branch of physics that deals with the study of motion and what produces and affects motion. Mechanics is usually divided into two parts: (1) *Kinematics* and (2) *Dynamics*. Kinematics deals with the description of motion of bodies without consideration of what causes the motion. Dynamics analyses the causes of the motion.

Force

Force is that which *changes a body's state of rest or of uniform motion in a straight line.* It is measured in Newtons (N). One Newton is defined as *the force required to give a mass of one kilogram an acceleration of one metre per second square.*

A force will act on a body to do the following.

1. to get the body to start moving,
2. to increase or decrease the speed of a body in motion,
3. to change its direction of motion in a straight line,
4. to bring a moving body to rest,
5. to deform the body.

Types of Forces

There are several types of forces. These include:

1. *Frictional force:* Friction is the name given to the force which opposes the relative sliding motion of two surfaces in contact with each other. This type of force exists between two solid surfaces in contact. An example is the opposing force between the sole of your sandals and the ground.
2. *Viscous drag:* This is the frictional force experienced in fluids (gases and liquids).
3. *Tension:* This force is experienced in a stretched material. If you touch the

...material your fingers will feel a pull or tautness because of the potential ...ergy set up in it.

4. *Centripetal force:* This force acts on objects to keep them moving in circular ...ths. Its direction is always towards the centre of the circular path. It can ... be defined as that part of the total gravitational force which is required to constrain the body to move in its circle of latitude.

5. *Force of gravity:* It is that part of the total gravitational force which acts on a body and so enables it in turn to exert an equal force on its support. This force on the support is called the body's weight.

6. *Gravitational force:* It is the force of attraction existing between two masses.

7. *Electrostatic force:* This force is produced by electric charges and electric currents. Electric charges tend to exert attractive or repulsive forces on each other depending on the sign of the charges. Like charges attract and unlike charges repel each other.

8. *Magnetic force:* This force is produced by magnets, and current-carrying ...nductors (e.g. two wires in which current is flowing).

...ss and Weight

...ss is a property of matter equal to the measure of an object's resistance to changes in either speed or direction of its motion. Its SI unit is the kilogram (kg). Mass is a scalar quantity and can be measured with a beam balance. There are other devices like electronic balance which can be used to measure the mass of an object. *The mass of an object is the same irrespective of where it is measured.* This means that if the mass of an object is measured both on the moon and on the earth the measuring device will give the same readings.

The weight of a body is the force it exerts on anything which freely supports it. This is due to the Earth's pull on it by the force of gravity. It is measured in Newtons. Spring balance can be used to measure the weight of an object. *The weight of an object depends greatly on the gravitational pull on it,* therefore, it varies from one place to another place. This means that if the weight of an object is measured ...tween the moon and on the earth the measuring device will give different readings.

...elation between Mass and Weight

The relationship between mass and weight is given by the expression

Weight (W) = mass (m) x acceleration due to gravity (g)

$$W = mg \qquad 3.1$$

Newton's Laws of Motion

Newton's First Law

Newton's first law of motion is also known as the law of inertia. *The tendency of a body to remain at rest or if moving to continue its motion in a straight line* is called **inertia.** So we can say that inertia is the reluctance of a moving body to stop and a body at rest to start moving. Newton's first law of motion states that *everybody continues in its state of rest or uniform motion in a straight line unless compelled by a net external force to act otherwise.*

Newton's Second Law

Newton's Second Law of motion states that *the time rate of change of momentum of a body is directly proportional to the applied force acting on it and it takes place in the direction of the force.*

This law enable scientists to define the unit of a force i.e. the Newton, N. Suppose a force F acts on a body of constant mass m for a time t and causes its velocity to change from u to v, then according to the second law, for a constant acceleration in the time interval t,

$$F \propto \frac{mv - mu}{t}, \text{ from which}$$

$$F = k\frac{mv - mu}{t} \text{ and}$$

$$F = km\frac{(v-u)}{t} \qquad 3.2$$

$$\text{but } a = \frac{(v-u)}{t}$$

$$\Rightarrow F = kma \qquad 3.3$$

[Note: If m is not constant, then,

$$F \propto \frac{d}{dt}(mv) = m\frac{dv}{dt} + v\frac{dm}{dt}\ ]$$

Solution

This problem can be solved in two different ways: (1) Geometrical method using graph paper and protractor and (2) by calculation. The geometrical method is as in Figure 2.4b. The displacement $\mathbf{R}$ is the resultant when the two individual displacements $\mathbf{A}$ and $\mathbf{B}$ are added.

Algebraically, the magnitude of the resultant $\mathbf{R}$ can be obtained from the law of cosines as applied to the triangle. The angle is given as

$$\theta = 180^0 - 60^0 = 120^0 \text{ and } R^2 = A^2 + B^2 - 2AB\cos\theta,$$

so that

$$R = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

$$R = \sqrt{(20km)^2 + (35km)^2 - 2(20km)(35km)\cos 120^0}$$

$$R = 48.2km$$

From the law of sines, the direction of $\mathbf{R}$ is

$$\frac{\sin\beta}{B} = \frac{\sin\theta}{R}$$

$$\sin\beta = \frac{B}{R}\sin\theta = \frac{35km}{48.2km}\sin 120^0 = 0.629$$

$$\beta = 38.9^0$$

The resultant displacement of the car is $48.2km$ in a direction 38.9^0 west of north.

Example 2.3

Find the sum of two vectors $\mathbf{A}$ and $\mathbf{B}$ lying in the xy plane and given by $\mathbf{A} = (2\mathbf{i} + 2\mathbf{j})\,m$ and $\mathbf{B} = (2\mathbf{i} - 4\mathbf{j})\,m$.

Solution

The general expression for $\mathbf{A}$ is $\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j}$ and comparing, we see that $A_x = 2m$ and $A_y = 2m$. Also, $B_x = 2m$ and $B_y = -4m$. Using Equation 2.8 to obtain the resultant vector $\mathbf{R}$ we have

$$\mathbf{R} = \mathbf{A} + \mathbf{B} = (2+2)\mathbf{i}\,m + (2-4)\mathbf{j}\,m$$

$$= (4\mathbf{i} - 2\mathbf{j})\,m$$

or

$$R_x = 4m \text{ and } R_y = -2m$$

The magnitude of $\mathbf{R}$ is given by Equation 2.10:

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(4m)^2 + (-2m)^2} = 4.5m$$

From Equation 2.11 we find the direction of $\mathbf{R}$ as,

$$\tan\theta = \frac{R_y}{R_x} = \frac{-2m}{4m} = -0.5$$

$$\theta = \tan^{-1}(-0.5) = -27^0 \text{ clockwise from the x-axis.}$$

Graphical Addition and Subtraction of Vectors

Parallelogram Law of Vectors

The parallelogram law of vectors is another useful law for the addition of two vectors.

Consider two vectors $\mathbf{P}$ and $\mathbf{Q}$ as shown in Figure 2.5a. Displace vector $\mathbf{Q}$ parallel to itself till the tail of $\mathbf{Q}$ touches the tail of $\mathbf{P}$. Complete the parallelogram as shown in Figure 2.5b. Applying the triangle law of vectors to the vector triangle OAC, we get

$$\mathbf{OA} + \mathbf{AC} = \mathbf{OC} \quad \text{or} \quad \mathbf{P} + \mathbf{Q} = \mathbf{R}$$

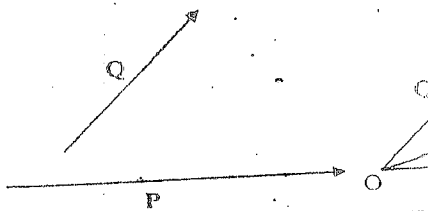


Fig. 2.5a: Vectors Q and P

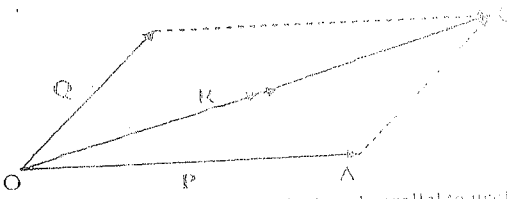


Fig. 2.5b: Vector Q displaced parallel to itself

So, the two vectors are represented completely by the two adjacent sides of the parallelogram and the diagonal drawn through the meeting point of the tails gives the resultant vector.

The parallelogram law of vectors states that

if two vectors, acting simultaneously at a point, can be represented both in magnitude and direction by the two adjacent sides of a parallelogram drawn from a point, then the resultant is represented completely both in magnitude and direction by the diagonal of the parallelogram passing through that point.

Now, let us analytically calculate the magnitude and direction of the resultant vector R.

Let θ be the angle between the two given vectors P and Q. Refer to Figure 2.6.

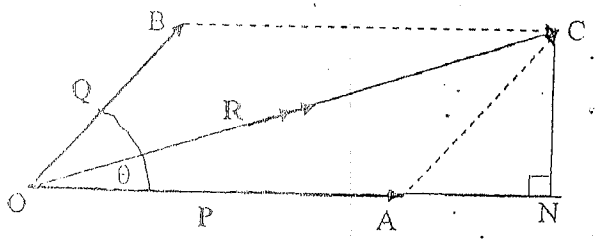


Fig. 2.6: Calculation of magnitude and direction of vector R.

From C we draw a perpendicular CN on an extension of OA to form a right-angled triangle ANC so that

$$\sin\theta = \frac{CN}{AC} \quad \text{or} \quad CN = AC\sin\theta \quad \text{or} \quad CN = Q\sin\theta, \qquad 2.14$$

$$AC = OB = Q \qquad 2.15$$

and also

$$\cos\theta = \frac{AN}{AC} \quad \text{or} \quad AN = AC\cos\theta \quad \text{or} \quad AN = Q\cos\theta \qquad 2.16$$

Now, $ON = OA + AN = P + Q\cos\theta$

Consider the right-angled triangle ONC.

$$OC^2 = ON^2 + CN^2 \qquad 2.17$$

or

$$R^2 = (P + Q\cos\theta)^2 + (Q\sin\theta)^2$$

$$R^2 = P^2 + Q^2\cos^2\theta + 2PQ\cos\theta + Q^2\sin^2\theta$$

$$R^2 = P^2 + Q^2\cos^2\theta + Q^2\sin^2\theta + 2PQ\cos\theta$$

$$R^2 = P^2 + Q^2(\cos^2\theta + \sin^2) + 2PQ\cos\theta$$

$$R^2 = P^2 + Q^2 + 2PQ\cos\theta$$

Since $\cos^2\theta + \sin^2\theta = 1$, then

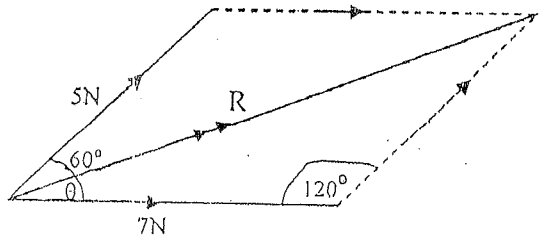
$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta} \qquad 2.18$$

Equation 2.18 is the required expression for the magnitude of the resultant of two vectors P and Q inclined to each other at an angle, and it is the cosine law.

Example 2.4

Two forces 5N and 7N act on a 10kg body. If the angle between the two forces is 60°, calculate the magnitude and direction of the resultant force on the body.

Solution



Completing the parallelogram gives R as the resultant force acting on the body.

Applying the cosine rule,

$R^2 = 5^2 + 7^2 + 2(5)(7)\cos 60^\circ$

$$R = \sqrt{25 + 49 + (0.5)70} = 10.44N$$

Using the sine rule to find the direction of the resultant force gives

$$\frac{\sin\theta}{5} = \frac{\sin 120}{10.44} \Rightarrow \sin\theta = \frac{5}{10.44}\sin 120$$

$\theta = \sin^{-1}(0.415) = 24.5^\circ$

Vector Multiplication

There are two ways to multiply a vector by a vector: one way produces a scalar and the other produces a new vector.

The Scalar Product of Two Vectors

In general, the scalar product of any vectors A and B is a scalar quantity equal to the product of the magnitudes of the two vectors and the cosine of the angle θ between them:

$$A.B = AB\cos\theta \qquad 2.19$$

This relationship is shown in Figure 2.7. Note that A and B need not have the same units. In figure 2.7, $B\cos\theta$ is the projection of B onto A. Therefore equation 2.19 says that A.B is the product of the magnitude of A and the projection of B onto A. The Scalar product, which is also called the *Dot Product*, is commutative. That is,

$$A.B = B.A \qquad 2.20$$

The scalar product also obeys the distributive law of multiplication, so that

$$A.(B+C) = A.B + B.C \qquad 2.21$$

The dot product is simple to evaluate from equation 2.19 when A is either perpendicular or parallel to B. If A is perpendicular to B ($\theta = 90^0$), then A.B = 0. If vector A is parallel to vector B and the two point in the same direction ($\theta = 0$), then A.B = AB. If vector A is parallel to vector B but the two point in opposite directions ($\theta = 180^0$), then A.B = $-AB$. The scalar product is negative when $90^0 < \theta < 180^0$

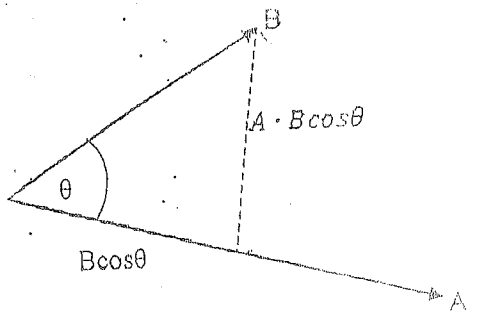


Fig. 2.7: Graphical method for the distributive law of multiplication

The unit vectors i, j, and k, lie in the positive x, y, and z directions, respectively, of a right-handed coordinated system. Therefore, it follows from the definition of A.B that the scalar products of these unit vectors are

$$i.i = j.j = k.k = 1 \qquad 2.22$$
$$i.j = i.k = j.k = 0 \qquad 2.23$$

From equation 2.12, the scalar product of A and B reduces to

$$A.B = A_xB_x + A_yB_y + A_zB_z \qquad 2.24$$

If A = B, then

$$A.A = A_x^2 + A_y^2 + A_z^2 = A^2 \qquad 2.25$$

The Vector Product

Given any two vectors A and B, the vector product AxB is defined as a third vector C, the magnitude of which is $AB\sin\theta$, where θ is the angle between A and B. That is, if C is given by.

$$C = AxB \qquad 2.26$$

Then its magnitude is

$$C = AB\sin\theta \qquad 2.27$$

Example 2.5

Vector A has x and y components of –8.70 cm and 15.0 cm, respectively; vector B has x and y components of 13.2 cm and –6.60 cm, respectively. If A–B+3C = 0, what are the components of C?

Solution

A=–8.70i+15.0j B=13.2i–6.60j

A–B +3C= 0

3C = B–A = 21.9i –21.6j

C = 7.30i – 7.20j or

C_x = 7.30 cm

C_y = –7.20 cm

Example 2.6

Consider two vectors A = 3i – 2j and B = –i – 4j. Calculate (a) A+B, (b) A–B, (c) |A+B|, (d) |A-B|, (e) the directions of A+B and A–B.

Solution

(a) (A+B) = (3i – 2j) + (–i – 4j) = (2i – 6j)

(b) (A – B) = (3i – 2j) – (–i – 4j) = (4i + 2j)

(c) |A+B| = $\sqrt{2^2 + 6^2}$ = 6.32

(d) |A–B| = $\sqrt{4^2 + 2^2}$ = 4.47

(e) θ|A+B| = $\tan^{-1}\left(\frac{6}{2}\right)$ = –71.6° = 288°

θ|A–B| = $\tan^{-1}\left(\frac{2}{4}\right)$ = 26.6°

Example 2.7

A boy runs 3 blocks north, 4 blocks northeast, and 5 blocks west. Determine the length and direction of the displacement vector that goes from the starting point to his final position.

Solution

Let i = east and j = north

R = 3.00bj + 4.00bcos45°i + 4.00bsin45°j – 5.00bi

R = –2.17bi + 5.83bj

R = $\sqrt{2.17^2 + 5.83^2}$ b at Arctan $\left(\frac{5.83}{2.17}\right)$ N of W

= 6.22 blocks at 110° counterclockwise from east

Example 2.8

You are standing on the ground at the origin of a coordinate system. An airplane flies over you with constant velocity parallel to the x-axis and at a constant height of 7.60×10^3 m. At t = 0, the airplane is directly above you, so that the vector from you to it is given by $\mathbf{P}_0 = (7.60\times10^3\ \text{m})\mathbf{j}$. At t = 30 s, the position vector leading from you to the airplane is $\mathbf{P}_{30} = (8.04\times10^3\ \text{m})\mathbf{i} + (7.60\times10^3\ \text{m})\mathbf{j}$. Determine the magnitude and orientation of the airplane's position vector at t = 45 s.

Solution

The y coordinate of the airplane is constant and equal to 7.60×10^3 m whereas the x coordinate is given by by $x = v_i t$ where v_i is the constant speed in the horizontal direction. At t = 30 s we have $x = 8.04\times10^3$, so v_i = 268 m/s.

The position vector as a function of the time is P = (268 m/s)ti + $(7.60\times10^3\ \text{m})$j.

At t = 45 s, P = $[1.21\times10^4\,\mathbf{i} + 7.60\times10^3\,\mathbf{j}]$ m. The magnitude is

P = $\sqrt{(1.21\times10^4)^2 + (7.60\times10^3)^2}$ m = 1.43×10^4 m and the direction is

$\theta = Arctan\left(\frac{7.6\times10^3}{1.21\times10^4}\right) = 32.2^0$ above the horizontal

The Newton is defined as *the force needed to give 1kg of mass an acceleration of 1ms^{-2}* and therefore $k = 1$. Therefore Equation 3.3 becomes

$$F = ma \qquad 3.4$$

Impulse

The effectiveness of a force which produces motion depends on both the magnitude of the force and the time for which the force acts. When a large force acts for an extremely short duration, neither the magnitude of the force nor the time for which it acts is important. In such a case, the total effect of force is measured. The total effect of force is called *impulse*. It is a vector quantity and is denoted by $\mathbf{J}$. It is the product of a force and the time for which the force acts. Therefore in SI units, impulse is measured in Newton second (Ns).

Suppose a force $\mathbf{F}$ acts for a short time dt. The impulse of this force is given by

$$d\mathbf{J} = \mathbf{F}dt \qquad 3.5$$

If we consider a finite interval of time from t_1 to t_2, then the impulse is given by

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\, dt \qquad 3.6$$

The right hand side of Equation 3.6 represents the impulse of a varying force. If the force is constant both in magnitude and in direction, then

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\, dt = \mathbf{F}\int_{t_1}^{t_2} dt = \mathbf{F}\,[t]_{t_1}^{t_2} = \mathbf{F}\,(t_2 - t_1) = \mathbf{F}\,\Delta t \qquad 3.7$$

From the above expression, impulse of a constant force $\mathbf{F}$ is equal to the product of the force and time interval Δt for which the force acts.

Impulse and Momentum

According to Newton's second law of motion, the applied force is,

$$\mathbf{F} = \frac{\textit{Change in momentum}}{\textit{time taken}} = \frac{d\mathbf{P}}{dt} \qquad 3.8$$

where $\mathbf{P}$ is the momentum of the body under consideration and t is the time.

$$\mathbf{F}dt = d\mathbf{P} \qquad 3.9$$

Integrating both sides of Equation 3.9 gives

$$\mathbf{F}\int_{t_1}^{t_2} dt = \int_{P_1}^{P_2} d\mathbf{P} \qquad 3.10$$

where P_1 and P_2 are momenta at times t_1 and t_2 respectively.

$$F\,[t]_{t_1}^{t_2} = [P]_{P_1}^{P_2} \quad \Rightarrow \quad F\,(t_2 - t_1) = P_2 - P_1$$

$$\mathbf{F}\,\Delta t = \mathbf{P}_2 - \mathbf{P}_1 \qquad 3.11$$

So the impulse is equal to the change in momentum produced by the force.

Example 3.1

A force of 75N is used to pull a body of mass 55kg through a distance of 35m in 6s. Find the change in momentum of the body.

Solution

Applying Newton's second law of motion, $F = \dfrac{mv - mu}{t}$, we can write $Ft = mv - mu$

mu = momentum change.

Therefore momentum change $(mv - mu) = Ft = 75 \times 6 = 450Ns$.

Example 3.2

While waiting in his car at a spotlight, an 80 kg man and his car are suddenly accelerated to a speed of 5 m/s as a result of a rear-end collision. Assuming the time taken to be 0.3s, find (a) the impulse on the man and (b) the average force exerted on him by the back of the seat of his car.

Solution

(a) Impulse = momentum change

$M=80kg,\ v=5m/s,\ u=0m/s$

$I = \Delta(mv - mu) = 80(5-0) = 400Ns$

(b) $F = \frac{\Delta(mv-mu)}{\Delta t} = \frac{400}{0.3} = 1333.3N$

Example 3.3

A golfer hits a golf ball of mass 51g and the ball leaves the club with a velocity of 80m/s. Assuming that the ball and the club are in contact for 0.006s, find the ball's final momentum and the average force exerted by the club on the ball.

Solution

Since impulse is equal to the change in momentum,

$I = mv - mu$

$F = \frac{mv-mu}{\Delta t} = \frac{0.051(80-0)}{0.006} = 680N$

The final momentum is, $mv = 0.051(80) = 4.08kgms^{-1}$

Example 3.4

A 5.0g bullet moving at 100m/s strikes a log. Assume that the bullet undergoes a uniform deceleration and stops in 6cm. Find (a) the time taken for the bullet to stop, (b) the impulse on the log and (c) the average force experienced by the log.

Solution

$v^2 = u^2 + 2aS$, where v, u and S have their usual meanings,

$u=100m/s,\ v=0m/s,\ S=0.06m.$

(a) $a = \frac{v^2-u^2}{2S} = \frac{0^2-100^2}{2x0.06} = -\frac{10^4}{0.12} ms^{-2}$

Now, the time taken, $t = \frac{v-u}{a} = \frac{0-100}{-10^4} x0.12 = 1.2x10^{-3}s$

(b) impulse $J = m(v-u) = 100(0.005) = 0.5Ns$

(c) $F = \frac{\Delta(mv)}{t} = \frac{0.5}{1.2x10^{-3}} = 417N$

Collision

A collision is said to occur *when two or more bodies physically exert relatively strong forces against each other* or *when the path of one body is changed by the influence of another body.* Collision ranges from the microscopic scale of sub-atomic particles to the astronomic scale of colliding Stars and Galaxies. The forces acting in the system of colliding forces are *Action* and *Reaction forces.* Also, collision causes the momentum and kinetic energy of the interacting bodies to change. But the total momentum is conserved and so also is the total energy, but mechanical energy may not be conserved.

There are two types of collisions namely *Elastic* and *Inelastic* collisions.

Elastic Collisions

A collision is said to be elastic *if both the momentum and kinetic energy are conserved in the collision.* In an elastic collision, deformed bodies regain their original shapes completely. So deformation is temporary. Colliding bodies always separate after collision. The mechanical energy is not converted to any other forms of energy. The forces of interaction are conservative in nature.

Perfectly elastic collision is an extremely rare physical phenomenon. Collisions between atomic and sub-atomic particles are assumed to be the nearest perfectly elastic collision.

Inelastic Collision

A collision is said to be inelastic *if the kinetic energy is not conserved in the collision.* However, momentum is conserved. In an inelastic collision, deformed bodies do not recover their original shapes. So deformation is permanent. Part of the mechanical energy is converted to other forms of energy such as sound, heat, light etc. The interacting forces are non-conservative in nature. Most of the collisions between macroscopic bodies are inelastic.

Perfectly inelastic (Plastic) collision occurs when the interacting bodies

stick together after the collision. In this type of collision the loss of kinetic energy is maximum but not complete. Examples of such a collision are (1) a lump of wet mud thrown against a wall (2) collision between two putty balls.

Momentum

The Momentum of a body is *the product of its mass m and the velocity v with which it is moving.* i.e., Momentum, $\mathbf{P} = mass \times velocity = mv$.

Momentum has both magnitude and direction, thus it is a vector quantity. Its direction is the same as that of the velocity since the mass has no direction. The SI unit for momentum is $kgms^{-1}$.

When two bodies, a heavy one and a light one are acted upon by the same force for the same time, the light body builds up a higher velocity than the heavy one. In both cases the momentum gained by the bodies are the same.

The Law of Conservation of Momentum

When two or more bodies act upon one another, their total momentum remains constant provided no external forces are acting. The law can also be stated as follows.

In a system of colliding bodies, total momentum before collision is equal to total momentum after collision. The condition under which the law holds is that no external forces should act on the bodies.

One-dimensional Collisions

Conservation of momentum for interacting bodies moving in the same straight line.

1. **Elastic collision between two bodies moving in the same direction.**

Let m_1 and m_2 be the masses of the two bodies. Also let their initial velocities be u_1 and u_2 respectively along the same direction. After collision they separate and move with different velocities v_1 and v_2.

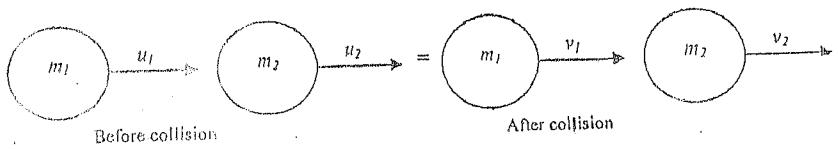


Fig. 3.1: Elastic collision between two bodies moving in the same direction.

Applying the law of conservation of momentum,

Total momentum before collision = Total momentum after collision

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad 3.12$$

Since the collision is elastic, kinetic energy will be conserved. Therefore, Total kinetic energy before collision = Total kinetic energy after collision.

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 3.13$$

2. *Elastic collision between two bodies moving in opposite direction*

If after the collision they move in opposite directions then the law of conservation of momentum becomes,

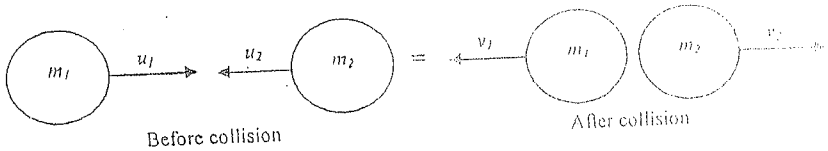


Fig. 3.2: Elastic collision between two bodies moving in opposite directions.

$$m_1u_1 - m_2u_2 = -m_1v_1 + m_2v_2 \qquad 3.14$$

Also, total kinetic energy before collision = total kinetic energy after collision

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 3.15$$

On the other hand if the momentum of one of the bodies involved is so great that after collision they separate and move in the same direction then,

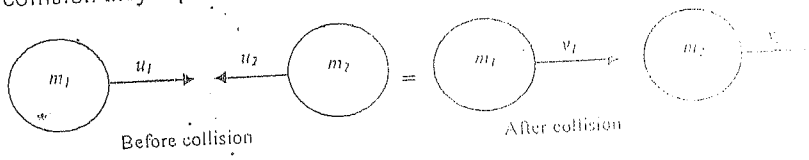


Fig. 3.3: Elastic collision between two bodies moving in opposite directions.

$$m_1u_1 - m_2u_2 = m_1v_1 + m_2v_2 \qquad 3.16$$

The conservation of kinetic energy expression will be

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 3.17$$

3. **Elastic collision between a moving body and a stationary body.**

After collision they separate and move with different velocities v_1 and v_2 along the same direction.

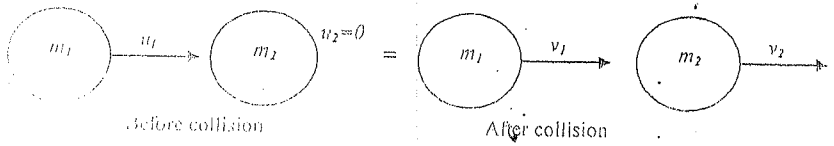


Fig. 3.4: Elastic collision between a moving body and stationary body with after collision moving in the same direction.

$$m_1u_1 = m_1v_1 + m_2v_2 \qquad 3.18$$

The conservation of kinetic energy expression becomes

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 3.19$$

On the other hand if after collision they separate and move in opposite directions with different velocities v_1 and v_2 then,

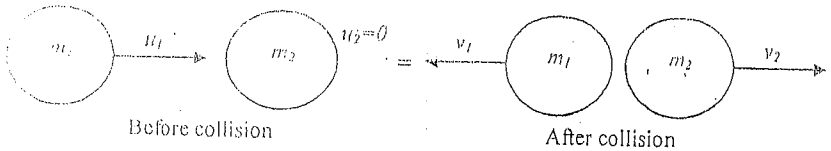


Fig. 3.5: Elastic collision between a moving body and a stationary body with after collision moving in opposite directions.

$$m_1u_1 = -m_1v_1 + m_2v_2 \qquad 3.20$$

The conservation of kinetic energy expression changes to

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad 3.21$$

Example 3.5

A ball of 0.4kg mass and a speed of 3m/s has a head-on, completely elastic collision with a 0.6kg mass initially at rest. Find the speeds on both bodies after the collision.

Solution

Apply conservation of momentum.

Let v_1 and v_2 be final velocities for the 0-4kg and 0-6kg masses respectively. Then,

$$(0.4x3) + (0.6x0) = 0.4v_1 + 0.6v_2$$

$$1.2 = 0.4v_1 + 0.6v_2 \qquad 1$$

For Elastic collision, velocity of separation = – velocity of approach

$$v_1 - v_2 = -v_1$$
$$-v_1 + v_2 = 3 \qquad 2$$

Solving Equations 1 and 2 simultaneously yields

$$v_1 = -0.6ms^{-1} \text{ and } v_2 = 2.4ms^{-1}$$

Example 3.6

A proton of mass 1.66×10^{-27}kg collides head-on with a helium atom at rest. The helium atom has a mass of 6.64×10^{-27}kg and recoils with a speed of 5×10^5m/s. If the collision is elastic, what are the initial and final speeds of the proton and the fraction of its initial energy transferred to the helium atom?

Solution

Mass of He = 4 x (mass of proton).

From conservation of momentum

$$m_p u_p + 0 = m_p v_p + m_{He} v_{He} \quad 1$$

$$m_p u_p + 0 = m_p v_p + (4m_p) v_{He}$$

$$u_p = v_p + 4v_{He}$$

From elastic condition,

velocity of approach = – velocity of separation

$$u_p = -v_p + v_{He} \quad 2$$

Adding Equations 1 and 2 gives,

$$2u_p = 5v_{He} = 5(5x10^5) ms^{-1} = 25x10^5 ms^{-1}$$

$$u_p = 1.25x10^6 ms^{-1}$$

From the momentum equation

$$-v_p = 4v_{He} - u_p$$

$$v_p = u_p - 4v_{He} = 1.25x10^6 - 4(5x10^5)$$

$$v_p = 1.25x10^6 - 2.00x10^6 = -7.5x10^5 ms^{-1}$$

$$\text{Fraction of Energy} = \frac{KE\ of\ He}{initial\ KE} = \frac{(4m_p)(5x10^5)}{m_p(1.25x10^6)} = 0.64$$

4. **Inelastic collision between two bodies moving in the same direction.**
After collision they stick together and move with a common velocity v

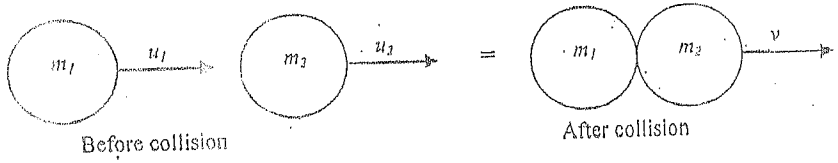


Fig. 3.6: Inelastic collision between two moving boides in the same direction.

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v \quad 3.22$$

5. **Inelastic collision between two bodies moving in opposite directions.**

After collision they stick together and move with a common velocity v

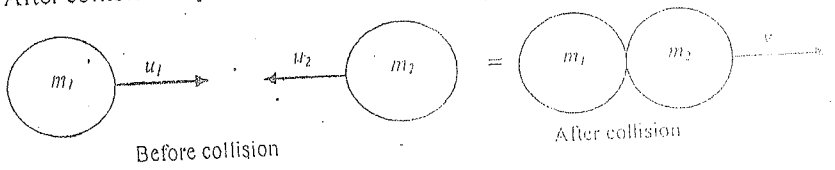


Fig. 3.7: Inelastic collision between two moving body in the opposite direction

$$m_1 u_1 - m_2 u_2 = (m_1 + m_2) v \quad 3.23$$

6. **Inelastic collision between a moving body and a stationary body.**

After collision they stick together and move with a common velocity v

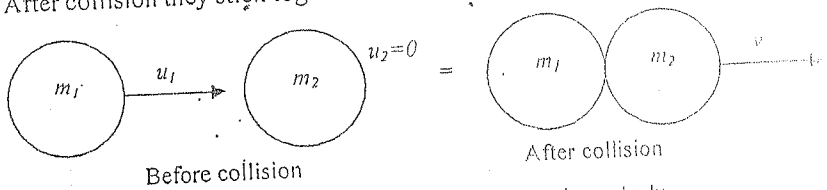


Fig. 3.8: Inelastic collision between a moving body and a stationary body.

$$m_1 u_1 = (m_1 + m_2) v$$

Example 3.7

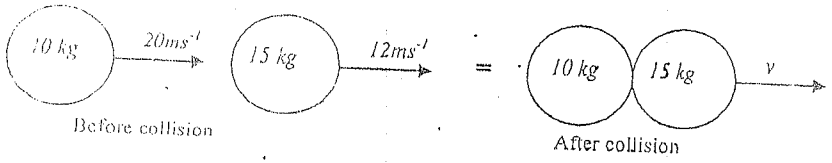
A ball of mass 10kg moving with a velocity of $20ms^{-1}$ collides inelastically with a 15kg ball moving with a velocity of $12ms^{-1}$. Calculate their common velocity after the collision if they initially moved in

(a) the same direction
(b) opposite direction.

Solution

From the law of conservation of linear momentum,

Total momentum before collision = Total momentum after collision



(a) if the balls initially moved in the same direction then,

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2, \text{ but } v_1 = v_2 = v$$

$$m_1u_1 + m_2u_2 = m_1v + m_2v = (m_1 + m_2)v$$

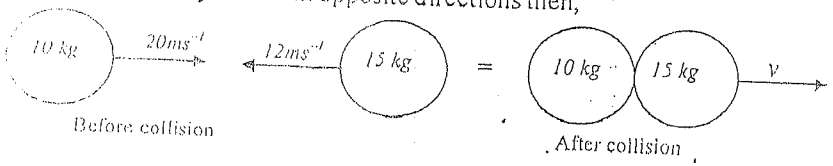
$$(10 \times 20) + (15 \times 12) = (10 + 15)v$$

$$200 + 180 = 25v$$

$$380 = 25v$$

Therefore common velocity in the same direction is $v = \frac{380}{25} = 15.2ms^{-1}$

(b) If the balls initially moved in opposite directions then,



Total momentum before collision = Total momentum after collision

(a) $m_1u_1 - m_2u_2 = m_1v_1 + m_2v_2$, but $v_1 = v_2 = v$

$$m_1u_1 - m_2u_2 = m_1v + m_2v = (m_1 + m_2)v$$

$$(10 \times 20) - (15 \times 12) = (10 + 15)v$$

$$200 - 180 = 150v$$

$$20 = 150v$$

Therefore common velocity in the opposite direction is $v = \frac{20}{150} = 0.13ms^{-1}$

Example 3.8

A 100kg object moving with a velocity of 25ms^{-1} collides with a stationary object of mass 45kg.

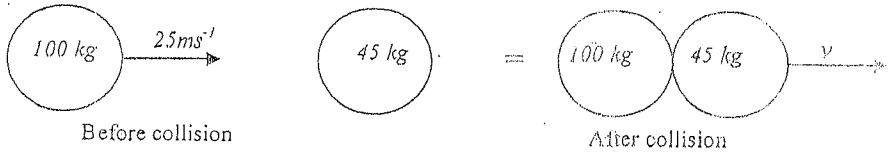
(a) Calculate their common velocities after collision if they stick together and moved in one direction.

(b) Assuming the two objects separated but moved in the same direction, find the velocity of the 45kg mass if after impact the 100kg object moved with a velocity of 15ms^{-1}.

Solution

From the law of conservation of linear momentum,

Total momentum before collision = Total momentum after collision



(a) $m_1u_1 = m_1v_1 + m_2v_2$, but $v_1 = v_2 = v$

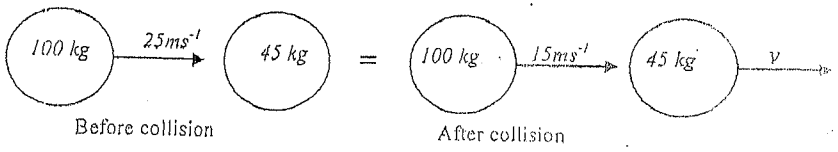
$$m_1u_1 = (m_1 + m_2)v$$

$$(100 \times 25) = (100 + 45)v$$

$$2500 = 145v$$

Therefore, their common velocity is $v = \frac{2500}{145} = 17.24ms^{-1}$

(b) From the law of conservation of linear momentum,
Total momentum before collision = Total momentum after collision



$$m_1u_1 = m_1v_1 + m_2v_2$$

$$m_1u_1 = m_1v + m_2v$$

$$(100 \times 25) = (100 \times 15) + 45\,v$$

$$2500 - 1500 = 45v$$

$$1000 = 45\,v$$

$$\text{is } v = \frac{1000}{45} = 22.22ms^{-1}.$$

Example 3.9

An 8g bullet is fired horizontally into a 9kg block of wood and sticks in it. The block, which is free to move, has a velocity of 40 cm/s after impact. Find the initial velocity of the bullet.

Solution

Let's consider the system (block + bullet). The velocity, and hence the momentum of the block before impact is zero. The momentum conservation law says.

Momentum of system before impact = momentum of system after impact

(mass) × (velocity of bullet) + 0 = (mass) × (velocity of block + bullet)

$$(0.008kg)v + 0 = (9.008kg)(0.40ms^{-1})$$

$$v = \frac{9.008x0.4}{0.008} = 450ms^{-1}$$

Example 3.10

A 16g mass is moving in the +x direction at 30 cm/s while a 4g mass is moving in the –x direction at 50 cm/s. They collide head-on and stick together. Find their velocity after collision.

Solution

Applying the law of conservation of momentum to the system consisting of two masses,

Momentum before impact = momentum after impact

$$(0.016kg)(0.30ms^{-1}) + (0.004kg)(-0.50ms^{-1}) = (0.016 + 0.004)kg \times v$$

$$v = \frac{(0.016kg)(0.30ms^{-1}) + (0.004kg)(-0.50ms^{-1})}{(0.020)kg} = 0.14ms^{-1}$$

Elastic Collision in Two Dimensions (Oblique Collision)

Oblique collision occurs *when the two bodies separate and move at angles to each other after collision.*

Let a body m_1 moving with initial velocity v_1 collides obliquely with a stationary body m_2. It is not a head-on collision so their lines of action do not coincide. After collision, the bodies m_1 and m_2 travel with velocities v_1 and v_2 at angles θ and ϕ respectively to the horizontal.

Note that in head-on collisions the lines of action of the bodies involved in the collision coincide.

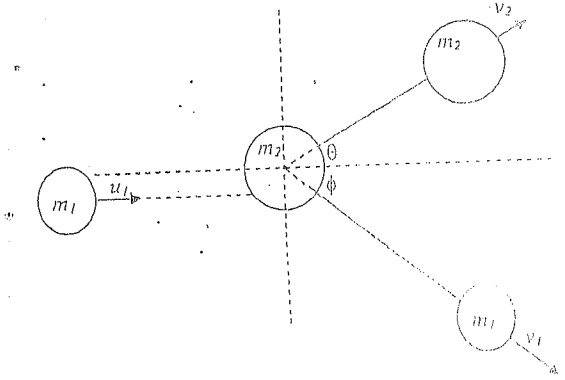


Fig. 3.9: Oblique collision between bodies' m_1 and m_2.

In the present case we shall apply the law of conservation of momentum to both the x-component (horizontal) and the y-component (vertical).

X-component

$$m_1u_1 = m_1v_1\cos\phi + m_2v_2\cos\theta \qquad 3.7$$

$$m_1 v_1 \sin\phi = m_2 v_2 \sin\theta \qquad 3.26$$

The collision is elastic, therefore, kinetic energy will be conserved

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_1^2 \qquad 3.27$$

Example 3.11

A 1.0 kg object, A, with a velocity of 4.0 m/s to the right, strikes a second object, B, of 0.3 kg, originally at rest. In the collision, A is deflected from its original direction through an angle of 50°; its speed after the collision is 2.0 m/s. Find the angle between B's velocity after the collision and the original direction of A, and also find the speed of B after the collision.

Solution

We choose our x-axis to the right. Let m_A=*1.0 kg*, m_B=*3.0 kg*, u_A=*4 m/s*, u_B=0 m/s; v_A and v_B are the magnitudes of the final velocities, and v_A=*2.0 m/s* aimed 50° above the x-axis.

By conservation of momentum we have

x – direction (Horizontal):

$m_A u_A + 0 = m_A v_A \cos 50^o + m_B v_B \cos\phi$, where f = angle of v_B below the x–axis.

$$m_B v_B \cos\phi = m_A u_A - m_A v_A \cos 50^o$$

$$v_B \cos\phi = \frac{m_A u_A - m_A v_A \cos 50^o}{m_B} = \frac{(1.0)(4) - (1.0)(2.0)\cos 50^o}{3.0} = 0.907 ms^{-1}$$

$$v_B \cos\phi = 0.907 ms^{-1} \qquad 1$$

y – direction (vertical):

$$0 = m_A v_A \sin 50^o - m_B v_B \sin\phi$$

$$v_B \sin\phi = \frac{m_A v_A \sin 50^o}{m_B} = \frac{(1.0)(2.0)\sin 50^o}{3.0} = 0.511 ms^{-1}$$

$$v_B \sin\phi = 0.511 ms^{-1} \qquad 2$$

Squaring and adding equations 1 and 2 gives

$$(v_B \sin\phi)^2 + (v_B \cos\phi)^2 = (0.511)^2 + (0.907)^2$$

Solving, we get

$v_B = 1.04 ms^{-1}$

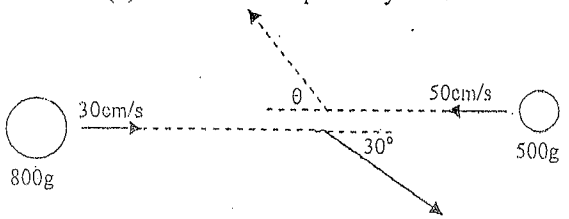
Dividing equation 2 by 1, we get,

$$\tan\phi = \frac{v_B \sin\phi}{v_B \cos\phi} = \frac{0.511}{0.907}$$

$\phi = 29.4°$

Example 3.12

The two spherical balls shown below collide and bounce off each other. (a) What is the final velocity of the 500g ball if the 800g ball has a speed of 15 cm/s after collision? (b) Is the collision perfectly elastic?



Solution

From the law of conservation of momentum,

X – axis

Momentum before collision = momentum after collision.

$$(0.80)(0.3) - (0.50)(0.5) = (0.80)(0.15\cos 30^o) + (0.5)v_x$$

$$\frac{(0.80)(0.3) - (0.50)(0.5) - (0.80)(0.15\cos 30^o)}{(0.5)} = v_x$$

$$v_x = -0.228 ms^{-1}$$

Y – axis

Momentum before collision = momentum after collision.

$$0 = -(0.80)(0.15\sin 30^o) + (0.5)v_y$$

$$v_y = \frac{(0.80)(0.15\sin 30^o)}{(0.5)} = 0.120 ms^{-1}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{-0.228^2 + 0.120^2} = 0.26 ms^{-1}$$

Also for the angle θ shown above

$$\theta = \tan^{-1}\left(\frac{0.120}{0.228}\right) = 28^o$$

Also for the angle θ shown above

(b) total KE before collision $= \frac{1}{2}(0.8)(0.3)^2 + \frac{1}{2}(0.5)(0.5)^2 = 0.0985J$

total KE before collision $\frac{1}{2}(0.8)(0.15)^2 + \frac{1}{2}(0.5)(0.26)^2 = 0.026J$

As KE is lost in the collision, the collision is not perfectly elastic.

Newton's Third Law

Newton's third law states that *to every action there is equal and opposite reaction.* Whenever a force acts on one body, an equal and opposite force acts on some other body. It is important to note that the action and reaction forces take place on different bodies. The two forces act normally to each other at the surface on which they act.

Examples of the Third Law are:

1. The recoil of a gun. When a bullet is fired from a gun, equal and opposite forces are exerted on the bullet and the gun.
2. Gases coming out of an inflated balloon. As the gas is forced out of the nozzle the reactional force propels the balloon in the opposite direction.
3. A rocket in motion. As the hot gases are forced out of the tail the reactional force propels the rocket forward.

Inertial Frames of Reference

An Inertial reference frame is one which is stationary or moves with constant velocity and therefore does not accelerate. It is a reference frame in which Newton's [illegible] law holds. If a bus travels at constant speed on a straight highway, then the [illegible] be considered as an inertia frame of reference.

Frames translating at constant velocity along a curved trajectory or rotating at a constant angular velocity are non-inertia frames due to centripetal acceleration. This means that a bus accelerating in a straight path or curving at constant speed is a non-inertia frame of reference. Rotating carousel is also an example of a non-inertia frame of reference.

Strings, Pulleys and Inclines

Let us consider two bodies A and B connected with an inextensible string passed over a pulley system. Body A slides on an inclined plane while body B hangs vertically downwards (Refer to Figure 3.10).

First, let us consider the motion of body A along the inclined plane.

$$T - (F + Mg\sin\theta) = Ma$$

T is the tension in the string, F is frictional force, M is the mass of body A, m is mass of body B, R is reactional force and a is the acceleration.

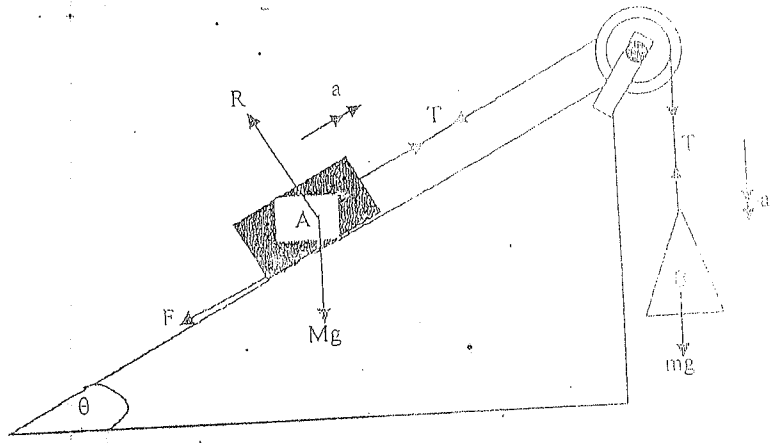


Fig. 3.10: The downward movement of body B causes body A to slide along the inclined plane because of the inextensible string connecting them.

Resolving the forces acting on body A parallel and perpendicularly to the inclined plane gives

$$R = Mg\cos\theta \qquad 3.29$$

$$T - (\mu Mg\cos\theta + Mg\sin\theta) = Ma \qquad 3.30$$

Therefore,

$$F = \mu R = \mu Mg\cos\theta \qquad 3.31$$

Applying the equation of motion to body B,

$$Mg - T = ma \qquad 3.32$$

Solving Equations 3.30 and 3.32 simultaneously gives,

$$a = \frac{mg - Mg(\mu\cos\theta + \sin\theta)}{m + M} \qquad 3.33$$

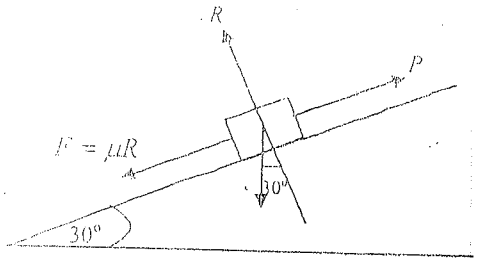
and

$$T = \frac{mMg(1 + \mu\cos\theta + \sin\theta)}{m + M} \qquad 3.34$$

Example 3.13

A body of mass 8kg is on the point of slipping down a plane of inclination 30°, what force applied parallel to the plane will just move it up the plane? [Take g=10ms^{-2}]

Solution



Let P be the force required to move the body up the plane.
Let $F = \mu R$ be the limiting frictional force.
Resolving along the plane, for equilibrium:

$$P = \mu R + mg\sin 30^0 \qquad 1$$

Resolving perpendicular to plane:

$$R = mg\cos 30^0 \qquad 2$$

Substituting 2 in 1, we get

$$P = \mu mg\cos 30^0 + mg\sin 30^0$$

Also since the body was about to slip down the plane of slope by 30°,

$$\mu = \tan 30^0 = \frac{1}{\sqrt{3}}$$

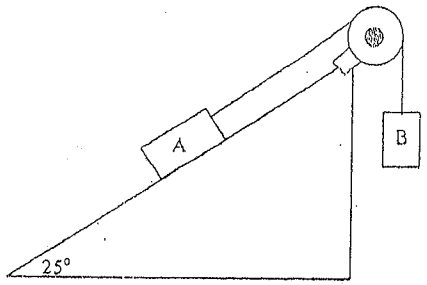
Therefore,

$$P = \left(8x\frac{1}{\sqrt{3}}x10x\frac{\sqrt{3}}{2}\right) + \left(8x10x\frac{1}{2}\right) = 80N$$

Example 3.14

An inclined plane making an angle of 25° with the horizontal has a pulley at its top. A 30kg block on the plane is connected to a freely hanging 20kg block by means of a cord passing over the pulley. Compute the distance the 20kg block will fall in 2s starting from rest. Neglect friction.

Solution



The situation is as shown above. We apply Newton's second law to each block separately. For block B we choose downward as positive, while for block A we choose our x-axis along the incline with the positive sense upward. This choice allows us to use the same symbol, a, for the acceleration of each block.

Then for block B:

$$W_B - T = m_B a \qquad 2$$

where $m_B = 20kg$ and $W_B = 20 \times 9.8 = 196N$ and T is the tension in the cord. Since the pulley is frictionless, the same tension T will exist on both sides of the pulley.

Then for block A:

$$T - W_A \sin 25^0 = m_A a \qquad 2$$

where $m_A = 30kg$ and $W_A = 30 \times 9.8 = 294N$. The tension T is eliminated by adding Equations 1 and 2, which yields

$$W_B - W_A \sin 25^0 = (m_B + m_A)a$$
$$196 - 294 \sin 25^0 = (20 + 30)a$$
$$a = 1.44ms^{-2}$$

The equation for fall from rest is $S_y = u_y t + \frac{1}{2} a_y t^2$.

$$S_y = (0x2) + \frac{1}{2}(1.44)(2)^2 = 2.88m$$

The distance covered by the 20kg block in 2s is 2.88m.

Friction

Friction is the retarding force which is called to play when a body actually moves or tend to move over the surface of another body. The surfaces of the bodies in contact are rough and, therefore, have irregular projections. Some surfaces appear to be smooth when viewed with the naked eye but a powerful microscope can expose all the irregularities. The resisted motion is caused by the interlocking of the large number of irregularities on the surfaces in contact. The opposing force called friction is always directed oppositely to the direction of motion.

Static Friction

It is the force of friction which exactly balances the applied force during the stationary state of the body. This means that static friction exists only when the bodies in contact are at rest relative to each other, but with a tendency to move.

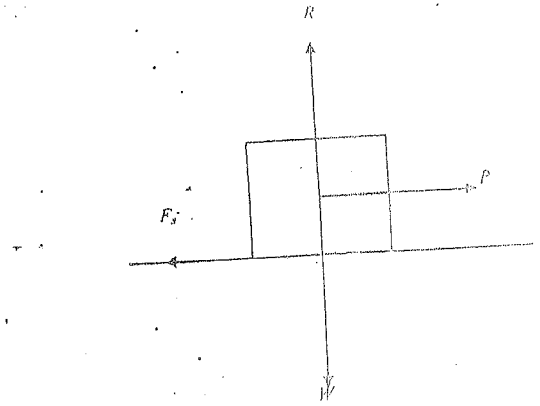


Fig. 3.11: A small force P is applied to move the block but it is opposed by the static friction F_s.

Consider a block of wood resting on a horizontal surface. Let a small force P be applied to try moving the block as shown in Figure 3.11. Also let F_s be the resulting force of static friction. Then in the equilibrium position, the weight W of the block will be balanced by the reaction force R and the pulling force P will be balanced by F_s.

Limiting Friction

It is the maximum value of static friction which is called into play when a body is just about to start sliding over the surface of another body. When the applied force P increases there is a corresponding increase in the static friction F_s. However, there is a particular limit up to which the static friction F_s can increase. Beyond this limit the applied force P will overcome the opposing force and produce motion in the body.

Dynamic or Kinetic Friction

This opposing force comes into play when the two surfaces in contact are in relative motion. It acts in a direction opposite to the direction of the instantaneous velocity

There are two types of dynamic or kinetic friction namely *Sliding* and *Rolling frictions.*

Sliding friction comes into play when a solid body slides over the surface of another body.

Rolling friction comes into play when a body rolls over the surface of another body.

Coefficient of Static Friction

For any two surfaces in contact, μ is the ratio of the limiting friction F_{ls} and the normal reaction R between them. It is denoted by m_s. m_s has no units and its value depends upon the state of polish of the two surfaces in contact. m_s is small for a highly polished surface and large for rough surfaces and also $0 \leq \mu \leq 1$.

$$\mu_s = \frac{F_{ls}}{R} \qquad 3.35$$

The force of static friction F_s is equal to the applied force. So F_s can have any value from 0 to F_{ls}. Therefore $F_s \leq F_{ls}$. The equality sign holds only when F_s has its maximum value.

$$\therefore F_s \leq \mu_s R \qquad 3.36$$

Coefficient of Kinetic Friction

The coefficient of Kinetic Friction is defined as the ratio of the kinetic friction and the normal reaction. It is denoted by m_k.

$$\mu_k = \frac{F_k}{R} \qquad 3.37$$

Now,

$$\frac{\mu_s}{\mu_k} = \frac{F_{ls}}{R} x \frac{R}{F_k} = \frac{F_{ls}}{F_k} \qquad 3.38a$$

$$\mu_s = \frac{F_{ls}}{R} x \frac{R}{F_k} = \frac{F_{ls}}{F_k} \mu_k, \qquad 3.38b$$

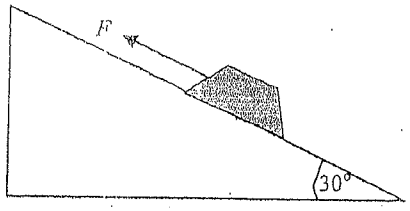
Since $\mu_s > \mu_k$, then from equation 3.38a,

$$\frac{\mu_s}{\mu_k} > 1, \quad \text{and } F_{ls} > F_k$$

Therefore, $F_{ls} > F_k$ and $\mu_s > \mu_k$.

Example 3.15

A body of mass 8kg is on the point of slipping down a plane which is inclined at 30° to the horizontal. What force parallel to the inclined force will just move it up the plane?



Solution

At point of slipping, F_{ls} (= μR) must be balanced by $mg\sin\theta$

i.e. $\mu R - mg\sin\theta = 0$

But $R = mg\cos\theta$

so that

$\mu mg\cos\theta - mg\sin\theta = 0$

$$\mu = \frac{mg\sin\theta}{mg\cos\theta} = \tan\theta$$

Resolving along the plane for equilibrium

$$T = mR + mg\ sin\ 30° \quad \ldots \quad \ldots \quad 1$$

Resolving perpendicular to the plane

$$R = mg\ cos\ 30° \quad \ldots \quad \ldots \quad 2$$

Substituting equation 2 into 1
$T = m\,mg\cos 30° + mg \sin 30°$
Since the body is about to slip down the plane at 30°,

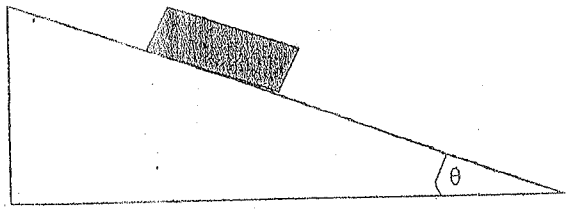
$$\mu = \tan 30° = 0.5774$$

therefore,

$$T = (0.5774 \times 8 \times 10 \times 0.866) + (8 \times 10 \times 0.5)$$
$$= 40.0 + 40.0 = 80.0\text{N}$$

Example 3.16

Suppose that in the figure below the block is at rest. The angle of inclination of the plane is slowly increased until the block begins to slide at $\theta = 42°$. What is the coefficient of static friction between the block and the incline?



Solution

At the point when the block begins to slide, $F\ (= mR)$ just balances $mg\sin\theta$

Along the plane $mR = mg\sin\theta$ so that

$$\mu mg\cos\theta - mg\sin\theta = 0$$

$$\mu = \frac{mg\sin\theta}{mg\cos\theta} = \tan\theta$$

therefore, $\mu = \dfrac{F}{R} = \tan\theta$
$= \tan 42° = 0.90$

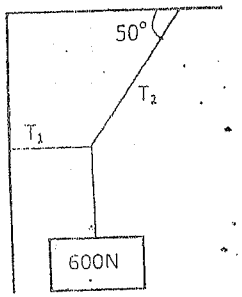
Exercise 3

3.1 A 110g hockey puck slides on the ice for 15m before it stops. If its initial speed was 6.0m/s, what was the magnitude of the frictional force on the puck during the sliding? **(Ans: 0.31N)**

3.2 A 68kg crate is dragged across a floor by pulling on a rope inclined 15° above the horizontal. If μ_k=0.35 what is the magnitude of the initial acceleration of the crate? **(Ans: 1.3ms^{-2})**

3.3 A bedroom bureau with a mass of 45kg, including drawers and clothing, rests on the floor. (a) If the coefficient of static friction between the bureau and the floor is 0.45, what is the minimum horizontal force a person must apply to start the bureau moving? (b) If the drawers and clothing, with 17kg mass are removed before the bureau is pushed, what is the new minimum magnitude? (Ans: [a] 200N, [b] 120N)

3.4 The coefficient of static friction between Teflon and scrambled eggs is about 0.04. What is the smallest angle from the horizontal that will cause the eggs to slide across the bottom of a Teflon-coated Skillet? (Ans: 2°)

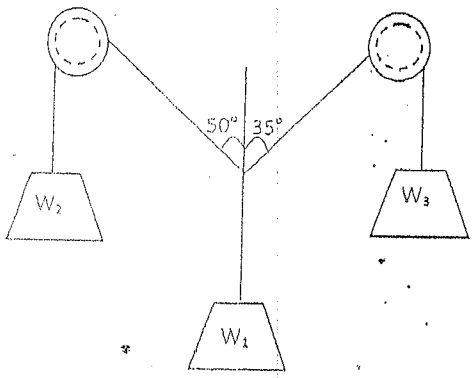


3.5 A horizontal force of 12N pushes a block weighing 5.0N against a vertical wall. The coefficient of static friction between the wall and the block is 0.60, and the coefficient of kinetic friction is 0.40. Assume that the block is not moving initially. (a) Will the block start moving? (b) In a unit-vector notation, what is the force exerted on the block by the wall? (Ans: [a] No, [b] $(-12\,\underline{i} + 5\,\underline{j})$N)

3.6 For the situation shown in the figure below, find the values of T_1 and T_2 if the object's weight is 600N. (Ans: 503N; 783N).

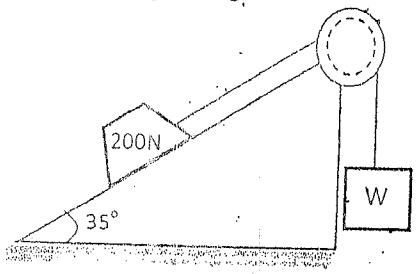
3.7 The following coplanar forces pull on a ring: 20N at 30°, 500N at 80°, 300N at 240°, and an unknown force. Find the magnitude and direction of the unknown force if the ring is to be in equilibrium. **(Ans: 350N at 252°)**

3.8 In the figure below, the pulleys are frictionless and the system hangs at equilibrium. If w_3, the weight of the object on the right, is 200N, what are the values of w_1 and w_2? **(Ans: 260N; 150N)**



Use the diagram below to answer questions 3.9& 3.10.

3.9 If in the figure the friction between the block and the incline is negligible, how much must the object on the right weigh if the 200N block is to remain at rest? (Ans: 115N)



3.10 The system in the figure remains at rest when w=220N. What are the magnitude and direction of the frictional force on the 200N block? **(Ans: 105N down the incline)**

3.11 A rough inclined plane is 100m long and is inclined to the horizontal at an angle $\sin^{-1}(3/5)$, the coefficient of friction being 0.5. A body slides down it from rest at the highest point. Find its velocity on reaching the bottom.

3.12 A particle is placed on an inclined plane of inclination, a, the coefficient of friction between the particle and the plane at all points of the plane is μ. Prove that the particle cannot rest on the plane unless $m > \tan a$.

3.13 A body A of mass, 2m and a body of mass, m move in the same direction along a common line. The velocity of A is 40cm/s and that of B is 10cm/s. if A and B undergo a perfect elastic collision, calculate (a) the velocity of A and B after the collision (b) the impulse on A and B and show that Newton's third law of motion is obeyed.

Chapter Four

KINEMATICS

Distance and Displacement

Distance and speed are scalar quantities.

Distance

Distance is simply the total path length traversed in moving from one position to the other. In general, the distance between two points depends on the path traveled.

Displacement

Displacement is defined as a straight line distance between two points, along the direction from the starting point to the final point. Displacement is a vector quantity.

Speed and Velocity

Speed

When a body is in motion, its position changes with time i.e., it moves a certain distance in a given time. This implies that both length and time are important quantities when describing motion. Imagine that two cars — car1 and car 2 — are moving in the same direction but car 2 is moving faster than car1. Car 2 will cover a definite distance x in a shorter time than car1 will do. This relation can be expressed by using the length and time to give the rate at which distance is travelled or speed, for each. Average speed is the distance d travelled, divided by the total time t elapsed in travelling that distance:

$$\text{Average speed, } s, = \frac{distance}{time} = \frac{d}{t} \qquad 4.1$$

Velocity

Velocity is defined as how fast a body is moving and in what direction. Velocity is also another quantity used to describe motion. We have the average velocity defined

as displacement divided by time taken:

$$\text{Average velocity} = \frac{displacement}{time} = \frac{\Delta x}{\Delta t} = \frac{x - x_0}{t - t_0} = \bar{v} \qquad 4.2$$

Where x_0 is the initial point, x is final position, t is initial time, t_0 is final time, Dx is the change in position or displacement and Dt is the change in t.

The SI units of velocity is metres per unit second (m/s). Instantaneous velocity on the other hand is defined as

$$\text{Instantaneous velocity} = \frac{dx}{dt} = \lim_{t \to 0} \frac{x - x_0}{t - t_0} \qquad 4.3$$

Acceleration

Analogous to average velocity we have average acceleration. Average acceleration is defined as *change in velocity divided by time taken*:

$$\text{Average acceleration} = \frac{velocity}{time} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{t - t_0} = \bar{a} \qquad 4.4$$

Accelerated Motion

Since velocity is a vector quantity with both magnitude and direction, an acceleration can occur when there is

1. a change in magnitude of the velocity but not the direction
2. a change in the direction of the velocity but not the magnitude
3. a change in both the magnitude and direction

Example 4.1

Mr. Tetteh and his driver are traveling to Cape Coast in a Toyota pickup which is moving at 90 km/h down a straight highway. The driver of the pickup sees an accident in the distance and slows down to 40 km/h in 5.0 s. What is the average acceleration of the Toyota pickup?

Solution

Given the initial and final velocities and the time duration,

$$v_0 = (90 km/h) = (25 m/s), \; v = (40 km/h) = (11 m/s), \; t = 5.0 s$$

We are solving to find: average acceleration, $\bar{a}$, which from Equation 4.4 is

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{\Delta t} = \frac{11m/s - 25m/s}{5.0s} = -2.8m/s$$

Average Velocity and Instantaneous Velocity in Components

In knowing how fast a particle moves from one point to the other one has to deal with

1. Average velocity and
2. Instantaneous velocity

These are vectors and so vector notation has to be used.

If a particle moves through a displacement $\Delta \mathbf{r}$ in a time interval Δt, then its average velocity $\bar{v}$ is

$$\bar{v} = \frac{\text{displacement}}{\text{time}} = \frac{\Delta \mathbf{r}}{\Delta t} \qquad 4.5$$

This tells us that, the direction of $\bar{v}$ is the same as that of. can also be written in unit vector form as

$$\bar{v} = \frac{\Delta x\mathbf{i} + \Delta x\mathbf{j} + \Delta x\mathbf{k}}{\Delta t} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j}\frac{\Delta z}{\Delta t}\mathbf{k} \qquad 4.6$$

The velocity of a particle usually refers to the instantaneous velocity which is the value that $\bar{v}$ approaches in the limit as we shrink the time interval to 0 about the instant. Using the language of calculus, we may write as the derivative

$$v = \frac{d\mathbf{r}}{dt} \qquad 4.7$$

The direction of the instantaneous velocity $\mathbf{v}$ of a particle is always tangent to the particle's path at the particle's position.

In substituting $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ into Equation 4.7 we get

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k} \qquad 4.8$$

$$\mathbf{v} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k} \qquad 4.9$$

where, $v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$

Since $\frac{dx}{dt}$ is the scalar component of $\mathbf{v}$ along the x-direction, it means we can find the scalar components of $\mathbf{v}$ by differentiating the scalar components of $\mathbf{r}$.

Average Acceleration and Instantaneous Acceleration in Components

As a particle's velocity changes from $\mathbf{v}_1$ to $\mathbf{v}_2$ in a time interval Δt, its average acceleration $\bar{a}$ during Δt is

$$\bar{a} = \frac{\text{change in velocity}}{\text{time interval}} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{\Delta t} = \frac{\Delta \mathbf{v}}{\Delta t} \qquad 4.10$$

If Δt is zero about some instant, then in the limit approaches the instantaneous acceleration (or acceleration) at the instant; that is

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}) = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k} \qquad 4.11$$

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k} \qquad 4.12$$

where $a_x = \frac{dv_x}{dt}, a_y = \frac{dv_y}{dt}$ and $a_z = \frac{dv_z}{dt}$. We can therefore find the scalar components of $\mathbf{a}$ by differentiating the scalar component of $\mathbf{v}$.

Example 4.2

If a rabbit's velocity is $v = -0.62t^2\mathbf{i} + 7.2\mathbf{j}$, find the acceleration $\mathbf{a}$ at the time $t = 5s$, in unit-vector notation and in magnitude notation.

Solution

Given that $v = -0.62ti + 7.2\mathrm{j}$, $a = \dfrac{dv}{dt} = -0.62(2t)i = -1.24ti$

At time $t=5s$

$$a = \frac{dv}{dt} = -1.24(5)i = -6.2i\ \mathrm{ms}^{-2}$$

$$|a| = \left|\frac{dv}{dt}\right| = |-1.24(5)i| = \sqrt{(-6.2)^2} = 6.2\ \mathrm{ms}^{-2}$$

Kinematic Equations and Problem Solving

Any rectilinear motion can be defined by the position of that body in space at any given instant. In order to make such a definition to describe a motion, we need to find the relationships between distance and displacement, speed and velocity, acceleration and time. The kinematic equations are equations of motion that relate the above-mentioned quantities with each other. The assumption of a uniform motion is made in all cases for use of these kinematics equations of motion. The following kinematics equations will not work, for example, for an object that is accelerating erratically. Below is the list of the kinematic equations.

Kinematic Equations

Average velocity, $\mathrm{v} = \dfrac{\mathrm{s}}{t}$	4.13
Displacement, $\mathrm{s} = \dfrac{1}{2}(\mathrm{v}-\mathrm{u})t$	4.14
Average acceleration, $\mathrm{a} = \dfrac{\Delta \mathrm{v}}{\Delta t} = \dfrac{\mathrm{v}-\mathrm{u}}{\Delta t}$	4.15
$\mathrm{v} = \mathrm{u} + \mathrm{a}t$	4.16
$\mathrm{s} = \mathrm{x} = \mathrm{u}t + \frac{1}{2}\mathrm{a}t^2$	4.17
$v^2 = u^2 + 2as$	4.18

Note that in some of the equations above we have replaced the time interval Δt with t, representing total time.

Example 4.3

A car traveling at a speed of +12 m/s accelerates at a rate of +4 m/s² for a time of 5s. What is the car's speed at the end of 5s?

Solution

Given u = 12m/s, a = 4m/s, t = 5s, v = ?. We use Equation 4.16 to find v

$$v = u + at = 12m/s + (4m/s^2 \times 5s) = 12m/s + 20m/s = 32m/s$$

Derivation of the Equations of Motion

Suppose that an object is initially moving with a constant acceleration, and increases its velocity from u to v and its displacement from 0 to s in the time interval t, then since acceleration is the rate of change of velocity, we have

$$a = \frac{v-u}{t}$$

Rearranging gives

$$v = u + at. \qquad 4.16$$

The average velocity is $\frac{1}{2}(u+v)$ and, therefore, since displacement is *average velocity* × *time*, we have

$$s = \frac{1}{2}(v-u)t \qquad 4.14$$

But from Equation 4.16, we can write

$$s = \tfrac{1}{2}(u+u+at)t = ut + \tfrac{1}{2}at^2 \qquad 4.17$$

From Equation 4.15 $t = (v-u)/a$ and substituting into Equation 4.17 we have

$$s = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right)$$

$$v^2 - u^2 = 2as$$

$$\therefore v^2 = u^2 + 2as \qquad 4.18$$

Graphs of Motion

The motion of a body or an object in a straight line (rectilinear motion) can be examined by graphical analysis. Graphical analysis is often helpful in understanding motion and its related quantities. For example, the motion of a runner can be graphed as a plot of position x versus time t.

We have

1. displacement-time graph
2. velocity-time graph and
3. acceleration-time graph

Displacement-time Graph

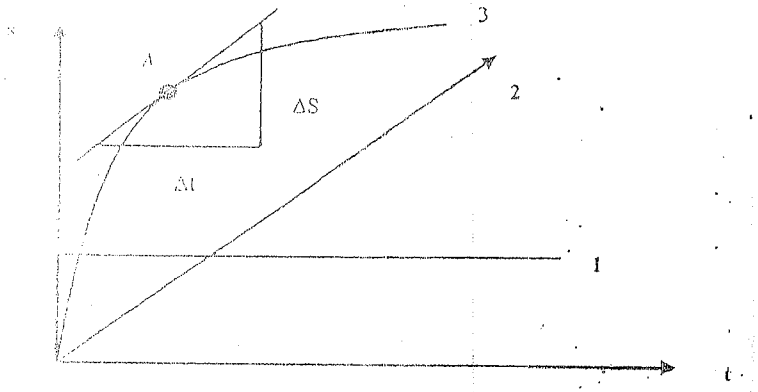


Fig. 4.1: Displacement-time graph.

From Figure 4.1 we can make the following deductions about the following curves:

(1) slope = 0: velocity is zero (i.e. the body or object is stationary);
(2) slope = constant : The body moves with uniform velocity;
(3) The slope varies at different points and the gradient at a given point say A is the instantaneous velocity. This implies the body moves with a non-uniform velocity.

Velocity-time Graph

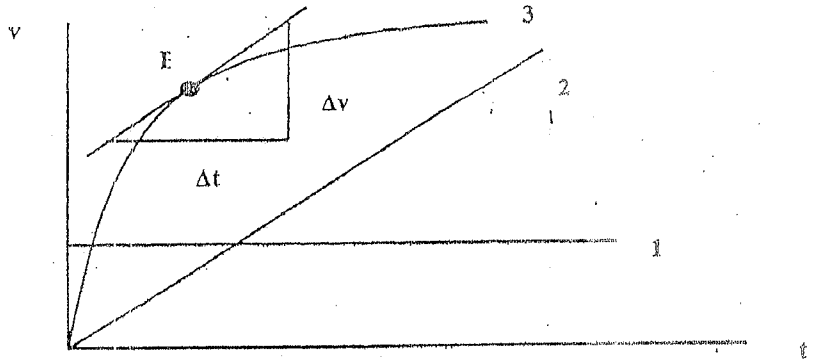


Fig. 4.2: The Velocity-time graph.

From Figure 4.2:

(1) slope = 0: acceleration is zero and hence velocity is constant
(2) slope = constant: The body moves with uniform acceleration
(3) The slope varies at different points and the gradient at a given point say B is the instantaneous acceleration, The body moves with a non-uniform acceleration. Note that for Velocity-Time graph the area under the graph is equal to displacement.

Acceleration-time Graph

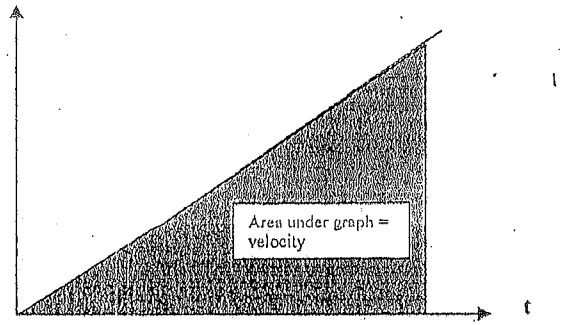


Fig. 4.3: The Acceleration-time graph.

For acceleration-Time graph the area under the graph is equal to the velocity.

Free Fall and Acceleration due to Gravity

An object is in free fall if it falls freely (i.e. in conditions similar to a vacuum without any resistance to motion) under the influence of gravity. Any object, regardless of its mass which falls freely near the surface of earth falls with an average acceleration of 9.8 m/s². Since the free fall acceleration is constant at a given point, we can use the kinematic equations of motion to solve problems involving free fall.

Example 4.4

A ball is dropped from rest from a high window of a tall building and falls for 3s. Neglecting air resistance (a) what is the speed of the ball? (b) What is the distance the ball has fallen from where it was released? (c) What is the ball's acceleration? [Assume that g = 10 m/s]. What distance will it travel in the last second (i.e. between t = 2 and t = 3)?

Solution

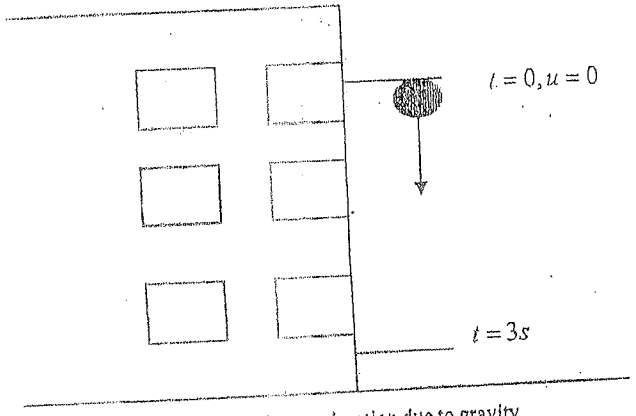


Fig. 4.4: A ball falling under acceleration due to gravity.

We start solving by listing the quantities already given in this problem:

$u = 0$ since the ball was dropped from rest, $t = 3s$, $a = g = 10m/s^2$ since the ball is falling freely under the influence of gravity.

(a) We are looking for the speed v after 3s, so we find an equation that includes v u, a and t. Equation 4.16 gives $v = u + at = 0 + (10m/s^2)(3s) = 30m/s$

(b) The distance fallen can be found by using Equation 4.17, choosing the downward direction as positive:

$$s = ut + \tfrac{1}{2}gt^2 = (0)(3s) + \tfrac{1}{2}(10m/s^2)(3s)^2 = 45m$$

Or use

$$\frac{v^2 - u^2}{2a} = s$$

(c) After 3s of free fall, the ball is still accelerating at 10m/s². The ball's speed is increasing, but the rate at which its speed is increasing (i.e. its acceleration) is constant.

$$S_2 = \frac{1}{2}.10.2^2; \quad S_3 = \frac{1}{2}.10.3^2$$

$$S_2 = 20, \quad S_3 = 45; \quad S_3 - S_2 = 45 - 20 = 25\,\text{m}$$

Relative Motion and Velocity in one dimension

Suppose that P and Q are moving points. Then PQ represents the position vector of Q relative to P, and $\frac{d}{dt}(PQ)$ represents its rate of change, which we shall call the velocity of Q relative to P and is written as v_{PQ}. So

$$v_{PQ} = \frac{d}{dt}(PQ) \qquad (4.18)$$

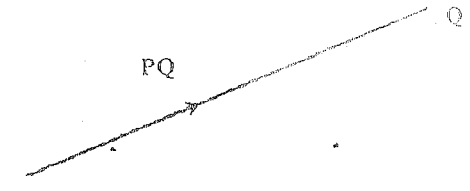


Fig. 4.5: Vector PQ

If we consider three moving points P, Q and R

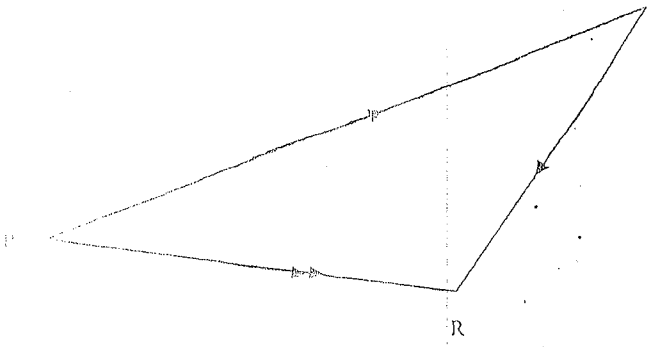


Fig. 4.5:

$$PQ + QR = PR \qquad 4.20$$

Hence differentiating with respect to time we have

$$\frac{d}{dt}(\mathbf{PQ}) + \frac{d}{dt}(\mathbf{QR}) = \frac{d}{dt}(\mathbf{PQ} + \mathbf{QR}) = \frac{d}{dt}(\mathbf{PR}) \qquad 4.21$$

$$\Rightarrow \mathbf{v}_{PQ} + \mathbf{v}_{QR} = \mathbf{v}_{PR} \qquad 4.22$$

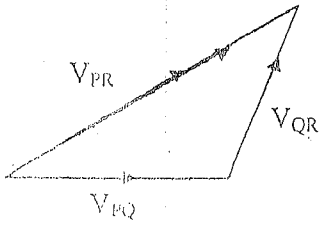


Fig. 4.6:

Furthermore, the rate of change of $\mathbf{v}_{PQ}$ can be called the acceleration of Q relative to P and is written as

$$\mathbf{a}_{PQ} = \frac{d}{dt}(\mathbf{v}_{PQ}) \qquad 4.23$$

and it immediately follows that

$$\mathbf{a}_{PQ} + \mathbf{a}_{QR} = \mathbf{a}_{PR} \qquad 4.24$$

Circular Motion

Uniform Circular Motion

A particle is in uniform circular motion if it travels around a circle or a circular arc at constant speed. A car moving in a circular path with linear constant speed v is in uniform circular motion. Although the speed does not vary, the car is accelerating. Since the car's direction of motion changes, the car has acceleration. This fact may be surprising because acceleration is usually thought of as an increase or decrease in speed. But actually v is a vector, not a scalar. If v changes, even only in direction, there is acceleration, as happens in uniform circular motion. For any motion, the velocity vector is tangent to the path. Consequently, when an object moves in a circular path, its velocity vector is perpendicular to the radius of the circle.

To find the magnitude and direction of the acceleration, consider Figure 4.7a which shows a particle first at point A and then at point B. Let the particle be at point A at time t_i, with velocity $\mathbf{v}_i$ and at point B at time t_f with velocity $\mathbf{v}_f$. Assume that $\mathbf{v}_i$ and $\mathbf{v}_f$ have the same magnitudes (speeds) but differ only in direction.

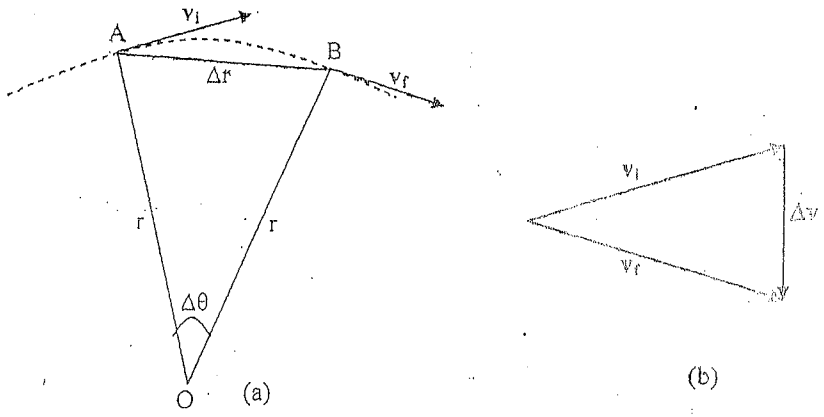


Fig. 4.7: (a) As a particle moves from A to B, its velocity vector changes from $\mathbf{v}_i$ to $\mathbf{v}_f$.
(b) The construction for determining the direction of the change in velocity $\Delta\mathbf{V}$, which is toward the centre of the circle for small Δr

We know from the definition of average acceleration that:

$$\bar{\mathbf{a}} = \frac{\mathbf{v}_f - \mathbf{v}_i}{t_f - t_i} = \frac{\Delta \mathbf{v}}{\Delta t} \qquad 4.25$$

Consider the triangle in Figure 4.7b with sides Δr and r. This triangle is similar with the one in Figure 4.7a having sides Δv and v. From the two triangles we can write

$$\frac{\Delta v}{v} = \frac{\Delta r}{r} \qquad 4.26$$

Solve for and substitute in the expression $\bar{a} = \Delta v / \Delta t$ to get

$$\bar{a} = \frac{v \Delta r}{r \Delta t} \qquad 4.27$$

If points A and B are extremely close together, then Δv points toward the centre of the circular path, and because the acceleration is in the direction of Δv, the acceleration also points toward the centre. In the limit as Δt approaches zero, the ratio $\Delta r / \Delta t$ also goes to the speed and the magnitude of the acceleration is

$$a_r = \frac{v^2}{r} \qquad 4.28$$

We then conclude that, when a particle moves at constant speed v in a circle of radius r, the acceleration of the particle is directed toward the centre of the circle and has a constant magnitude v^2/r.

Consider a particle moving along a curved path where the velocity changes in both direction and magnitude, as shown in Figure 4.8 As always, the velocity vector is tangent to the path, but the direction of the acceleration vector a changes from point to point. This vector can be resolved into two components: *a radial component* a_r and *a tangential component* a_t. Thus a becomes

$$\mathbf{a} = \mathbf{a}_r + \mathbf{a}_t \qquad 4.29$$

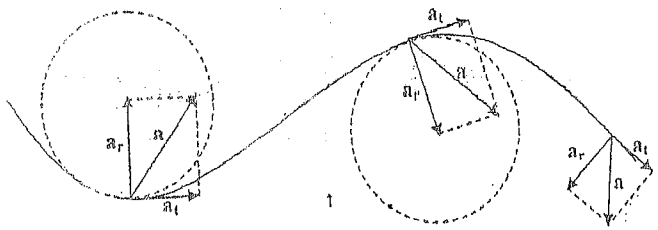


Fig. 4.8: The motion of a particle along an arbitrary curved path lying in the xy plane. If the velocity vector v (always tangent to the path) changes in direction and magnitude, the component vectors of the acceleration a are a tangential component a_t and a radial component a_r.

The tangential acceleration causes the change in the speed of the particle and it is parallel to the instantaneous velocity, and its magnitude is

$$a_t = \frac{d|v|}{dt} \qquad 4.30$$

The radial acceleration arises from the change in direction of the velocity vector and has an absolute magnitude given by

$$a_r = \frac{v^2}{r} \qquad 4.31$$

where r is the radius of the curvature of the path at the point under consideration.

The acceleration associated with uniform circular motion is called a centripetal (centre seeking) acceleration. During this acceleration at constant speed, the particle travels the distance $2\pi r$ (circumference) of the circle in time T given by

$$T = \frac{2\pi r}{v} \qquad 4.32$$

which is called the period of revolution, or the period of the motion. In general, T is the time for a particle to go round a closed path exactly once.

Circular Motion

Consider an object moving in a circle with a uniform speed round a fixed point O as centre as shown in Figure 4.9. Assume the object moves from A to B so that the radius OA moves through an angle θ. A particle at B is at a fixed distance r from the origin and rotates about it in a circle of radius r. We represent the position of B with its polar coordinates (r, θ), where r is the distance from the origin to B and θ is measured anticlockwise from some preferred direction like the positive x-direction.

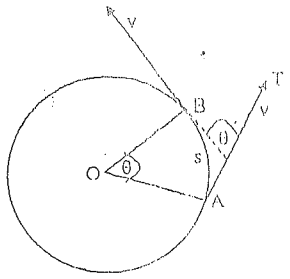


Fig. 4.9: Circular motion

In this way only the angle è changes with time; r remains constant. If s is the length of the arc AB, then s/r = è, by definition of an angle in radians.

$$\therefore s = r\theta \qquad 4.33$$

and

$$\theta = s/r \qquad 4.34$$

One radian is *the angle subtended by an arc length equal to the radius of the arc.*

Since the circumference of a circle is $2\pi r$, it follows from Equation 4.34 that 360^0 corresponds to an angle of $2\pi r/r\ rad = 2\pi\ rad$. Hence $1rad = 360^0/2\pi \approx 57.3^0$.

To convert an angle in degrees to an angle in radians, we have: $2\pi rad = 360^0$

$$\theta(rad) = \frac{\pi}{180^0}\theta(\deg) \qquad 4.35$$

As an example, $60^0 = \pi/3\ rad$ and $90^0 = \pi/2\ rad$. As the object travels from position P to position Q in a time Δt as shown in Figure 4.10, the radius vector sweeps out an angle $\Delta\theta = \theta_f - \theta_i$ called the **angular displacement** of the body. That is

$$\Delta\theta = \theta_f - \theta_i$$

The angular speed, ù, about O is defined as the change of the angle per second. If t is the time taken to move from A to B as in Figure 4.9, then

$$\omega = \frac{\theta}{t} \qquad 4.36$$

expressed in 'radians per second' (rad s^{-1}).

Define the average angular speed $\bar{\omega}$ as the ratio of the angular displacement to the time interval Δt (Figure 4.10)

$$\bar{\omega} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\Delta\theta}{\Delta t} \qquad 4.37$$

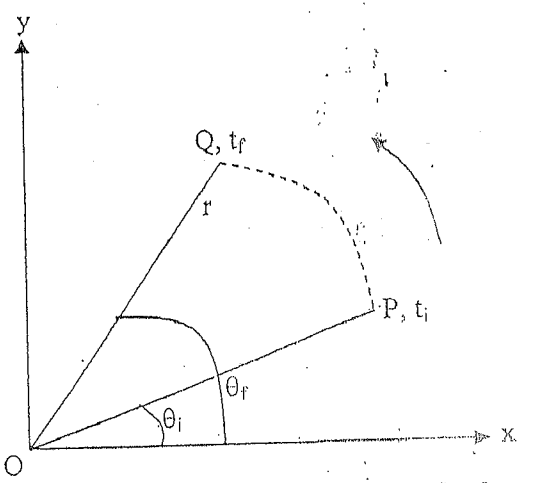


Fig. 4.10: A particle on a rotating rigid object moves from P to Q along the arc of a circle. In the time interval $\Delta t = t_f - t_i$, the radius vector sweeps out an angle $\Delta\theta = \theta_f - \theta_i$.

The instantaneous angular speed is defined as the limit of the ratio $\Delta\theta/\Delta t$ as Δt goes to zero:

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \qquad 4.38$$

ω is taken to be positive when θ is increasing (anticlockwise motion) and negative when è is decreasing (clockwise motion).

Divide Equation 4.33 by t, the time to move from A to B in Figure 4.9,

$$\therefore \frac{s}{t} = r\frac{\theta}{t} \qquad 4.39$$

$$\therefore v = r\omega \qquad 4.40$$

If the angular velocity of a rotating body is not constant, then the body has an angular acceleration. Let ω_f and ω_i be the angular velocities at time t_f and t_i respectively. The average angular acceleration of the rotating body in the interval from t_i to t_f (Δt) is defined as

$$\bar{\alpha} = \frac{\omega_f - \omega_i}{t_f - t_i} = \frac{\Delta\omega}{\Delta t} \qquad 4.41$$

where $\Delta\omega$ is the change in the angular velocity that occurs during the time interval Δt. The instantaneous angular acceleration is defined as the limit of the ratio $\Delta\omega/\Delta t$ as Δt goes to zero:

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} \qquad 4.42$$

The unit of angular acceleration is the radian per second-squared (rad/s²) or the revolution per second –squared (rev/s²).

Divide Equation 4.40 by t to get:

$$\frac{v}{t} = r\frac{\omega}{t} \qquad 4.43$$

v/t is the tangential component a_t of the linear acceleration and ù/t is the angular acceleration a. Then

$$a_t = \alpha r \qquad 4.44$$

As has been seen already, the radial component of the acceleration is:

$$a_r = \frac{v^2}{r} = \omega^2 r \qquad 4.45$$

Example 4.5

Express each of the following in terms of the other angular measures: (a) 28°, (b) $\frac{1}{4}rev/s$, (c) 2.18 rad/s^2.

Solution

(a) $28^0 = (28\,deg)\left(\frac{1\,rev}{360\,deg}\right) = 0.078\,rev$

$= (28\,deg)\left(\frac{2\pi\,rad}{360\,deg}\right) = 0.49\,rad$

(b) $\frac{1}{4}\frac{rev}{s} = \left(0.25\frac{rev}{s}\right)\left(\frac{360\,deg}{1\,rev}\right) = 90\frac{deg}{s}$

$= \left(0.25\frac{rev}{s}\right)\left(\frac{2\pi\,rad}{1\,rev}\right) = \frac{\pi}{2}\frac{rad}{s}$

(c) $2.18\frac{rad}{s^2} = \left(2.18\frac{rad}{s^2}\right)\left(\frac{360\,deg}{2\pi\,rad}\right) = 125\frac{deg}{s^2}$

$= \left(2.18\frac{rad}{s^2}\right)\left(\frac{1\,rev}{2\pi\,rad}\right) = 0.35\frac{rev}{s^2}$

Example 4.6

A car moves round a circular track of radius 0.3 m at 2 revolutions per second. What is

(a) the angular speed ω,
(b) the period T,
(c) the speed v of the car? Find also
(d) the angular speed of the car if it moves with a uniform speed of 2 m/s round a circle of radius 0.4 m.

Solution

(a) For 1 revolution, angle turned $\theta = 2\pi\,rad\,(360^0)$. So that

$\omega = 2 \times 2\pi = 4\pi\,rads^{-1}$

(b) Period T = time for 1 rev. $= \frac{2\pi}{\omega} = \frac{2\pi}{4\pi} = 0.5\,s$

(c) Speed $v = r\omega = 0.3 \times 4\pi = 1.2\pi = 3.8\, ms^{-1}$

(d) From $v = r\omega$

$$\omega = \frac{v}{r} = \frac{2\, ms^{-1}}{0.4\, m} = 5\, rads^{-1}$$

Simple Harmonic Motion (SHM)

Any motion that repeats itself at regular intervals is called **periodic motion** or **harmonic motion**. It turns out that for such motion the displacement x of the particle from the origin is given as a function of time.

Consider a physical system that consists of a block of mass m attached to the end of a spring, with the block free to move on a horizontal, frictionless surface as shown in Figure 4.11. When the spring is neither stretched nor compressed, the block is at the position $x = 0$, which is the equilibrium position of the system as shown in Figure 4.11b. When the block is displaced a distance x, which is small, from equilibrium, the spring, with a spring constant k, exerts a force on the block which is proportional to the displacement and given by Hooke's law as

$$F = -kx \qquad 4.46$$

The force is always directed towards the equilibrium position and, therefore, *opposite* the displacement and it is called a **restoring force**. This means that when the block is displaced to the left of $x = 0$ in Figure 4.11c then the displacement is negative and the restoring force is directed to the right. When the block is displaced to the right of, then the displacement is positive and the restoring force is directed to the left.

Let us apply Newton's second law to the motion of the block, together with Equation 4.46 to obtain

$$F = -kx = ma \qquad 4.47$$

$$a = -\frac{k}{m}x \qquad 4.48$$

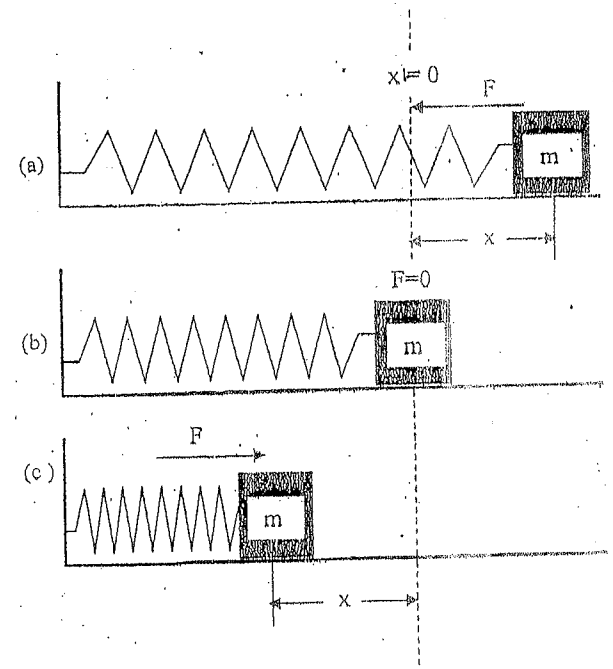


Fig. 4.11: A block attached to a spring moving on a frictionless surface. (a) when the block is displaced to the right of equilibrium (x > 0), the force exerted by the spring acts to the left. (b) when the block is at its equilibrium position (x = 0), the force exerted by the spring is zero. (c) when the block is displaced to the left of equilibrium (x < 0), the force exerted by the spring acts to the right.

But the acceleration is also given by

$$a = -\omega^2 x \qquad 4.49$$

which means that

$k = m\omega^2$ and

$$\omega = \sqrt{k/m} \qquad 4.50$$

From Equation 4.49 the acceleration is proportional to the displacement but opposite in sign, and the two quantities are related by the angular frequency-squared. Systems that behave this way are said to show *simple harmonic motion*. *Simple*

harmonic motion is the motion executed by a particle of mass m subject to a force that is proportional to the displacement of the particle but opposite in sign.

In general, it turns out that for such motion, the displacement x of the particle from the origin is given as a function of time by

$$x = A\cos(\omega t + \phi) \qquad 4.51$$

where A, ω, and ϕ are constants. The quantity A, a positive constant whose value depends on how the motion was started, is called the *amplitude* of the motion. The cosine function in Equation 4.51 varies between the limits ±1, so the displacement x varies between the limits $\pm A$, as shown in Figure 4.12.

The time-varying quantity $(\omega t + \phi)$ is called the *phase* of the motion, and the constant ϕ is called the *phase constant* (or *phase angle*). If the particle is at its maximum position $x = A$ at $t = 0$, then $\phi = 0$ as shown in Figure 4.12b. Let us interpret the constant ω. The displacement x must return to its initial value after one period T of the motion. That is, x must equal $x(t + T)$ for all. Let us put $\phi = 0$ in Equation 4.51 to simplify the analysis. From Equation 4.51 we have

$$A\cos\omega t = A\cos[\omega(t + T)] \qquad 4.52$$

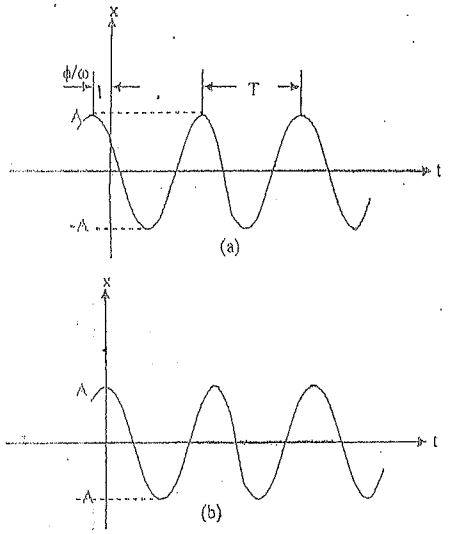


Fig. 4.12: (a) An x – t curve for a particle undergoing simple harmonic motion. The amplitude of the motion is A, the period is T, and the phase constant is f. (b) The x – t curve in the special case in which $x = A$ at $t = 0$ and hence $\phi = 0$.

The cosine function first repeats itself when its argument (the phase) has increased by 2ð rad, so that we should have

$$\omega(t + T) = \omega t + 2\pi$$

$$\therefore \omega T = 2\pi$$

Thus

$$\omega = \frac{2\pi}{T} \qquad 4.54$$

The constant ù is called the angular frequency of the motion and has units of radians per second.

The linear velocity of a particle undergoing simple harmonic motion can be found by differentiating Equation 4.51 with respect to time,

$$v = \frac{dx}{dt} = \frac{d}{dt}\left[A\cos(\omega t + \phi)\right] \qquad 4.55$$

which means

$$v = -\omega A\sin(\omega t + \phi) \qquad 4.56$$

From Equation 4.56, the acceleration of the particle is

$$a = \frac{dv}{dt} = \frac{d}{dt}\left[-\omega A\sin(\omega t + \phi)\right]$$

which gives

$$a = -\omega^2 A\cos(\omega t + \phi) \qquad 4.57$$

But $x = A\cos(\omega t + \phi)$ so Equation 4.57 becomes

$$a = -\omega^2 x \qquad 4.58$$

From Equation 4.57, because the sine function oscillates between ±1, the extreme values of v are $\pm\omega A$. From Equation 4.58, and since the cosine function oscillates between ±1, the extreme values of a are $\pm\omega^2 A$. Therefore, the maximum speed and the magnitude of the maximum acceleration of a particle moving in simple harmonic motion are

$$v_{max} = \omega A \qquad 4.59$$

$$a_{max} = \omega^2 A \qquad 4.60$$

The importance of the phase constant can be seen when comparing the motion of two or more oscillating objects. Let us imagine two identical pendulum bobs swinging side by side in simple harmonic motion, with one being released later than the other. Then the bobs have different phase constants. We can also show how the phase constant and the amplitude of any particle moving in simple harmonic motion can be determined if we know the particle's initial speed and position and the angular frequency of its motion.

Now suppose that at $t = 0$ the initial position of a single oscillator is $x = x_i$ and its initial speed is $v = v_i$. Under such conditions; Equation 4.51 and Equation 4.56 give

$$x_i = A\cos\phi \qquad 4.61$$

$$v_i = -\omega A\sin\phi \qquad 4.62$$

Equation 4.62 divided by Equation 4.61 eliminates A, to give $v_i/x_i = -\omega\tan\phi$, or

$$\tan\phi = \frac{-v_i}{\omega x_i} \qquad 4.63$$

Square Equations 4.61 and 4.62, divide Equation 4.62 by ω^2, and then add terms to obtain,

$$x_i^2 + \left(\frac{v_i}{\omega}\right)^2 = A^2\cos^2\phi + A^2\sin^2\phi.$$

But $\sin^2 + \cos^2 = 1$, then

$$A = \sqrt{x_i^2 + \left(\frac{v_i}{\omega}\right)^2} \qquad 4.64$$

Recall that the period of any simple harmonic oscillator is $T = 2\pi/\omega$ and that the frequency is the inverse of the period. Also $\omega = \sqrt{k/m}$, so the period and the frequency of the spring-block system become

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \qquad 4.65$$

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad 4.66$$

Equations 4.65 and 4.66 tell us that a large angular frequency and thus a small period go with a stiff spring (i.e. large k) and a light block (small m). Furthermore, the frequency and period are independent of the amplitude of the motion.

Energy in Simple Harmonic Motion

We examine the mechanical energy of the block-spring system illustrated in Figure. 4.13. Since the surface is assumed frictionless, it is expected that the total mechanical energy will be constant. The potential energy of a linear oscillator like that of Figure 4.13 is associated entirely with the spring. Its value depends on how much the spring is stretched or compressed, that is, on x.

The elastic potential energy stored in the spring of any elongation x is given by $\frac{1}{2}kx^2$. From Equation 4.51, we have

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \phi) \qquad 4.67$$

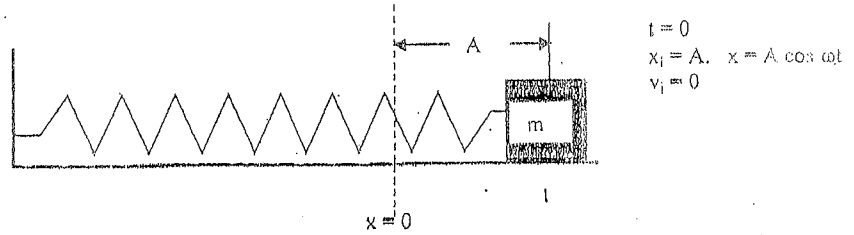


Fig. 4.13: A block – spring system that starts from rest at $x_i = A$. In this case f = 0 and thus $x = A\cos\omega t$.

The kinetic energy of the system is associated with the block only. Its value depends on how fast the block is moving, that is, on v. From Equation 4.56, the kinetic energy can be expressed as

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \phi) \qquad 4.68$$

U and K are always positive quantities. Since $\omega^2 = k/m$, the mechanical energy of the simple harmonic oscillator can be expressed as

$$E = U + K = \frac{1}{2}kA^2\left[\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)\right] \qquad 4.69$$

But the identity $\sin^2\theta + \cos^2\theta = 1$, so that if we let $\theta = \omega t + \phi$ for the quantity in bracket, then

$$E = \frac{1}{2}kA^2 \qquad 4.70$$

The mechanical energy of a linear oscillator is indeed constant, independent of time. *The total mechanical energy of a simple harmonic oscillator is a constant of the motion and is proportional to the square of the amplitude.* Note also that U is small when K is large, and vice versa, because the sum must be constant.

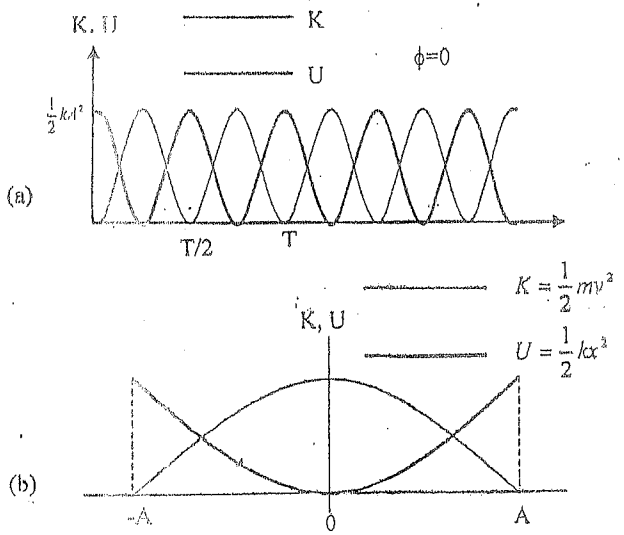


Fig. 4.14: (a) Kinetic energy and potential energy verses time for a simple harmonic oscillator with $\phi = 0$. (b) Kinetic energy and potential energy verses displacement for a simple harmonic oscillator. In either plot, note that K + U = constant

The total mechanical energy is equal to the maximum potential energy stored in the spring when $x = \pm A$ because $v = 0$ at these points and thus there is no kinetic energy. At the equilibrium position, where $U = 0$ because $x = 0$, the total energy, all in the form of kinetic energy, is again $\frac{1}{2}kA^2$. That is

$$E = \frac{1}{2}mv_{max}^2 = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}m\frac{k}{m}A^2 = \frac{1}{2}kA^2 \qquad 4.71$$

when $x = 0$.

Plots of the kinetic and potential energies against time are shown in Figure 4.14a, with $\phi = 0$. The variations of K and U with the displacement x of the block are plotted in Figure 4.14b. Energy is continuously being transformed between potential energy stored in the spring and kinetic energy of the block.

It can now be understood why an oscillating system contains an element of springiness and an element of inertia: it uses the former to store its potential energy and the latter to store its kinetic energy.

Example 4.7

A spring makes 12 vibrations in 40 s. Find the period and frequency of the vibration.

Solution

$$T = \frac{elapsed\ time}{vibrations\ made} = \frac{40s}{12} = 3.3s$$

$$f = \frac{\text{vibrations made}}{\text{elapsed time}} = \frac{12}{40s} = 0.30Hz$$

Example 4.8

When a 400 g mass is hung at the end of a vertical light spring, the spring stretches 35 cm. What is the spring constant, k, and how much further will it stretch if an additional 400 g mass is hung from it?

Solution

We know that

$F = ky$, since the spring is vertical, where

$F = mg = (0.400kg)(9.8m/s^2) = 3.92N$ so that

$$k = \frac{F}{y} = \frac{3.92N}{0.35m} = 11.2\,N/m$$

With an additional 400 g load, the total force stretching the spring is 2F = 7.84 N. Then

$$y = \frac{F}{k} = \frac{7.84N}{11.2\,N/m} = 0.70m = 2\times 35cm$$

So that each 400 g load stretches the spring by the same amount whether the spring is loaded or not.

Example 4.9

A 200 g mass vibrates horizontally without friction at the end of a horizontal spring for which k = 7.0 N/m. The mass is displaced 5.0 cm from equilibrium and released. Find (a) its maximum speed and (b) its speed when it is 3.0 cm from equilibrium. (c) What is its acceleration in each of these cases?

Solution

Using the conservation of energy law in this case gives,

$\frac{1}{2}x_0^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$, where

$k = 7.0\,N/m$, $x_0 = 0.050m$, and $m = 0.200kg$. Then v becomes

$$v = \sqrt{\frac{k}{m}\left(x_0^2 - x^2\right)}$$

(a) For maximum speed $x = 0$; i.e., when the mass is passing through the equilibrium position:

$$v = x_0\sqrt{\frac{k}{m}} = (0.05m)\sqrt{\frac{7.0\,N/m}{0.200kg}} = 0.296\,m/s$$

(b) When $x = 0.030m$,

$$v = \sqrt{\frac{7.0\,N/m}{0.200kg}\left[(0.050)^2 - (0.030)^2\right]m^2} = 0.237\,m/s$$

(c) Using $F = ma$ and $F = kx$, we have

$a = \frac{k}{m}x = (35s^{-2})(x)$ which gives a = 0 when the mass is at $x = 0$ and

$a = 1.05\,m/s^2$ when $x = 0.030m$

Example 4.10

A 50 g mass vibrates in simple harmonic motion (SHM) at the end of a light spring. The amplitude of the motion is 12 cm, and the period is 1.70 s. Find: (a) the frequency, (b) the spring constant, (c) the maximum speed of the mass, (d) the maximum acceleration of the mass, (e) the speed when the displacement is 6 cm, and (f) the acceleration when $x = 6\,cm$.

Solution

(a) $f = \frac{1}{T} = \frac{1}{1.70s} = 0.588\,H_z$

(b) We know that $T = 2\pi\sqrt{m/k}$, so that

$$k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2(0.050kg)}{(1.70s)^2} = 0.68\,N/m$$

(c) $v_0 = x_0\sqrt{\frac{k}{m}} = (0.12m)\sqrt{\frac{0.68\,N/m}{0.050kg}} = 0.44\,m/s$

(d) From $a=-\left(k/m\right)x$, a has a maximum magnitude when x is maximum, i.e., at the endpoints $x=\pm x_0$. So that

$$a_0=\frac{k}{m}x_0=\frac{0.68\,N/m}{0.050kg}(0.12m)=1.63\,m/s^2$$

(e) We know $|v|=\sqrt{\left(x_0^2-x^2\right)}\left(k/m\right)$,

$$|v|=\sqrt{\left(0.12^2-0.06^2\right)\frac{0.68}{0.050}}=0.38\,m/s$$

(f) $$a=-\frac{k}{m}x=-\frac{0.68\,N/m}{0.050kg}(0.06m)=-0.82\,m/s^2$$

Example 4.11

A 20 kg electric motor is mounted on four vertical springs, each having a spring constant of $30\,N/cm$. Find the period with which the motor vibrates vertically.

Solution

The four springs can be replaced by an equivalent spring and its spring constant becomes $4\left(3000\,N/m\right)$ or $12000\,N/m$. Then

$$T=2\pi\sqrt{\frac{m}{k}}=2\pi\sqrt{\frac{20kg}{12000\,N/m}}=0.256\,s$$

Pendulums

Let us turn now to a class of simple harmonic oscillators in which the springiness is associated with the gravitational force rather than with the elastic properties of a twisted wire or a compressed or stretched spring.

The Simple Pendulum

It is another mechanical system that exhibits periodic motion. It consists of a particle-

like bob of mass m suspended from an unstretchable, massless string of length L fixed at the upper end as shown Figure 4.15a. The bob is free to swing back and forth in the plane of the page, to the left and right of a vertical line through the point at which the upper end of the string is fixed. The forces acting on the particle, shown in Figure 4.15b are its weight mg and the tension T in the string. The weight can be resolved into a radial component $mg\cos\theta$ and a component $mg\sin\theta$ tangent to the path taken by the particle. This tangential component is a restoring force, since it always acts opposite the displacement of the particle so as to bring the particle back toward its central location, the equilibrium position ($\theta=0$), where it would be at rest were it not swinging.

We use Newton's second law to write the restoring force as

$$F_t=-mg\sin\theta=m\frac{d^2s}{dt^2}$$

where s is the bob's displacement measured along the arc and the minus sign indicates that the tangential force acts toward the equilibrium (vertical) position. Since $s=L\theta$ and L is constant, Equation 4.72 reduces to

$$\frac{d^2\theta}{dt^2}=-\frac{g}{L}\sin\theta \qquad 4.73$$

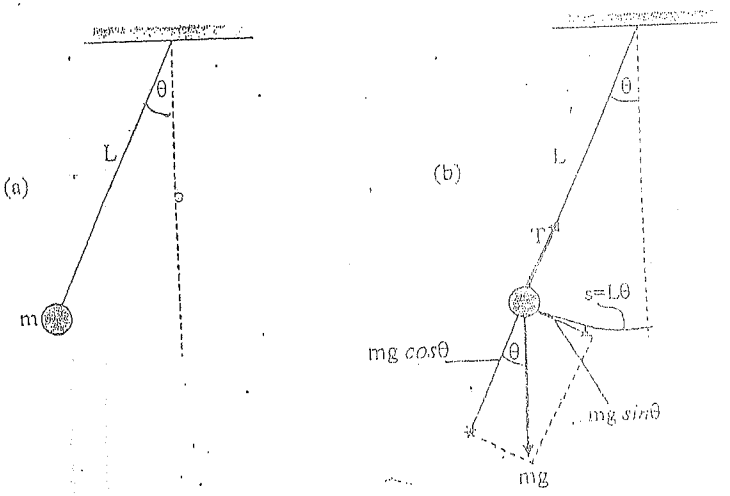


Fig. 4.15: (a) A simple penduluim. (b) The forces acting on the bob are its weight mg and the tension T in the string. The tangential component mg sin q of the weight is a restoring force that brings the pendulum back to the central position.

For small angles (i.e. sin $\theta \approx \theta$) the equation of motion for the simple pendulum becomes

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta \qquad 4.74$$

Equation 4.74 is in the form of Equation 4.58, and we conclude that the motion for small amplitudes of oscillation is a simple harmonic motion. Therefore, θ can be written as $\theta = \theta_{max}\cos(\omega t + \phi)$, where θ_{max} is the maximum angular displacement and the angular frequency ω is

$$\omega = \sqrt{\frac{g}{L}} \qquad 4.75$$

The period of the motion of a simple pendulum is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}} \qquad 4.76$$

We see that, the period and frequency of a simple pendulum depend only on the length of the string and the acceleration due to gravity. Because the period is independent of the mass, we conclude that all simple pendulums that are of equal length and are at the same location (so that g is constant) oscillate with the same period.

The Conical Pendulum

Suppose a small object of mass m is tied to a string OA of length *l* and then whirled round in a horizontal circle of radius *r*, with O fixed directly above the centre B of the circle as shown in Figure 4.16. If the circular speed of A is constant, the string turns at a constant angle θ to the vertical. This is called a conical pendulum.

Since A moves with a constant speed v in a circle of radius r, there must be a centripetal force, mv^2/r, acting towards the centre B. The horizontal component, $T\sin\theta$ of the tension in the string provides this force along AB.

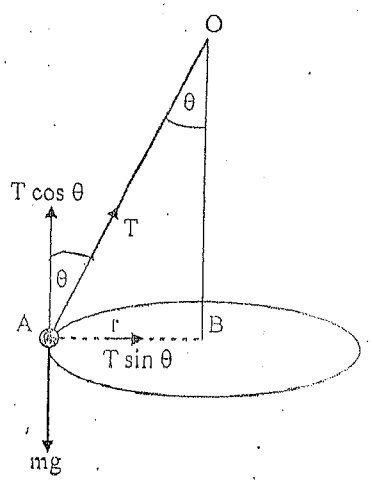


Fig. 4.16: Conical pendulum

So,

$$T\sin\theta = \frac{mv^2}{r} \qquad 4.77$$

Also, since the mass does not move vertically, its weight mg must be counterbalanced by the vertical component $T\cos\theta$ of the tension. So

$$T\cos\theta = mg \qquad 4.78$$

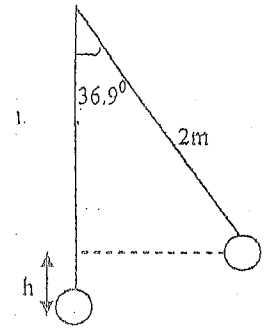
Dividing Equation 4.77 by Equation 4.78 gives

$$\tan\theta = \frac{v^2}{rg} \qquad 4.79$$

If a parallel beam of light is used to cast a shadow of the bob A on a screen, the motion of the shadow will be simple harmonic, maximum displacement is r, and $\omega = v/r$.

Example 4.12

pulled back through an angle of 36.9° and released. Find (a) its maximum potential energy relative to its lowest position, (b) its potential energy when the cord makes an angle of 10° with the vertical, (c) its maximum speed, and (d) its speed when the cord makes an angle of 10° with the vertical.



Solution

$h = 2 - 2\cos 36.9° = 2 - 1.6 = 0.4\text{m}$

(a) $PE_{max} = (0.5 \times 9.8)0.4 = 1.96\ \text{J}$

(b) When $\theta = 10°$, $h_{10} = 2 - 2\cos 10° = 2(1 - 0.9848) = 0.0304\text{m}$

$PE_{10°} = (0.5 \times 9.8) \times 0.0304 = 0.149J$

(c) At bottom $\frac{1}{2}\left(\frac{1}{2}\right) v_{max}^2 = 1.96$ $\qquad v_{max} = 2 \times 1.4 = 2.8m/s$

(d) $(1.96 - 0.149)\text{J} = \frac{1}{2}\left(\frac{1}{2}\right) v_{10}^2$

$v_{10}^2 = 7.24$ $\qquad v_{10} = 2.69\text{m/s}$

Projectile Motion

Anyone observing the motion of a stone or any object thrown into the air is observing projectile motion. The stone moves in a curved path, and the motion is simple to analyze if two assumptions are made: (1) the acceleration of free-fall g is constant over the range of motion and is directed downward, and (2) the effect of air resistance is negligible. Using these assumptions, it is found that the path of a projectile, also

called *trajectory*, is always a parabola. Projectile motion looks complicated but we have the following simplifying feature: *The horizontal motion and vertical motion are independent of each other; that is, neither motion affects the other.*

To show that the trajectory of a projectile is a parabola, we choose our reference frame such that the y direction is vertical and positive is upward. Neglecting air resistance, we have $a_y = -g$ and that $a_x = 0$ (the projectile has *no* horizontal acceleration). Furthermore, let us assume that at $t = 0$, the projectile leaves the origin ($x_0 = y_0 = 0$) with speed v_0 as shown in Figure 4.17.

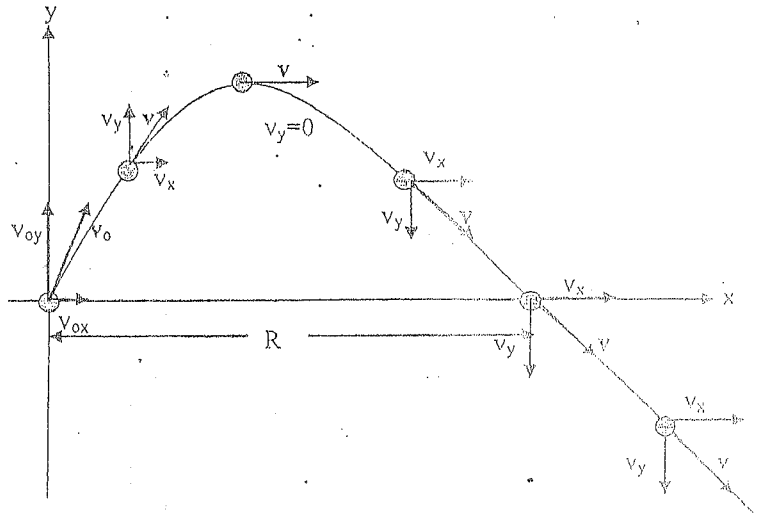


Fig. 4.17: The path of a projectile that is launched at $x_0 = 0$ and $y_0 = 0$, with an initial velocity v_0. The initial velocity and the velocities at various points along its path are shown along with their components. Note that the horizontal velocity component remains constant but the vertical velocity component changes continuously. The range R is the horizontal distance the projectile has traveled when it returns to its launch height.

The vector $\mathbf{v}_0$ makes an angle θ_0 with the horizontal, where θ_0 is the angle at which the projectile leaves the origin. The initial velocity $\mathbf{v}_0$ can be written as

$$\mathbf{v}_0 = v_{0x}\mathbf{i} + v_{0y}\mathbf{j} \qquad 4.80$$

The components v_{0x} and v_{0y} can be found if the angle between $\mathbf{v}_0$ and the positive x direction θ_0 is known:

$$v_{0x} = v_0 \cos\theta_0 \quad \text{and} \quad v_{0y} = v_0 \sin\theta_0 \tag{4.81}$$

As stated earlier for projectile motion, the horizontal motion and vertical motion are independent of each other; that is neither motion has an effect on the other. This allows breaking up a problem involving two-dimensional motion into two separate and easier one-dimensional problems, one for the horizontal motion and one for the vertical motion.

Analysis of Projectile Motion

The Horizontal Motion .

Since there is no acceleration in the horizontal direction, the horizontal component v_{ox} of the projectile's initial velocity remains unchanged throughout the motion. The horizontal displacement $x - x_0$ from an initial position x_o is given as

$$x - x_0 = v_{0x} t \tag{4.82}$$

since $a = 0$.

But $v_{0x} = v_0 \cos\theta_0$, so that

$$x - x_0 = (v_0 \cos\theta_0) t \tag{4.83}$$

The Vertical Motion

The vertical motion is the motion for a particle in free fall, i.e.

$$\begin{aligned} y - y_0 &= v_{0y} t - \frac{1}{2} g t^2 \\ &= (v_0 \sin\theta_0) t - \frac{1}{2} g t^2 \end{aligned} \tag{4.84}$$

where the initial vertical velocity component v_{0y} is replaced with the equivalent $v_o \sin\theta_o$.

The Equation of the Path

To find the equation of the path (the trajectory) of the projectile, eliminate t between Equations 4.83 and 4.84. Solve Equation 4.83 for t and substitute into Equation 4.84 and obtain after a little rearrangement,

$$y = (\tan\theta_0) x - \frac{g x^2}{2(v_0 \cos\theta_0)^2} \tag{4.85}$$

This is the equation of the path shown in Figure 4.17.

For simplicity in the derivation, we let $x_o = 0$ and $y_o = 0$ in Equations 4.83 and 4.84, respectively. Since g, θ_o, and v_o are constants, Equation 4.85 is of the form $y = ax + bx^2$, in which a and b are constants. This is the equation of a parabola, so the path is parabolic.

The Horizontal Range .

The horizontal range R of the projectile is the horizontal distance the projectile has traveled when it returns to its initial (launch) height as shown in Figure 4.17. To find the range R of the projectile, put $x - x_0 = R$ in Equation 4.83 and $y - y_0 = 0$ in Equation 4.84, obtaining

$$x - x_0 = (v_0 \cos\theta_0) t = R \tag{4.86}$$

and

$$y - y_0 = (v_0 \sin\theta_0) t - \frac{1}{2} g t^2 = 0 \tag{4.87}$$

Eliminating t between Equations 4.86 and 4.87 yields

$$R = \frac{2v_0^2}{g} \sin\theta_0 \cos\theta_0 \tag{4.88}$$

Using the identity $\sin 2\theta_0 = 2\sin\theta_0 \cos\theta_0$ we obtain

$$R = \frac{v_0^2}{g} \sin 2\theta_0 \tag{4.89}$$

Note that R has a maximum value when $\sin 2\theta_0 = 1$, which corresponds to

$$2\theta_o = 90 \text{ or } \theta_o = 45^0.$$

The horizontal range R is a maximum for a launch angle of 45^0.

We have assumed throughout that the air through which the projectile moves has no effect on its motion, which is a reasonable assumption at low speeds. However, at higher speeds, the disagreement between our calculations and the actual motion of the projectile can be large because of air resistance offered to the motion.

Example 4.13

A marble with speed 20 cm/s rolls off the edge of a table 80 cm high. How long does it take to drop to the floor? How far, horizontally, from the table edge does the marble strike the floor?

Solution

Choose downward as positive with origin at edge of table top.

$v_{ox} = v_0 = 20\text{cm/s} \qquad v_{oy} = 0 \qquad a_y = +g = +980\ \text{cm/s}^2 \qquad a_x = 0$

To find time of fall, $y = v_{oy}t + \frac{1}{2}gt^2$, or 80 cm = 0 + (490 cm/s2)t^2; t = 0.40s. The horizontal distance is gotten from $x = v_{ox}t$ = (20 cm/s) (0/40s) = 8.0cm.

Example 4.14

A marble traveling at 100 cm/s rolls off the edge of a level table. If it hits the floor 30 cm away from the spot directly below the edge of the table, how high is the table?

Solution

This is a projectile problem with v_{ox} = 100 cm/s. For the horizontal motion:

$s_x = v_{ox}t \qquad 30 = 100t \qquad t = 0.30\ s$

For the vertical motion:

$s_y = v_{oy}t + \frac{1}{2}at^2 = 0 + \frac{1}{2}(980)(0.30)^2 = 44.1\text{cm}$

Example 4.15

A golf ball leaves the golf club at an angle of 60° above the horizontal with a velocity of 30 m/s. (a) How high does it go? (b) Assuming level fairway, determine how far away it hits the ground.

Solution

(a) Taking up as positive, $v_{oy} = 30sin60° = 15\sqrt{3}$ m/s and, at maximum height, v_y equals zero. Thus,

$v_y^2 = v_{oy}^2 + 2ah \qquad 0 = 675 - 19.6h \qquad h = \frac{675}{19.6} = 34.4\text{m}$

(b) $x = \frac{v_0^2 sin2\theta}{9} = \frac{900(\sqrt{3}/2)}{9.8} = 79.5\text{m}$

Exercise 4

4.1 A driver moving with a velocity of 30.0ms^{-1} is approaching a traffic light. The light turns yellow, and the driver applies the brakes and skids to a stop. If his acceleration is 8.00ms^{-2}, determine the displacement of the car during the skidding process. **(Ans: 56.3m)**

4.2 A car moving with a velocity of 15ms^{-1} accelerates uniformly at the rate of 20ms^{-1}. Find (a) the time taken (b) the distance traveled in this time. (Ans: [i] **t = 2.5s, [ii] s = 43.75)**

4.3 A particle moves along a circular path of radius 3.0m with an angular velocity of 20 rads^{-1}. Calculate: (a) the linear speed of the particle (b) the angular velocity in revolution per second (c) the time for one revolutions (d) the centripetal acceleration. (Ans: [a] 60ms^{-1} , [b] 3.2revs^{-1} , [c] 0.32s , [d] **1.2×10^3ms^{-2})**

4.4 A particle of mass 0.2kg moves in a circular path with an angular velocity of 5rads^{-1} under the action of a centripetal force of 4N. What is the radius of the path? **(Ans: 0.8m)**

4.5 A particle of mass 0.20kg is attached to one end of a light inextensible string of length 50cm. The particle moves in a horizontal circle with an angular velocity of 5.0rads^{-1} with the string inclined at è to the vertical. Find the value of θ. **(Ans: 37 °)**

4.6 A flywheel with a moment of inertia of 5.0kgm^2 moves from rest under the action of a torque of 3.0Nm. Find: (a) the angular acceleration (b) the angular velocity after10 revolutions. (Ans: [a] 0.60rads^{-2}, [b] 8.7rads^{-1})

4.7 A particle is moving with SHM of period 16s and amplitude 10m. Find the speed of the particle when it is 6.0m from its equilibrium position. (Ans: $3.1ms^{-1}$)

4.8 A particle moving with SHM has a speed of $8.0ms^{-1}$ and an acceleration of $12ms^{-2}$ when it is 3.0m from its equilibrium position. Find: (a) the amplitude of the motion (b) the maximum velocity (c) the maximum acceleration. (Ans: [a] 5.0m, [b] $10ms^{-1}$, [c] $20ms^{-2}$)

4.9 A long jumper leaves the ground at an angle of 20° and jumped a distance of 7.94m. With what speed did he leave the ground? (Ans: $11ms^{-1}$)

4.10 A long jumper leaves the ground at an angle of 20° to the horizontal and at a speed of $11ms^{-1}$. (a) How far does he jump? (b) what is the maximum height. (Assume that motion of the long jumper is equivalent to that of a particle) (Ans: [a] 7.94m, [b] 0.722m)

Chapter Five

MECHANICAL WORK, ENERGY AND POWER

Introduction

The term *work* as understood in everyday life has a different meaning in scientific sense. If a woman is carrying a load on her head and waiting for the arrival of the bus, she is not performing any work in the scientific sense. In this chapter, we shall look at the scientific meaning of the term *work* which is commonly used by ordinary people. In addition to this, we shall also discuss into details the term *energy*. How many forms of energy are there? How is energy related to work? What are the laws which govern energy?

Work

The physical meaning of the term work is entirely different from the meaning attached to it in everyday life. In everyday life, the work is considered to be synonymous to 'labour', 'toil', 'effort' etc. In physics there is a specific way of defining work, that is

Work is said to be done by a force when the force produces a displacement in the body which it acts on in any direction except perpendicular to the direction of the force.

For work to be done, the following two conditions must be fulfilled.

1. A force must be applied,
2. The applied force must produce a displacement in any direction except perpendicular to the direction of the force.

Suppose a force **F** is applied on a body in such a way that the body suffers displacement **S** in the direction of the force. Then the work done is given by

$$W = \mathbf{F}.\mathbf{S} \qquad 5.1$$

However, the displacement does not always take place in the direction of the force.

Suppose a constant force F applied to a body produces a displacement S in such a way that S is inclined to F at an angle θ as shown Figure 5.1. Now the work done will be given by the dot product of force and displacement defined as

W = component of force in the direction of displacement x magnitude of the displacement.

$$\therefore W = \mathbf{F}.\mathbf{S} = FS\cos\theta = (F\cos\theta)S \qquad 5.2$$

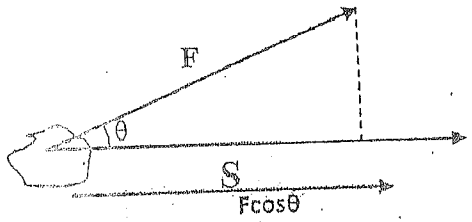


Fig. 5.1: Force F acts on the object at an angle θ on the horizontal.

So, work is the product of the component of force in the direction of displacement and the magnitude of the displacement. In terms of vector components, F and S may be written as:

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} \qquad 5.3$$

$$\mathbf{S} = S_x\mathbf{i} + S_y\mathbf{j} + S_z\mathbf{k} \qquad 5.4$$

$$\mathbf{F}.\mathbf{S} = \left(F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}\right)\left(S_x\mathbf{i} + S_y\mathbf{j} + S_z\mathbf{k}\right) \qquad 5.5$$

$$= F_xS_x(\mathbf{i}.\mathbf{i}) + F_xS_y(\mathbf{i}.\mathbf{j}) + F_xS_z(\mathbf{i}.\mathbf{k}) + F_yS_x(\mathbf{j}.\mathbf{i}) + F_yS_y(\mathbf{j}.\mathbf{j}) + F_yS_z(\mathbf{j}.\mathbf{k}) + F_zS_x(\mathbf{k}.\mathbf{i}) + F_zS_y(\mathbf{k}.\mathbf{j}) + F_zS_z(\mathbf{k}.\mathbf{k})$$

But $\mathbf{i}.\mathbf{j} = \mathbf{i}.\mathbf{k} = \mathbf{j}.\mathbf{i} = \mathbf{j}.\mathbf{k} = \mathbf{k}.\mathbf{i} = \mathbf{k}.\mathbf{j} = 0$

and $\mathbf{i}.\mathbf{i} = \mathbf{j}.\mathbf{j} = \mathbf{k}.\mathbf{k} = 1$

$$\therefore \mathbf{F}.\mathbf{S} = F_xS_x + F_yS_y + F_zS_z \qquad 5.6$$

Since work done is the dot product of two vectors it is therefore considered as a scalar quantity.

Example 5.1

A ladder 3.0m long and weighing 200N has its centre of gravity 1.2m from the bottom. At its top end is a 50N weight. Calculate the work required to raise the loaded ladder from a horizontal position on the ground to a vertical position.

Solution

The work done (against gravity) consists of two parts.

(1) The work done to raise the center of gravity of the ladder to a height of 1.2m.
(2) The work done to raise the weight at the end of the ladder through 3.0m.
The work done to raise the centre of gravity of the ladder = 200N × 1.2m = 240 J.
The work done to raise the weight at the end of the ladder = 50N × 3.0 = 150J
Total work done = 240 + 150 = 390 J.

Special Cases

Case 1

When $\theta = 90^0$, then $W = FS\cos 90^0 = 0$ (since $\cos 90^0 = 0$)

So, work done by a force is zero if the body is displaced in a direction perpendicular to the direction of the force.

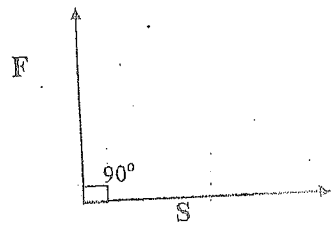


Fig: 5.2: Work done by force F perpendicular to displacement S

Example 5.2

Consider a body sliding over a smooth horizontal surface. The work done by the force of gravity and the reactional force of the surface will be zero. This is because both force of gravity and the reaction act normally to the displacement.

The same argument can be applied to a man carrying a load on his head and walking on a level street.

Example 5.3

Consider a body moving in a circle with constant speed. At every point on the circular path, the *centripetal force* and the *displacement* are mutually perpendicular. So the work done by the centripetal force is zero. The same argument can be applied to a satellite moving in a circular orbit. In this case, the gravitational force is always perpendicular to the displacement. So, work done by the gravitational force is zero.

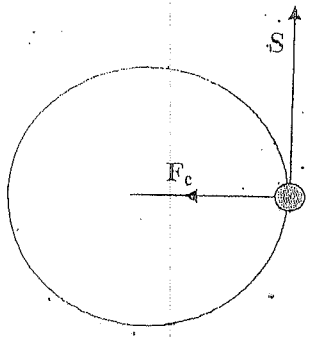


Fig. 5.3: A body moving in a circular path

Example 5.4

The tension in the string of a simple pendulum is always perpendicular to the displacement. So, work done by the tension is zero.

Case 2

When $S = 0$, then $W = 0$.

So, work done by a force is zero if the body suffers no displacement on the application of a force.

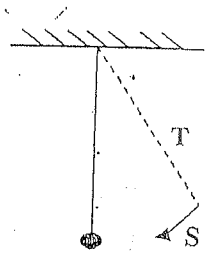


Fig. 5.4: Simple pendulum

Example 5.5

A man carrying a load on his head and standing at one place does no work.

Case 3

When $0^0 \leq \theta \leq 90^0$, then $\cos\theta$ is positive.

Therefore, $W = FS\cos\theta$ is positive.

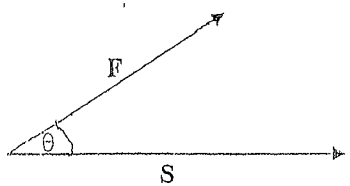


Fig. 5.5: Work done by force F inclined at an acute angle θ to the displacement S

Definition

Work done by a force is said to be positive if the applied force has a component in the direction of the displacement.

Example 5.6

When a horse pulls a cart, the applied force and displacement are in the same direction. So work done by the horse is positive.

Example 5.7

When a body is lifted, the lifting force and the displacement act in the same direction. So work done is positive.

Example 5.8

When a spring is compressed, both the compressing force and the displacement act in the same direction therefore work done is positive.

Case 4

When $90^0 \leq \theta \leq 180^0$, then $\cos\theta$ is negative. Therefore $W = FS\cos\theta$ is negative.

Definition

Work done by a force is said to be negative if the applied force has a component in a direction opposite to that of the displacement.

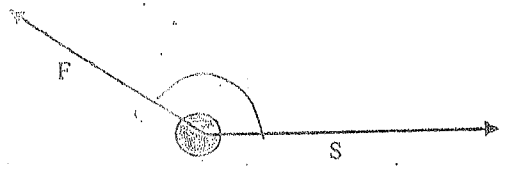


Fig. 5.6: Work done by force F inclined at an obtuse angle to the displacement S

Example 5.9

When brakes are applied to a moving vehicle, the work done by the braking force is negative. This is because the braking force and the displacement act in opposite directions.

Example 5.10

When a body is dragged along a rough surface, the work done by the friction force is negative. This is because the frictional force acts in a direction opposite to that of the displacement.

Example 5.11

When a body is lifted, the work done by the gravitational force is negative. This is because the gravitational force acts vertically downwards while the displacement is in the vertical upward direction.

Example 5.12

A boy drags a bag of rice along a horizontal, smooth floor, with a force, F = 2N applied horizontally for a distance S = 3m. What work has the body done? If the direction of F is not horizontal but an angle $\theta = 60^{\circ}$ to the floor, what work has the boy done in this case?

Solution

In the first case,

$$W = FS = 2x3 = 6J$$

In the second case,

$$W = F\cos\theta \, x \, S = 2\cos 60^{0} x3 = 3J$$

Hence, for a given force and displacement, maximum work is done when the force is applied in the direction of the displacement.

Work Done by a Variable Force

When the applied force varies both in magnitude and direction, then the total work done by the variable force is calculated by integration as described below and in Figure 5.7.

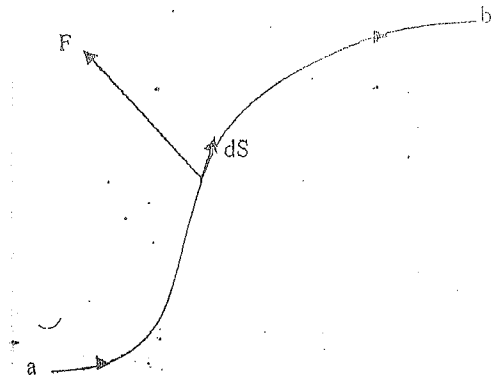


Fig. 5.7: A path taken by a variable force

Suppose the variable force displaces a body from position *a* to position *b* along a curved path. Let us divide the path of the particle into a large number of infinitesimally small displacements. Each displacement should be so small that the force, during that displacement, may be regarded as constant.

Let dW be the small amount of work done in giving an infinitesimally small displacement ds to the body. Let F be the corresponding force. Then,

$$dW = \mathbf{F}.ds \qquad 5.7$$

The total work done W_{ab}, on the body as it moves from a to b can be obtained by adding up the elements of work done over each of the infinitesimally small displacements between a and b. This can be done by integrating both sides of Equation 5.7 to get

$$\int dW = \int \mathbf{F}.ds$$

or

$$W_{ab} = \int F ds \cos\theta \qquad 5.8$$

or

$$W_{ab} = \int (F\cos\theta) ds \qquad 5.9$$

Equation 5.9 is the required expression for work done by a variable force.

Units of Work

Unit of work is the amount of work done when a unit force displaces a body through a unit distance in the direction of the force.. So, the units of work will depend upon the units of force and distance.

Absolute Units of Work

Work done is said to be *one absolute unit if an absolute unit of force moves a body through unit distance in the direction of the force.*

In SI units, the absolute unit of work is joule (abbreviated as J).

One Joule of work is said to be done *when a force of one Newton displaces a body through one metre in its own direction.*

$$\begin{aligned} 1\ joule &= 1Newton \times 1metre \\ &= 1kg \times 1ms^{-2} \times 1m \\ &= 1kgm^2s^{-2} \text{ or } Nm. \end{aligned} \qquad 5.10$$

Energy

Energy is the capacity or ability of a body to do work.

It is a scalar quantity. Its units and dimensions are the same as those of work.

There are many types of energy: mechanical energy, sound energy, heat energy, light energy, chemical energy, atomic energy, nuclear energy among others. Mechanical energy is of two types –kinetic energy and potential energy.

Kinetic Energy

Kinetic energy is *the energy possessed by a body by virtue of its motion*. It is logical that a body moving with higher speed should possess more kinetic energy than a body moving with lower speed. Examples of bodies possessing kinetic energy include

1. A ball in motion,
2. A moving vehicle,
3. Running water,
4. Air in motion.

Deriving Expression for kinetic Energy

Method I, Using Vectors

Consider a body of mass m lying on a smooth horizontal surface as shown in Figure 5.8. Let a constant force $\mathbf{F}$ displace this body in its own direction through $\mathbf{S}$. Let the velocity of the body, after traveling the distance S be v.

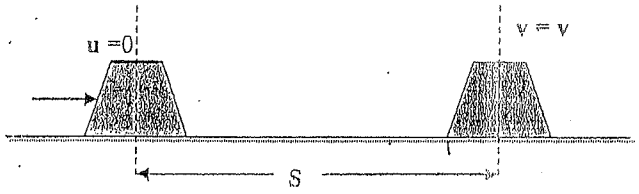


Fig. 5.8: A force F displaces a body of mass m through a distance S

If W is the work done by the force, then

$$W = \mathbf{F}.\mathbf{S} = FS\cos\theta \qquad 5.11$$

Both the force and the displacement are in the same direction

$$\Rightarrow \quad \theta = 0$$

Therefore,

$$W = FS\cos 0 = FS \qquad 5.12$$

If a is the acceleration produced in the body, then

$$a = \frac{F}{m} \qquad 5.13$$

From $v^2 - u^2 = 2aS$, with $u = 0$

$v^2 - 0^2 = 2aS$ so that

$v^2 = 2\left(\frac{F}{m}\right)S$ and using Equation 5.12

$$FS = W = \frac{1}{2}mv^2 \qquad 5.14$$

This work done becomes the kinetic energy of the body. So, the kinetic energy of a body of mass m moving with velocity v is $W = \frac{1}{2}mv^2$

Method 2, Using Calculus

If dW is the small work done in giving an infinitesimally small displacement ds to the body, then

$$dW = \mathbf{F}.ds$$

Here, F is the applied force. Suppose the force and the displacement have the same direction, then,

$$dW = \mathbf{F}.ds = Fds\cos 0 \qquad 5.15$$

Let m be the mass of the body and a its acceleration. Then $\mathbf{F} = m\mathbf{a}$ and

$$dW = (ma)ds = m\frac{dv}{dt}ds$$

$$dW = m\frac{ds}{dt}dv = mvdv \qquad 5.16$$

Let W be the total work done in increasing the velocity from zero to v. Then,

$$dW = \int_0^v mvdv = m\int_0^v vdv = m\left[\frac{v^2}{2}\right]_0^v$$

$$W = m\left(\frac{v^2}{2} - 0\right) = \frac{1}{2}mv^2 \qquad 5.17$$

This work done becomes the kinetic energy. The kinetic energy of a body is one half the product of the mass of the body and the square of its speed. It is directly proportional to the (a) mass m of the body and (b) square of the velocity v of the moving body

The velocity v may be acquired by the body in any manner. The kinetic energy of a group of particles or bodies is the algebraic sum of the kinetic energies of the individual particles. Consider a system consisting of n particles of masses $m_1, m_2, \ldots\ldots m_n$. Let $v_1, v_2, \ldots\ldots v_n$ be their respective velocities. Then the total kinetic energy E_k of the system is given by

$$E_k = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \ldots\ldots\ldots\ldots + \frac{1}{2}m_nv_n^2 \qquad 5.18$$

If m is measured in kg and v in ms^{-1}, then the kinetic energy is measured in joule. It may be noted that the units of kinetic energy are the same as those of work. In fact, this is true of all forms of energy since they are interconvertible.

Work-Energy Theorem

We know that $dW = mvdv$. If W is the work done in increasing the velocity from $v(0)$ to v, then

$$dW = \int_0^v mvdv = m\int_0^v vdv = m\left[\frac{v^2}{2}\right]_0^v$$

$$W = m\left(\frac{v^2}{2} - \frac{0^2}{2}\right) = \frac{1}{2}mv^2 - \frac{1}{2}mo^2 \qquad 5.19$$

So, the work done on a body by a resultant force is equal to the increase or change in the kinetic energy of the body. This is called the *Work-Energy Theorem* or *Work-Energy Principle*.

Discussion of Work-Energy Theorem

1. If there is no change in the speed of a particle, there is no change in kinetic energy. So, work done by the resultant force is zero.

Example 5.13

When a particle moves with constant speed in a circle, there is no change in the kinetic energy of the particle. So, according to the work-energy principle, the work done by the centripetal force is zero.

2. If the kinetic energy of the body decreases, then the work done on the body is negative. In this case, force and displacement are oppositely directed i.e. the force opposes the motion of the body.

3. In the above discussion, it has been assumed that the work done by the force is effective only in changing the kinetic energy of the body. However, it should be clearly noted that the work done on a body may be stored in the body as potential energy.

Potential Energy

Potential is the energy possessed by a body by virtue of its position in a force field or by its configuration. Potential energy is also known as mutual energy or energy of configuration. The energy of a system can be altered by the application of a force. Basically there are three types of potential energy. They are

1. gravitational potential energy,
2. elastic potential energy,
3. Electrostatic potential energy (Electromagnetic).

Examples of Bodies Possessing Potential Energy Due to Position

1. water stored in a reservoir,
2. lifted weight,
3. object hanging in space.

Examples of Bodies Possessing Potential Energy Due to Configuration

1. The coiled spring of a watch or a door lock,
2. A stretched bow or an elastic material,
3. Compressed or elongated spring.

Gravitational Potential Energy Near the Surface of Earth

As we go higher above the surface of the earth, the value of acceleration due to gravity decreases. But for those heights which are very small as compared to the radius of the earth, the acceleration due to gravity g and hence the weight mg of a body can be regarded as constant

Consider a body of mass m lying on the surface of the earth at a place where the value of acceleration due to gravity is g (Fig. 5.9). The weight of the body acts vertically downwards. In order to lift the body, a force is required to be applied in the upward direction. Let be the height through which the body is lifted. Work done W is given as

$$W = mg \times h$$

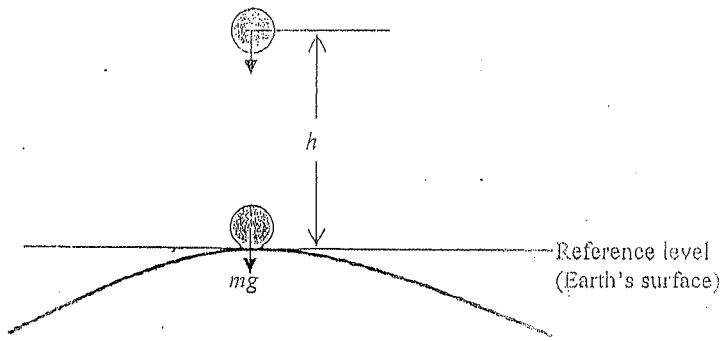


Fig. 5.9: A body of mass m lying on the surface of the Earth has gravitational potential energy stored in it.

This work done is stored in the body as gravitational energy. Therefore gravitational potential energy is. Strictly speaking, is the gravitational potential energy of earth-body system. The height can be taken from any level. The chosen level has to be fixed and considered as a reference level. The potential energy at the reference level is taken to be zero, and can be varied.

The potential energy at a point depends upon the reference level. However the difference of potential energies between two points is independent of reference level.

Conversion of Gravitational Potential Energy to Kinetic Energy

When a body falls from a certain height, its kinetic energy begins to increase. The gain in kinetic energy is at the expense of the gravitational potential energy. But the total mechanical energy remains conserved.

Consider a freely falling body of mass m to be at a position A. Let h be the height of the body above the reference level which is close to the ground.

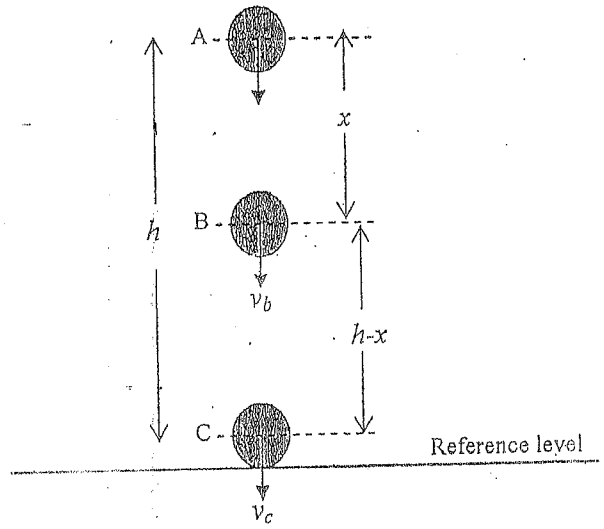


Fig. 5.10: A body falling freely from a height h above a reference level.

At the position A, kinetic energy of the body is zero (the body is stationary) Potential energy of the body is mgh

Total energy of the body is mgh

At position B, let the body be at the position B at any instant after having fallen through a distance x. Potential energy of the body at B is

$$mg(h-x)$$

If v_b is the velocity of the body at B, then from equation 3.18

$$v_b^2 - 0^2 = 2gx \Rightarrow v_b^2 = 2gx$$

Kinetic energy of the body becomes

$$\frac{1}{2}mv_b^2 = \frac{1}{2}m \times 2gx = mgx \qquad 5.20$$

The total energy of the body becomes

$$mg(h-x) + mgx = mgh \qquad 5.21$$

At position C, (just before impact with the ground), the height h of the body above the ground may be regarded as zero. The potential energy of the body at C is zero.

If v_c is the velocity with which the body just touches the ground, then

$$v_c^2 - 0^2 = 2gh \Rightarrow v_b^2 = 2gh$$

The kinetic energy of the body then is

$$\frac{1}{2}mv_c^2 = \frac{1}{2}m \times 2gh = mgh \qquad 5.22$$

Total energy of the body becomes mgh.

From the above deductions it is clear that the sum of the kinetic and potential energies of a freely falling body is constant at all stages of its motion. As the body falls, its potential energy decreases while its kinetic energy increases. However, the total mechanical energy is conserved. (Here, we have assumed zero air resistance and hence no friction loss of energy).

Example 5.14

A ball of mass 1.5kg is thrown vertically upwards from the ground with an initial velocity of 100ms^{-1}. Calculate (a) the potential energy at a point A, 50m above the ground; (b) the kinetic energy at the point A. [assume g = 10ms^{-2}]

Solution

Mass of ball, $m = 1.5kg$, initial velocity, $u = 100ms^{-1}$, height above ground, $h = 50m$.

(a) PE = mgh = 1.5 × 10 × 50 = 750 J

To calculate the KE we have to first find the velocity of the ball at point A.

From kinematic equation of motion,

$v^2 = u^2 + 2gh$, *but* $g = -10ms^{-2}$ (motion against gravity)

$= 100^2 + [2 \times (-10) \times 50] = 9000ms^{-1}$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}x1.5x9000 = 6750J$$

OR

Since p.e. + k.e. = constant

$\frac{1}{2}mu^2$ = p.e.(A) + k.e.(A)

= mgh + k.e.(A)

k.e.(A) = $(1/2mu^2 - mgh) = 6750$ J

after substitution of values.

Example 5.15

A body of mass 1000kg is released from a height of 10m. With what energy does it strike the ground? [Take g=10ms^{-2}]

Solution

$E = mgh = 1000 \times 10 \times 10 = 1 \times 10^5$ J

It strikes the ground with energy of 10^5J.

Transformation of Energy in Vibrating Simple Pendulum

The conversion of energy from one form to another is called transformation of energy. In Figure 5.11 a vibrating simple pendulum is shown. O is the point of suspension. When the centre of gravity of the bob is vertically below the point of suspension, the position of the pendulum is called the rest position or mean position

or equilibrium position. Suppose the bob is displaced to position E_1, and then released from this position, the pendulum will start vibrating between the two extreme positions E_1 and E_2. When the bob moves from the mean position to the extreme position E_1, its centre of gravity is raised vertically through a height h.

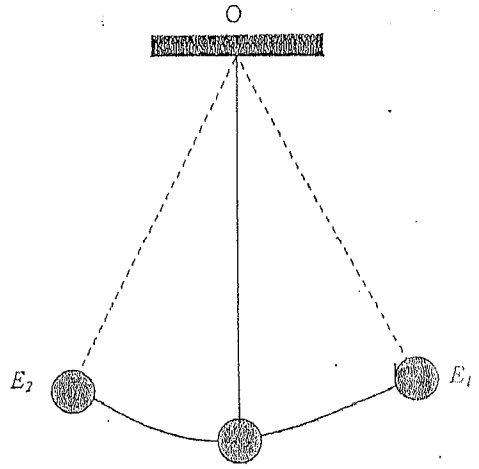


Fig. 5.11: A vibrating simple pendulum

The potential energy of the body at the extreme position is mgh, where m is the mass of the bob and g is the value of the acceleration due to gravity at the place where the pendulum is suspended. As the pendulum moves from the extreme position to the mean position, its potential energy goes on decreasing while the kinetic energy goes on increasing. However, the sum of the two energies is constant. At the mean position, the whole of the potential energy is converted into kinetic energy. If v is the velocity of the bob at the mean position, then the Kinetic energy at the mean position is

$$\frac{1}{2}mv^2$$

Applying the law of conservation of energy,

$$mgh = \frac{1}{2}mv^2 \text{ or } v^2 = 2gh \qquad 5.23$$

Practical Application of Conversion of Gravitational Potential Energy into Kinetic Energy

The conversion of gravitational potential energy into kinetic energy has many practical applications. The most important application is 'hydroelectric power generation'. A large amount of water is stored in dams which are built at high levels. The water stored in the dam possesses a very large amount of gravitational potential energy. When this water is made to run in the pipes, the gravitational potential energy is converted into kinetic energy. Now the fast-moving water is employed to run the turbines of the electric generators. This produces electrical energy.

Power

Power is defined as *the rate of doing work*. We consider the power of an agent doing work if the time taken to complete a given amount of work is important to us. The average power P delivered by any working device is given by

$$P = \frac{workdone}{time} = \frac{W}{t} \qquad 5.24$$

W is the work done in time t.

But $W = \mathbf{F}.\mathbf{S}$ and velocity, $v = S/t$

$$P = \frac{W}{t} = \frac{\mathbf{F}.\mathbf{S}}{t} = \mathbf{F}.\frac{\mathbf{S}}{t} = \mathbf{F}.\mathbf{v} \qquad 5.25$$

From the definition of dot product, we can write

$$P = \mathbf{F}.\mathbf{v} = Fv\cos\theta \qquad 5.26$$

where θ is the angle between $\mathbf{F}$ and $\mathbf{v}$.

If both $\mathbf{F}$ and $\mathbf{v}$ are in the same direction, then $\theta = 0$ and

$$P = \mathbf{F}.\mathbf{v} = Fv\cos 0 = Fv \qquad 5.27$$

Instantaneous Power, P is the power of a working device at any instant. Suppose a machine does an infinitesimally small amount of work dW in an infinitesimally small time dt, then

$$P = \frac{dW}{dt} = \frac{\mathbf{F}.d\mathbf{s}}{dt} = \mathbf{F}.\frac{d\mathbf{s}}{dt} = \mathbf{F}.\mathbf{v} \qquad 5.28$$

Therefore mechanical power of a working machine at any instant is equal to the dot product of the applied force and the velocity at that instant.

The unit of power is joule per second (J/s) or watts (W). The power of a machine is said to be *one watt if it does one joule of work in one second.*

Example 5.16

Find the work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by $\mathbf{F} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. Calculate the power dissipated if it took 4s to move the object to its final position.

Solution

Work done = (Force field). (Displacement vector)

$$= (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}.) . [(2\mathbf{i} - \mathbf{j} + 4\mathbf{k}) - (3\mathbf{i} + 2\mathbf{j} - \mathbf{k})]$$
$$= (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}.) . (-\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$$
$$= -4 + 9 + 10 = 15J$$

$$Power = \frac{workdone}{time\ taken} = \frac{15}{4} = 3.75W$$

Example 5.17

Calculate the power of a pump which lifts 200kg of water through a vertical distance of 2m in 4s. [g=10ms^{-2}]

Solution

$$Power = \frac{W}{t} = \frac{mgh}{t} = \frac{200x10x2}{4} = 1000W$$

Exercise 5

5.1 A particle moving in the x-y plane undergoes a displacement d=(2.0 i [illegible]

as a resultant force F=(5.0+3.0)N acts on the particle. Calculate the work done by F. (Ans: W=19J)

5.2 Calculate the work done by a 2.0N force (directed at a 30° angle to the vertical) to move a 500gram box a horizontal distance of 400cm across a rough floor at a constant speed of 0.5ms^{-1}. (Ans: W=4.0J)

5.3 How much work is done by an applied force to lift a 15N block 3.0m vertically at a constant speed? (Ans: 45J)

5.4 A cart is loaded with a brick and pulled at constant speed along an inclined plane to the height of a seat-top. If the mass of the loaded cart is 3.0kg and the height of the seat-top is 0.45m, then what is the potential energy of the loaded cart at the height of the seat-top? (Ans: 13.2J)

5.5 A 900kg compact car moving at 60mi/hr has approximately 320000J of kinetic energy. Estimate its new energy if it is moving at 30mi/hr. (Ans: 80000J)

5.6 A platform diver had a kinetic energy of 1200J just prior to hitting the surface of the water. If the diver has a mass of 40kg, what is her speed? (Ans: 24.5ms^{-1})

5.7 An escalator is used to move 20 passengers every minute from the first floor of a department store to the second. The second floor is located 5.20m above the first floor. The average passenger's mass is 54.9kg. Determine the power requirement of the escalator in order to move this number of passengers in this amount of time. (Ans: 933watts)

5.8 What is the power output of a cyclist moving at a steady speed of 5.0ms^{-1} along a level road against a resistance of 20N? (Ans: 1.0×10^2W)

5.9 A 15kg ball is thrown at a velocity of 20km/hr to a 60kg person who is at rest on ice. The person catches the ball and subsequently slides with the ball across the ice. Determine the velocity of the person and the ball after the collision. (Ans: V=1.1ms^{-1})

5.10. A 1kg ball moving at 12ms^{-1} collides head-on with a 2kg ball moving in the opposite direction at 24ms^{-1}. Find the velocity of each after impact if: (a) e=2/3 (b) the balls stick together.
(Ans: [a] $V_1 = -28$ms^{-1}; $V_2 = -4$ms^{-1}, [b] V= -12ms^{-1})

Chapter Six

ELASTICITY

When an external force acts on a body which is not free to move, the molecules of the body are shifted to new positions. This change results in the deformation of the body because its size and shape changes. The applied force which caused changes in the body is referred to as a *deforming force*. Some materials become permanently deformed even when the deforming force has been removed. There are certain materials which tend to recover their size and shape after the deforming force has been removed. Such materials are called *elastic materials* because the molecules tend to regain their original positions. When a deforming force is applied to an elastic body, an internal force or a restoring force is set up in the body which acts in opposite direction to the deforming force.

Elasticity is *the property of a material by virtue of which it regains its original length, volume and shape after the deforming forces have been removed.*

Stress

Stress is a measure of the strength of a force that is causing a deformation. Stress is defined as *the deforming force acting per unit area of a body.* Its SI unit is Nm^{-2}.

$$Stress = \frac{Deforming\ Force}{Cross\ sectional\ Area} \qquad 6.1$$

Types of Stress

Tensile Stress

In tensile stress the deforming force stretches the body in one direction. This causes the body to increase in length. This is also known as *longitudinal stress.*

Consider a long wire fixed to a rigid support at one end as shown in Figure 6.1. Let a deforming force *F* be applied normally outwards at the free end.

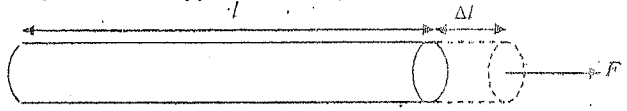


Fig. 6.1: Deforming force *F* stretching a body in one direction.

Let us also assume that the wire extends a length Δl in the direction of the force. Then,

$$Tensile\ stress = \frac{Deforming\ Force}{Cross\ sectional\ Area} \qquad 6.2$$

Compressional Stress

The restoring force developed per unit cross sectional area of a body when the body is compressed is called compressional stress. The deforming force causes the length of the body to decrease.

Tangential or Shearing Stress

When the deforming force acts tangentially over an area of a body, it causes the body to get sheared through an angle θ. Such a deformation results from a tangential stress.

Hydrostatic or Bulk Stress

In this type of stress, the deforming force acts uniformly and equally from all sides of the body's surface. Bulk stress causes the entire body's size to decrease.

Strain

It is the fractional deformation resulting from a stress. This means that any type of stress has a corresponding strain. Strain has no unit.

Tensile or Longitudinal Strain

When the deforming force produces a change in length, then the strain becomes the extension per unit length. Suppose that a wire of length l is stretched to a new length $l + \Delta l$ under stress. Then,

$$Tensile\ strain = \frac{change\ in\ length}{original\ length} = \frac{\Delta l}{l} \qquad 6.3$$

Therefore, within the *elastic limit*, tensile strain can be defined as *the ratio of the change in length to the original length.*

Shear Strain

If the deforming force produces a change in the shape of a body, then the strain is measured by the angle through which a line originally perpendicular to the fixed plane is turned. Within the elastic limit, shear strain is measured by the ratio of the relative displacement of one plane to its distance from the fixed plane.

So, shear strain can be defined as *the relative displacement between two parallel planes which are a unit distance apart.*

Bulk Strain

When the deforming force produces a change in volume, then the strain becomes the bulk strain. Suppose a body of volume V is compressed to a new volume $V - \Delta V$ under stress. Then,

$$Bulk\ strain = \frac{change\ in\ volume}{original\ volume} = \frac{\Delta V}{V} \qquad 6.4$$

Therefore, within the elastic limit, bulk strain can be defined as *the ratio of the change in volume to the original volume.*

Example 6.1

What are stress and strain?

Solution

Stress is a measure of the strength of the agent that is causing a deformation. If a force F is applied, possibly obliquely, to a surface of area A, then

$$stress = \frac{force}{area\ of\ surface\ on\ which\ force\ acts} = \frac{F}{A}$$

The units of stress are those of pressure.

Strain is the fractional deformation resulting from a stress. It is measured by the ratio of the change in some dimension of a body to the original dimension in which the change occurred.

$$strain = \frac{change\ in\ dimension}{original\ dimension}$$

Strain has no units because it is a ratio of like quantities.

Plasticity

Plasticity is *the property of a body remaining deformed even after the deforming force has been removed.* Plastic bodies do not show any tendency to recover their original shape and size when the force acting on them is removed. Forces exerted on plastic materials cause dislocation in their crystal lattices. If a metal is deformed beyond its elastic limit, it exhibits plastic behaviour.

Hooke's Law

The law states that *the extension is proportional to the force or tension in a wire if the elastic limit is not exceeded* OR *provided the elastic limit is not exceeded, stress is directly proportional to strain.*

Let us consider a wire supported at a fixed point as shown in Figure 6.2. When a load is placed on the wire, it extends and the extension e is read as change in length. Figure 6.3 shows a typical Stress — Strain graph if the thin wire is loaded with increasing weights. From the graph, within region OA, the wire obeys Hooke's law. However a further increase in load increases the tension in the wire until a point is reached where the proportionality limit is exceeded i.e., the wire cannot return to its initial state or position when it is off-loaded. This is the *yield point* which is indicated by kinks in the wire. A further increase in the load causes *plastic deformation* where the wire stretches and increases in length permanently and becomes thinner. Finally the wire breaks after reaching the *breaking point* with more loads on the wire. The break is due to the inability of the tension in the wire to support the applied load on the wire. The tensile stress in the wire is the force per unit area. Tensile strain is extension per unit length. The modulus of elasticity is a measure of the stiffness of the material. From the graph, three important

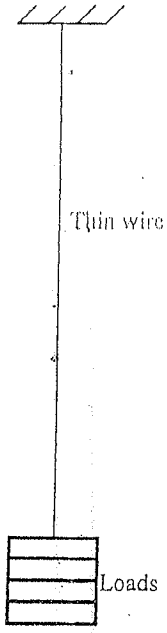


Fig. 6.2: A wire under tensile stress.

mechanical properties can be obtained: (1) yield strength (2) tensile strength (3) modulus of elasticity.

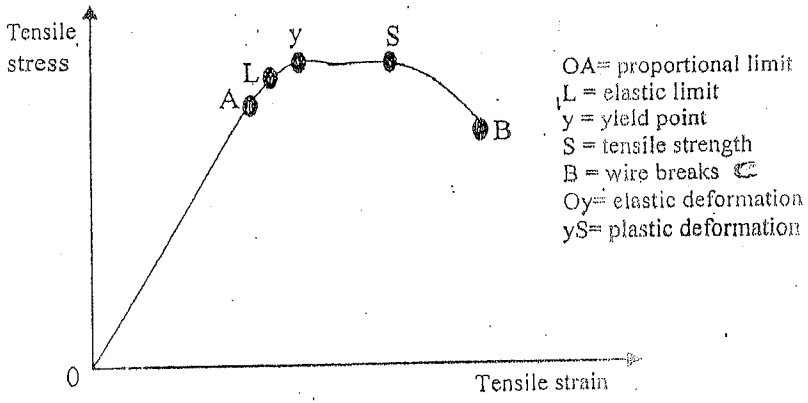


Fig. 6.3: A typical Stress verses Strain graph

Example 6.2

Define elasticity, Hooke's law, elastic limit, and ultimate strength.

Solution

Elasticity is that property by virtue of which a body returns to its original size and shape when the forces that deformed it are removed.

Hooke's law can be stated in terms of stress and strain. If the system obeys Hooke's law, then stress $\propto$ strain. We then define a constant the modulus of elasticity, by the relation

$$\text{modulus of elasticity} = \frac{\text{stress}}{\text{strain}}$$

The modulus has the same units as stress. In the case of pure tension, the modulus is called Young's modulus.

The elastic limit is the smallest value of the stress required to produce a permanent distortion in a body. When a stress in excess of this limit is applied, the body will not return exactly to its original state after the stress is removed.

The ultimate strength is the smallest value of stress required to break the body.

Young's Modulus of elasticity

From Hooke's law, within the elastic limit stress is proportional to strain.
That is, Force $\mathbb{F} \propto$ extension e.

$$\mathbb{F} = ke \qquad 6.5$$

Dividing both sides of Equation 6.5 by the area A

$$\frac{\mathbb{F}}{A} = \frac{k}{A}e$$

but the area $A = l \times l$

$$\frac{F}{A} = \frac{k}{l}\frac{e}{l}$$

$$stress = \frac{F}{A}, \quad \frac{k}{l} = constant, K, \quad strain = \frac{e}{l}$$

Therefore, $stress = K \times strain$

$$\text{Constant } K = \frac{Stress}{Strain} \qquad 6.6$$

K is a constant known as *Modulus of Elasticity*.

If the change is an increase in length, then K becomes *Young Modulus of Elasticity* denoted by Y.

Therefore, Young's Modulus of Elasticity is defined as

$$Y = \frac{Tensile\ Stress}{Tensile\ Strain} \qquad 6.7$$

Mathematically,

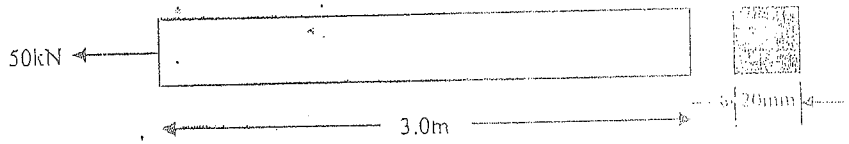
$$Y = \frac{F}{A} \div \frac{\Delta l}{l} = \frac{F}{A} \cdot \frac{l}{\Delta l} \qquad 6.8$$

Note that Young's Modulus applies to only solids.

Example 6.3

A prismatic bar with square cross section (20mm × 20mm) and length [illegible] subjected to an axial tensile force of 50kN. The measured elongation of the bar is DL= 1.5mm. Calculate the tensile stress and strain in the bar. Hence find the Young's modulus of the prismatic bar.

Solution



$$Tensile = \frac{Force}{cross\ sectional\ Area} = \frac{50kN}{(20 \times 20)mm^2} = 125MPa$$

$$Tensile\ strain = \frac{elongation}{original\ length} = \frac{1.5x10^{-3}m}{3.0m} = 5.0x10^{-4}$$

$$Young's\ modulus = \frac{Tensile\ stress}{Tensile\ strain} = \frac{125MPa}{5.0x10^{-4}} = 25MPa$$

Example 6.4

What is Young's modulus?

Solution

Young's modulus (or tensile modulus) describes the length elast[illegible]
Suppose a wire or rod of original length L and cross sectional a[illegible]
amount ΔL under a stretching force F applied to its end. Then

$$\text{tensile stress} = \frac{F}{A} \quad \text{tensile strain} = \frac{\Delta L}{L}$$

$$\text{and Young's modulus} = Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$$

The value of Y depends only on the material of the wire or rod not on its dimensions.

Example 6.5

An iron rod 4 m long and 0.5 cm² in cross section stretches 1 mm when a mass of 225 kg is hung from its lower end. Compute Young's modulus for the iron.

Solution

$Y = \dfrac{\text{stress}}{\text{strain}} = \dfrac{F/A}{\Delta L/L}$ where $\dfrac{\Delta L}{L} = (1\times10^{-3}\text{ m})/4\text{ m} = 0.25\times10^{-3}$, $A = 0.5\times10^{-4}\text{ m}^2$, 10^{-4}m^2, and $F = (225\text{kg})(9.8\text{m/s}^2) = 2205$ N. Substituting, we get $Y = 1.75 \times 1011$ N/m² = 176 MPa

Example 6.6

If we consider the rod of example **Example 6.5** to be a spring, within its elastic limit, what is the effective force constant of this spring?

Solution

$F = YA\Delta L/L = k\Delta L$, where k is, by definition, the force constant. Thus

$$k = \frac{YA}{L} = \frac{(1.76\times10^{11}\text{ N/m}^2)(0.5\times10^{-4}\text{ m}^2)}{4\text{ m}} = 2.2\text{ MN/m}$$

Shear Modulus of Elasticity or Modulus of Rigidity

If the deforming force produces a change in the shape of the body, then from Equation 6.6, the constant K becomes *Shear Modulus of Elasticity,* which is denoted by G. Therefore Equation 6.6 becomes,

$$G = \frac{\textit{Shear Stress}}{\textit{Shear Strain}} \qquad 6.9$$

Let us consider a rectangular block with the front vertical face labeled QRST as shown in Figure 6.4a and b. If a tangential force **F** is applied on the upper parts, say ST, of the block, a reaction force will also be set up at the bottom face, QR. These

two tangential forces will cause the shape of the block to change but the volume will be unaltered.

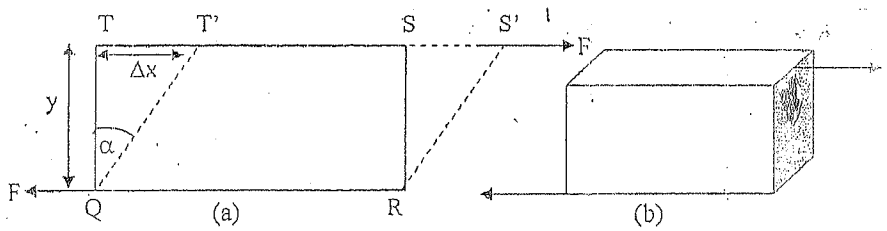


Fig. 6.4: Shear deformation of a rectangular block by a tangential force, seen in 2 dimensions in (a) and 3 dimensions in (b).

As you go up from the bottom each layer slides past the neighbour by the same margin making the block to suffer a tangential shear. Therefore, the shear stress is given by the tangential force per unit area as F/A.

Also, the corresponding shear strain is given by the tangent of the angle of shear as $\tan\alpha = \Delta x/y$ where α is the angular displacement from the initial position.

$$G = \frac{\textit{Shear Stress}}{\textit{Shear Strain}} = \frac{F}{A} \div \frac{\Delta x}{y} = \frac{F}{A}\cdot\frac{y}{\Delta x} \qquad 6.10$$

So, the Shear Modulus or Modulus of rigidity is defined *as the ratio of the shear stress to the shear strain.*

Note that Shear Modulus applies to solids and liquids.

Bulk Modulus

Bulk Modulus applies to a change in volume of a body if a deforming force acts uniformly over the surface of the body. Suppose we apply a force normally and uniformly over an area of a sphere of volume V such that the volume decreases by ΔV. In this case there will be no change in shape of the sphere. Such a uniform pressure acting normally to the entire surface of the body is termed *Hydrostatic pressure*.

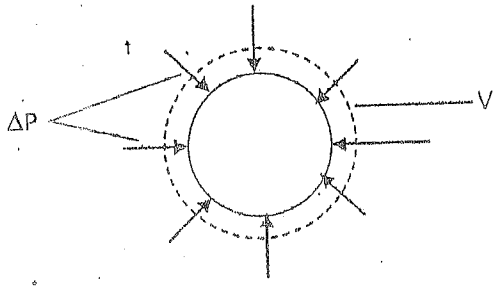


Fig. 6.5: A uniform pressure acting normally to the entire surface of the body.

Considering the volume change, the constant K in Equation 6.6 becomes the Bulk Modulus of elasticity. Therefore, Bulk Modulus of elasticity is,

$$K = \frac{Bulk\ Stress}{Bulk\ Strain} \qquad 6.11$$

The pressure applied to compress the body is called the Bulk Stress which is given by $\Delta P = F/A$.

The corresponding fractional change in volume is called the Bulk strain and it is given by $\left(-\frac{\Delta V}{V}\right)$. The negative sign denotes the decrease in volume.

$$K = \frac{Bulk\ Stress}{Bulk\ Strain} = \frac{P}{A} \div \frac{\Delta V}{V} = \frac{F}{A} \cdot \frac{V}{\Delta V} = \frac{\Delta P}{\Delta V/V} \qquad 6.12$$

So, the Bulk Modulus is defined as *the ratio of normal stress to the volumetric strain.*

Note that the Bulk Modulus applies to all phases of matter.

Example 6.7

Determine the fractional change in volume as the pressure of the atmosphere (0.1 MPa) around a metal block is reduced to zero by placing the block in vacuum. The bulk modulus for the metal is 125000 MPa.

Solution

$$B = \frac{-\Delta p}{\Delta V/V} \text{ or } \frac{\Delta V}{V} = -\frac{\Delta p}{B} = \frac{-(-0.1)}{125000} = 8 \times 10^{-7}$$

Example 6.8

How large a pressure must be applied to water if it is to be compressed by 0.1 per cent? What is the ratio of this pressure to atmospheric pressure, 101 kPa? The bulk modulus of water is 2100 MPa.

Solution

The volume strain $\frac{\Delta V}{V} = 1.0 \times 10^{-3}$; set $B = p/(\Delta V/V_0)$ to find

$p = (2.1 \times 10^9)(1.0 \times 10^{-3}) = 2100$ kPa.

Dividing by the atmospheric pressure, the ratio is 21.

Exercise 6

6.1 A metal wire 75cm long and 0.13cm in diameter stretches 0.035cm when a load of 8.0kg is hung on its end. Find the stress, the strain, and the Young's modulus (Y) for the material of the wire. (Ans: stress=5.91 × 10⁷Pa; strain=4.6710⁻⁴; y=127GPa)

6.2 A solid cylindrical steel column is 4.0m long and 9.0cm in diameter. What will be its decrease in length when carrying a load of 80,000kg? (Y=1.910¹¹Pa) (Ans: ΔL=2.6mm)

6.3 A 60kg woman stands on a light, cubical box that is 5cm on each edge. The box sits on the floor. What pressure does the box exert on the floor? (Ans: P=240kPa)

6.4 An iron rod 4m long and 0.5cm² in cross section stretches 1mm when a mass of 225kg is hung from its lower end. Compute Young's modulus for the iron. (Ans: 176GPa)

6.5 Find the density and specific gravity of ethyl-alcohol if 63.3g occupies 80.0mL. (Ans: 791kgm^{-3}; 0.791)

6.6 A load of 50kg is applied to the lower end of a steel rod 80cm long and 0.60cm in diameter. How much will the rod stretch? Y=190GPa for steel. (Ans: 73im)

6.7 Compute the volume change of a solid copper cube, 40mm on each edge, when subjected to a pressure of 20MPa. The bulk modulus for copper is 125GPa. (Ans: $-10mm^3$)

6.8 The compressibility of water is $5 \times 10^{-10}m^2/N$. Find the decrease in volume of 100mL of water when subjected to a pressure of 15MPa. (Ans: 7.5mL)

6.9 A body of mass 2kg is attached to the end of a vertical wire of length 2m and diameter 0.64mm and the extension is 0.60mm. Calculate (a) tensile stress (b) tensile strain (c) Young's Modulus (E) [Take $g = 9.8ms^{-2}$]

6.10 A vertical wire is suspended from one end. A weight of 20N is attached to the other end to stretch the wire. If the weight extends the wire by 1mm, calculate the energy gained by the wire.

6.11 If a force of 100N stretches a string by 0.1cm, find (a) the elastic constant (b) the work done in stretching the string 0.3cm if the elastic limit is not exceeded.

6.12 A uniform steel wire has cross sectional area $3mm^2$ and length 10m. A mass of 10kg is attached at one end. If the Young's Modulus (E) of the steel is 10^{-2} Nm^{-2}, calculate the extension produced. [Take $g = 9.8ms^{-2}$]

Chapter Seven

FLUIDS

States of Matter

Macroscopic matter is usually classified into three phases: *solid, liquid,* and *gas*. Solids tend to hold their shapes. Unlike solids, liquids and gases do not hold their shapes. A liquid flows and takes the shape of its container and a gas expands to fill its container. We can think of liquids as incompressible, that is, as having a fixed volume that is impossible to change. The shape of the liquid can be changed by pouring it from a container of one shape into a container of a different shape, but the volume of the liquid is the same.

Liquids and gases are both called fluids. Fluid atoms or molecules can move around fairly freely and that contributes to the range of shared properties. They are easily deformed by external forces. Before considering their behaviour at rest (or whether moving) certain basic properties need to be defined. This chapter deals mainly with these properties that are common to both liquids and gases. The atoms or molecules in a fluid do not have fixed positions, so a fluid does not have a definite shape. Thus there are still difficulties in defining what a fluid is or is not. Even diamond can be made to flow like soft wax at a high enough pressure ($>1.7\times10^{11}$Pa). It is easier to define a fluid as an aggregate of atoms that do not significantly resist shearing.

Liquids

In most liquids, the atoms or molecules are almost as closely packed as those in the solid phase of the same material. The intermolecular forces in a liquid are almost as strong as those in solids, but the molecules are not locked in fixed positions as they are in solids. This contradicts Brownian motion and it applies to solids. That is why the volume of the liquid can remain nearly constant while the shape is easily changed. Water is one of the exceptions: in cold water, the molecules in the liquid phase are actually *more* closely packed than those in the solid phase (ice).

Gases

Gases, on the other hand, cannot be characterized by a definite volume or by a definite shape. A gas expands to fill its container and can easily be compressed.

Hence gases are much more compressible than liquids. The molecules in a gas are very far apart compared to the molecules in liquids and solids. The molecules are almost free of interactions with each other except when they collide. As with many branches in physics there is the study of

1. Statics of fluids (known as Hydrostatics)
2. Dynamics of fluids (known as Hydrodynamics).

Hydrostatics

A static fluid does not flow. In the study of fluid statics (hydro-statics), we also assume that any solid object in contact with the fluid whether a vessel containing the fluid or an object submerged in the fluid is at rest. The atoms or mole-cules in a static fluid are not themselves static but are continuously moving.

Density

A fluid exerts a pressure on all bodies immersed in it. For a fluid at rest the difference in pressure between two points in it depends only upon the density of the fluid and the difference in depth between the two points. We define density ρ as

$$density = \frac{mass}{volume},$$

$$\rho = \frac{m}{V},$$

$$m = \rho V \qquad 7.1$$

The SI unit of density is kilograms per cubic metre (kg/m³). Density is the measure of compactness of a substance; the greater the density, the more matter or mass per unit volume.

Relative Density

Relative density, ρ_r (or Specific Gravity σ_s) of a material or substance is the ratio of its density to that of water and is, therefore, a pure number.

Relative density = density of material/density of water

$$\rho_r = \sigma_s = \frac{density\ of\ material}{density\ of\ water} = \frac{\rho}{\rho_{H_2O}} \qquad 7.2$$

Example 7.1

A solid sphere has a radius of 1.5cm and a mass of 0.038kg. What is the specific gravity of the sphere?

Solution

We are given: radius, $r = 1.5cm = 1.5\times10^{-2}m$. Thus we calculate the volume of the sphere as follows:

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(1.5\times10^{-2}m)^3 = 1.413\times10^{-5}m^3$$

Density is calculated as follows: $\rho = \frac{m}{V} = \frac{0.038kg}{1.413\times10^{-5}m^3} = 2690kgm^{-3}$

Specific gravity is calculated as

$$\therefore \sigma_s = \frac{density\ of\ material}{density\ of\ water} = \frac{\rho}{\rho_{H_2O}} = \frac{2690kgm^{-3}}{1000kgm^{-3}} = 2.69$$

Pressure, Force and Area

When the concept of hydrostatic fluid was introduced above, the weight of the fluid was neglected and the pressure was assumed to be the same everywhere. We can generalize the definition of pressure and define pressure at any point as the ratio of the uniform perpendicular force exerted on a small area dA as

$$p = \frac{\Delta F_\perp}{\Delta A} = \frac{dF_\perp}{dA},$$

$$\Rightarrow dF_\perp = pdA \qquad 7.3a$$

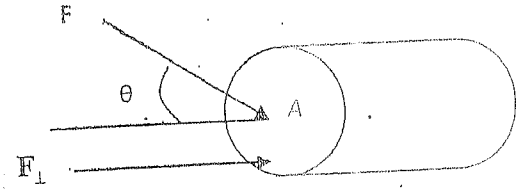


Fig. 7.1: Forces acting on the surface cylinder containing fluid.

If the force acts at an angle θ to the surface and the pressure is the same at all points of a finite plane of surface area A (see Figure 7.1), then Equation 7.3a reduces to the following

$$p = \frac{F_\perp}{A} = \frac{F\cos\theta}{A}, \qquad 7.3b$$

$$\Rightarrow F\cos\theta = pA \qquad 7.4$$

Example 7.2

Consider a gas confined to a container by means of a piston of area 40cm². If a perpendicular force of 20N is exerted on the piston to keep the gas from expanding, find the gas pressure.

Solution

The piston is in equilibrium, so the gas exerts a counterbalancing force of 20N (from Newton's third law of motion). Thus

$$p = \frac{F}{A} = \frac{20N}{(40\times10^{-4}m^2)} = 5kPa$$

Example 7.3

A 80kg metal cylinder 2m long and with each end of area 25cm² stands vertically on one end. What pressure does the cylinder exert on the floor?

Solution

We are given: $m = 80kg$, $g = 9.8kgms^{-2}$, $A = 25cm^2 = 25\times10^{-4}m^2$

$$p = \frac{normal\ force}{area} = \frac{F_\perp}{A} = \frac{mg}{A} = \frac{(80kg)(9.8kgms^{-2})}{25\times10^{-4}m^2}$$

$$= 3.14\times10^5 Nm^{-2} = 314Pa$$

Pressure in Liquids

It is well observed, however, that atmospheric pressure is greater at sea level than at mountain altitudes and that the higher up one goes in the air, the less the pressure experienced. We encounter a different experience with liquids in that the pressure in a lake or the ocean increases with depth i.e. the higher the depth the higher the pressure. Since liquids are nearly incompressible, their density does not change significantly with increasing depth. Therefore, the increase in pressure is caused solely by the increase in depth if we neglect temperature gradient.

Let us try and find how pressure in a fluid varies with depth by considering a rectangular column of water as shown in Figure 7.2

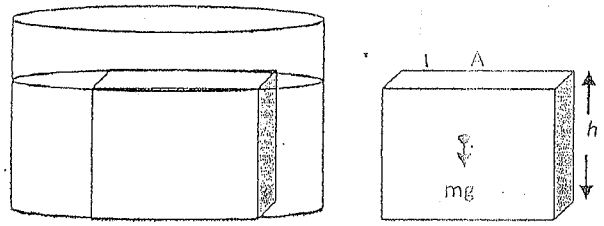


Fig. 7.2: Pressure in liquids

The force acting on the bottom of the container below the column is equal to the weight of the fluid making the column: $F = w = mg$, since the density $\rho = \frac{m}{V}$, the mass of the column is equal to the density times the volume; that is $m = \rho V$ (here the fluid is considered to be incompressible, so ñ is constant).

Volume of column = height of column × base area of column

$\Rightarrow V = hA$.

We can therefore write

$$F = w = mg = \rho Vg = \rho ghA \qquad 7.5$$

From Equation 7.4 with $p = F/A$, the pressure at the depth h due to the weight of the water column is

$$p = \rho gh \qquad 7.6$$

where, using SI units, p is the hydrostatic pressure (in Pascal), ñ is the water density (in kilograms per cubic metre), g is gravitational acceleration (in metres per second squared), h is the height of fluid column (in metres). Equation 7.6 holds for incompressible fluids. The pressure is the same everywhere on a horizontal plane at a depth h (with ñ and g constant).

The derivation of Equation 7.5 did not take into account the constant pressure being applied at the open end of the vessel. This factor adds to the pressure at the depth h to give a total pressure of

$$p = p_0 + \rho gh = p_a + \rho gh \qquad 7.7$$

where p_0 or p_a is most often the atmospheric pressure for open end containers.

$$p_0 = p_a = 1atm = 1.01325\times10^5\,Pa = 1.01325\times10^5\,N/m^2 \qquad 7.8$$

Pressure and Head

We can further find the general relation between pressure p at any point in a fluid of density ρ in a gravitational field and at an elevation of the point y (this point can either be in terms of altitude or depth). Consider the volume element of a thin slab of fluid as shown in Figure 7.3. The thickness of the slab is dz and the area of the surfaces on which the perpendicular forces act is A. If ρ is the density of the fluid, the mass of the volume element dw is $\rho A dz$. The force exerted on the element by the surrounding fluid is everywhere perpendicular to its surface. By symmetry, the resultant of the horizontal forces on its vertical sides is zero. The upward force on its lower face is pA, and the downward force on its upper surface is $(p+dp)A$.

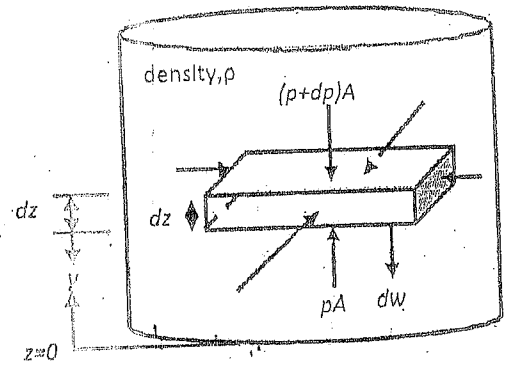


Fig. 7.3: Open vessel containing liquid of density ρ.

Since the volume element is in equilibrium, $\sum F_y = 0$ and

$$pA - (p+dp)A - \rho gA dz = 0 \qquad 7.9$$

Therefore

$$\frac{dp}{dz} = -\rho g \qquad 7.10$$

Since ρ and g are both positive quantities, it follows that if the change dz is positive (elevation is increased) it is accompanied by a negative change dp (decrease in pressure). Equation 7.10 can also be applied to gas, e.g. the atmosphere.

In a fluid of constant density, $dp/dz = -\rho g$ can be integrated immediately to give

$$p = -\rho gz + const. \qquad 7.11$$

which is equivalent to Equation 7.7. In a liquid, the pressure p at any depth z, can be measured downwards from the free surface so that $z = -h$ will be

$$p = \rho gh + const. \qquad 7.12$$

and, since the pressure at the free or open surface will normally be atmospheric pressure p_a,

$$p = \rho gh + p_a \qquad 7.13$$

It is often convenient to take atmospheric pressure as a datum. Pressures measured above atmospheric pressure are known as gauge pressures. Since atmospheric pressure varies with atmospheric conditions, a perfect vacuum is taken as the absolute standard of pressure. Pressures measured above perfect vacuum are called absolute pressures:

Absolute Pressure = Gauge pressure + Atmospheric pressure

Taking p_a as zero, equation 7.13 becomes

$$p = \rho gh \qquad 7.14$$

which indicates that, if g is assumed constant, the gauge pressure at point X in

Figure 7.4 can be defined by stating the vertical height h, called the head, of a column of a given fluid of mass density ñ which would be necessary to produce this pressure. Note that when pressures are expressed as head, it is essential that the mass density ρ is given or the fluid named. For example, from Equation 7.14, $h = p/\rho g$, a pressure of 100kN/m² can be expressed in terms of water ($\rho_{H_2O} = 10^3 kgm^{-3}$) as a head of $(100\times10^3)/(10^3\times9.81) = 10.19m$ of water. Alternatively in terms of mercury (relative density 13.6) a pressure of 100kN/m² will correspond to a head of $(100\times10^3)/(13.6\times9.81) = 0.75m$ of Mercury.

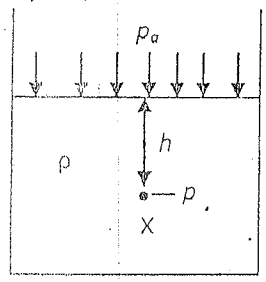


Fig. 7.4: Pressure and Head

Example 7.4

A cylinder contains a fluid at a gauge pressure of 350kN/m². Express this pressure in terms of head of (a) water (b) mercury. What would be the absolute pressure in the cylinder if the atmospheric pressure is 101.3kN/m²?

Solution

From Equation 7.14, head, $h = p/\rho g$

(a) Putting $p = 350\times10^3 Nm^{-2}$, $\rho = \rho_{H_2O} = 10^3 kgm^{-3}$,

$$\text{Equivalent head of water} = \frac{350\times10^3}{10^3\times9.81} = 35.68m$$

(b) For mercury, mercury $\rho_{Hg} = \sigma\rho_{H_2O} = 13.6\times10^3 kgm^{-3}$, where σ is the relative density of mercury.

$$\text{Equivalent head of water} = \frac{350\times10^3}{13.6\times10^3\times9.81} = 2.62m$$

Absolute Pressure = gauge Pressure + Atmospheric Pressure = 350+101.3 = 451.3 kN/m².

The *1atm* unit is a unit called a standard atmosphere and this exerts a pressure equivalent to a head of 76 cm (or 760mm) high of mercury at sea level when the temperature is 0°C. So we can state any pressure in atmospheric units.

Example 7.5

How high will water rise in the pipes of a building if the water pressure gauge shows the pressure at the ground floor to be 270kPa?

Solution

Water pressure gauges read the excess pressure due to the water, that is, the difference between the pressure in the water and the atmospheric pressure. The water pressure at the bottom of the highest column that can be supported is 270kPa. Therefore $p = h\rho g$ gives

$$h = \frac{p}{\rho g} = \frac{2.7\times10^5 Nm^{-2}}{(1000kgm^{-3})(9.8ms^{-2})} = 27.6m$$

Hydrostatic Paradox

From equation 7.14 it can be seen that the pressure exerted by a fluid is dependent only on the vertical head of the fluid and its mass density ρ; it is not affected by the weight of the fluid present. Thus, in Figure 7.5 the four vessels all have the same base area and are filled to the same height h with the same liquid of density ρ.

Pressure on the bottom of each vessel is is $p = \rho gh$. Thus force on bottom in each vessel is

$$\text{Force on bottom} = \text{Pressure} \times \text{Area} = pA = \rho ghA$$

Thus, although the weight of the fluid is obviously different in the four cases (vessels), the force on the bases of the vessels is the same, depending on the depth h and the base area A.

$$p - p_{atm} = \rho g h \qquad 7.17$$

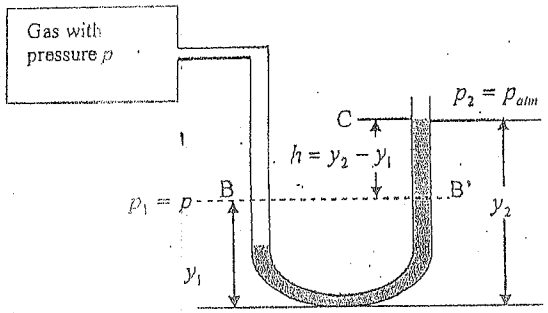


Fig. 7.8: A U-tube manometre

The pressure p is called absolute pressure, whilst the difference is called the gauge pressure (this is what most gauges measure). Thus the gauge pressure is directly proportional to the difference in height of the liquid column. This difference in mercury levels is commonly reported in millimetres of mercury (mmHg).

Pascal's Law

Due to their ability to resist deformation, fluids exert pressure normal to any contacting surface. In addition, when the fluid is at rest (static) that pressure is isotropic, i.e. it acts with equal magnitude in all directions. This characteristic allows fluids to transmit force through the length of pipes or a tube, i.e., a force applied to a fluid in a pipe is transmitted, via the fluid, to the other end of the pipe. If the forces are unequal, the fluid will move in the direction of the resultant force. This concept was first formulated, in a slightly extended form, by the French mathematician Blaise Pascal and would later be known as Pascal's law. This law has many important applications in hydraulics. It states that:

> *The pressure at a point in a fluid in equilibrium is distributed uniformly in all directions.*

The pressure at all points on the same level in a static continuous liquid is the same. Pressure is transmitted uniformly in a fluid in equilibrium if the effect of gravity is neglected. Pascal's principle can also be experienced also on the basis of the operation of a hydraulic lift.

In the hydraulic lever operation, a force F_1 is directed downward on the left-hand piston, whose surface area is A_1. An incompressible liquid in the device produces an upward force of magnitude F_2 on the right hand piston, whose surface area is A_2. To keep the system in equilibrium, there must be a downward force of magnitude F_0 on the output piston from an external load. The force F_1 applied on the left and the downward force F_0 from the load on the right produces a change Δp in the pressure of the fluid and that is given by

$$\Delta p = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\Rightarrow F_2 = F_1 \frac{A_2}{A_1} \qquad 7.18$$

Equation 7.18 shows that the output Force F_2 on the load must be greater than the input force F_1 if $A_2 > A_1$ as in the case of Figure 7.9. If the input piston is moved downwards a distance d_1, the output piston moves upwards a distance d_2, such that the same volume V of the incompressible liquid is displaced at both pistons.

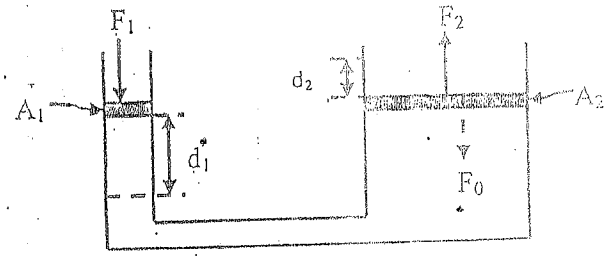


Fig. 7.9: A hydraulic lift demonstrating Pascal's Law

Then

$$V = A_1 d_1 = A_2 d_2 \qquad 7.19$$

which we can write as

$$d_2 = d_1 \frac{A_1}{A_2} \qquad 7.20$$

This shows that, if $A_2 > A_1$ (as in Figure 7.9), the output piston moves a smaller

distance than the input piston ($\Rightarrow d_1 > d_2$). From Equations 7.18 and 7.20, we write the output work as

$$W = F_2 d_2 = (F_1 \frac{A_2}{A_1})(d_1 \frac{A_1}{A_2}) = F_1 d_1 \qquad 7.21$$

Equation 7.21 shows that the work done W on the input piston by the applied force is equal to the work done W by the output piston in lifting the load placed on it.

Example 7.9

A hydraulic lift in a service station has a large piston 30cm in diameter and a small piston 2cm in diameter. (a) What force is required on the small piston to lift a load of 1500kg? (b) What is the pressure increase due to the force in the confined liquid?

Solution

(a) Pascal's principle states that the pressure change is uniformly transmitted throughout the oil, so $\Delta p = \frac{F_1}{A_1} = \frac{F_2}{A_2}$, where F_1 and F_2 are the forces on the small and on the large pistons respectively and A_1 and A_2 are the areas of the small and large pistons also respectively. Thus

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_1}{\pi(2^2/4)} = \frac{1500 \times 9.8}{\pi(30^2/4)}$$

Cancelling out $\pi/4$ and making F_1 the subject, we have

$$F_1 = \frac{(1500)(9.8)(2^2)}{30^2} = 65N$$

(b) $$\Delta p = \frac{F_1}{A_1} = \frac{65}{\pi(2^2/4)} = 21Ncm^{-2} = 210kPa$$

Buoyancy and Archimedes' Principle

If a body is immersed in a fluid it seems to have less weight than when immersed in air. This phenomenon is buoyancy. When a body's average density is less than that

of the fluid it floats or partially submerges at the free upper surface of the fluid. If the body is submerged and it is at rest with the entire fluid then all the forces on it cancel out or their sum is zero.

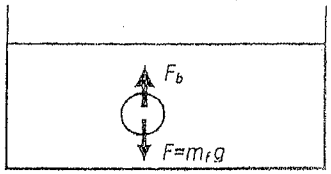


Fig. 7.10: The buoyant force F_b acts upwards and the weight $m_f g$ downwards.

The forces acting on such a body (shown in Figure 7.10) are the net upward buoyant force F_b which equals the weight $m_f g$ of the displaced liquid, where m_f is the mass of fluid displaced by the submerged body.

$$F_b = m_f g \qquad 7.22$$

In other words,

> *When a body is fully or partially submerged in a fluid, a buoyant force, F_b from the surrounding fluid acts on the body. The force is directed upward and has a magnitude equal to the weight mg of the fluid that has been displaced by the body'*

The above statement is known as Archimedes principle.

Example 7.10

A uniform cube of material 10cm on each side has a mass of 700g. (a) Will the cube float in water? (b) If so, how much of its volume will be submerged? ($\rho_w = 1.0 \times 10^3 kgm^{-3} = 1.0gcm^{-3}$)

Solution

Given: m=700g=0.700kg, L=10cm, $\rho_w = 1.0 \times 10^3 kgm^{-3} = 1.0gcm^{-3}$

[Hint for solution (a)]: The question is whether the density of the cube material is greater or less than that of water, so we have to calculate first the cube's density.

Thus we find whether the cube will float in water. Using Equation 7.1, the density of the cube material, ρ_c is given by

$$\rho_c = \frac{m}{V_c} = \frac{m}{L^3} = \frac{700g}{(10cm)^3} = 0.70gcm^{-3} < \rho_w = 1.00gcm^{-3}$$

Since ρ_c is less than ρ_w, the cube will float.

[Hint for solution (b)]: If the cube floats, then the buoyant force and the cube's weight are equal. Both of these forces are related to volume and so we can write them in terms of volume and equate them. From Equation 7.5, the weight of the cube is $w_c = \rho_c g V_c$. When the cube is floating (in equilibrium), its weight is balanced by the buoyant force. That is, $F_b = \rho_w g V_w$, where V_w is the volume of water the submerged part of the cube displaces. Equating the expressions for weight and buoyant force gives

$$\rho_w g V_w = \rho_c g V_c \text{ or } \frac{V_w}{V_c} = \frac{\rho_c}{\rho_w} = \frac{0.70gcm^{-3}}{1.00gcm^{-3}} = 0.70$$

Thus, $V_w = 0.70V_c$, and 70% of the cube is submerged.

Hydrodynamics

Fluid dynamics (or hydrodynamics) is the study of moving fluids. One difference between moving fluids and static fluids is that a moving fluid does exert a force parallel to any surface over or past which it flows. Since the fluid moving exerts a force on the surface over which it is moving, by Newton's 3rd law of motion the surface will also exert an equal but opposite force to the moving fluid. The opposing force is called the viscous force which opposes the flow of the fluid.

Viscosity

All real fluids in nature have an internal resistance to flow, therefore, an external force must act on a viscous fluid to keep it flowing. The force must be exerted to cause one layer of fluid to slide past another, or cause one surface to slide past another if there is a layer of fluid between the surfaces. Both gases and liquids exhibit viscosity although liquids are more viscous than gases. The simplest flow of a viscous fluid is the flow between two parallel plates as shown in Figure 7.11.

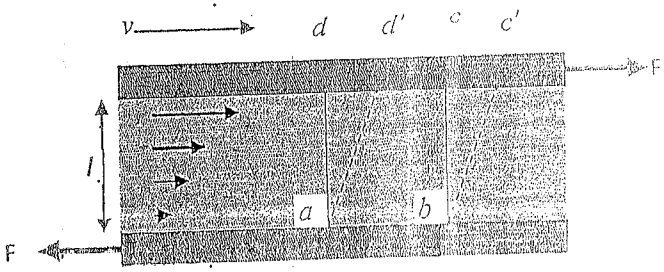


Fig. 7.11: Laminar flow of a viscous fluid.

The bottom plate is stationary whilst the top plate moves with constant speed v. It would be found that the fluid in contact with each has the same speed as that surface; thus at the top, the fluid has speed of v whilst the fluid at the bottom is at rest. The speeds of the middle layers of fluid increase uniformly from one surface to the other. The flow of this type is called laminar flow. The layers of fluid slide over each other and as a consequence of this motion, a parcel of the fluid that has shape *abcd* will, moments later, have the shape $abc'd'$ and thus become more distorted as the motion continues. This implies that the fluid is in a continuous state of increasing shear strain. In order for the motion to be maintained, a constant force must be exerted to the right of the moving upper surface plate and indirectly to the upper surface fluid. This force tends to drag the layers of the fluid and the lower surface plate to the right. Therefore, an equal but opposite force must be exerted on the lower surface plate to make it stationary.

The force is labeled F in Figure 7.11 and if A is the surface area of the plates then the ratio F/A is the shear stress exerted on the fluid. When shear stress is applied to a solid a displacement occurs such as dd' in Figure 7.11. The shear strain is defined as the ratio of this displacement to the transverse dimension, l and within elastic limits the shear stress is proportional to the shear strain. With the fluid on the other hand, the shear strain increases without limit so long as the stress is applied and the stress is found by experiment not to depend on the shear strain but on its rate of change. The strain in Figure 7.11 (at the moment the parcel of fluid has the shape $abc'd'$) is dd'/ad or dd'/l. Since l is constant, the rate of change of strain equals $1/l$ times the rate of change of of dd'. But the rate of change of dd' is simply the rate of change of point, i.e. the speed v of the moving plate. Thus rate of change of shear strain $= \frac{v}{l}$.

The rate of change of shear strain is sometimes referred to simply as the strain rate. The coefficient of viscosity of the fluid (or in other words viscosity η) is defined

as the ratio of the shear stress F/A to the rate of change of shear strain.

$$\eta = \frac{shear\ stress}{rate\ of\ change\ of\ shear\ strain} = \frac{F/A}{v/l} \qquad 7.23$$

or

$$F = \eta A \frac{v}{l} \qquad 7.24$$

The constant of proportionality η in equation 7.24 is known as the dynamic viscosity of the fluid. There is another form of viscosity known as the kinematic viscosity η_k which is given as

$$\eta_k = \frac{\mu}{\rho} \qquad 7.25$$

and it is the ratio of the dynamic viscosity to the mass density of the fluid.

Viscosities of gases at ordinary pressures and temperatures are much smaller than those of common liquids. Viscosities of all fluids are markedly dependent on temperature, increasing for gases and decreasing for liquids as the temperature is increased. An important consideration in the design of lubricating oils is to reduce as much as possible the temperature variation of viscosity.

Viscous Drag

When an object moves through a fluid, the fluid exerts a drag force on it. When the rela-tive velocity between the object and the fluid is low enough for the flow around the object to be laminar, the drag force derives from viscosity and is called viscous drag. The vis-cous drag force is proportional to the speed of the object. For larger relative speeds, the flow becomes turbulent and the drag force is proportional to the square of the object's speed.

The viscous drag force depends also on the shape and size of the object. For a spherical object, the viscous drag force is given by Stokes's law, $F = 6\pi\eta v r$, where r is the radius of the sphere, η is the viscosity of the fluid, and v is the speed of the object with respect to the fluid. An object's terminal velocity is the velocity that produces just the right drag force so that the net force is zero. An object falling at its terminal velocity has zero acceleration, so it continues moving at that constant velocity. Using Stokes's law, we can find the terminal velocity of a spherical object falling through a fluid. When the object moves at terminal velocity, the net force acting on

it is zero. If the object sinks, the terminal velocity is downward and the viscous drag force acts upward to oppose the motion. For an object, such as a helium balloon in air or an air bubble in oil, that rises rather than sinks, the terminal velocity is upward and the drag force is downward.

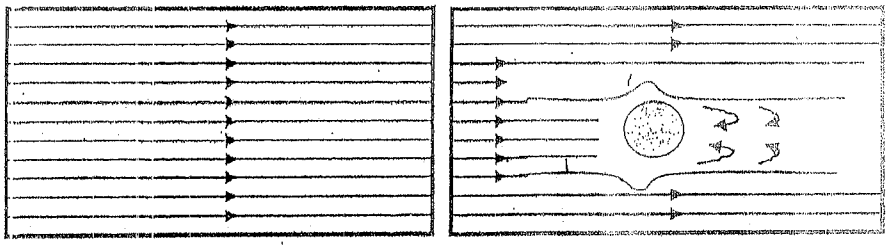


Fig. 7.12: Formation of eddies when fluid flows past a solid sphere or vice versa.

Example 7.11

An air track used in physics teaching demonstrations supports a cart that rides on thin cushion of air 1mm thick and $0.04m^2$ in area. If the viscosity of the air is 1.8×10^{-5} Pas, find the force required to move the cart at a constant speed of $0.2ms^{-1}$.

Solution

Given: $l = 1mm = 10^{-3}m$, $A = 0.04m^2$, $\eta = 1.8\times10^{-5}$, $v = 0.2ms^{-1}$,

The force required is calculated using Equation 7.24, $F = \eta A\frac{v}{l}$

and this gives $F = \eta A\frac{v}{l} = (1.8\times10^{-5}\,\text{Pa s})(0.04m^2)(\frac{0.2ms^{-1}}{10^{-3}m}) = 1.44\times10^{-4}$ N

Stokes Law-Motion in a Fluid

For a fluid flowing past a stationary solid sphere as shown in Figure 7.12, when the sphere moves slowly instead of the fluid, the fluid flow patterns are equal to that of a fluid moving past a stationary solid sphere. The layers of fluid in contact with the moving sphere moves, creating a velocity gradient between this layer and other layers of the fluid. Viscous forces are thereby brought into play and they result in resistance experienced by the moving sphere. The viscous retarding force F depends on (1) the dynamic viscosity η of the fluid (2) the velocity v and the radius r of the

solid sphere. Experiments first done by the English scientist Stokes gives the retarding force on the sphere as

$$F = 6\pi\eta v r \qquad 7.26$$

Equation 7.26 is known as Stokes law and holds only for steady motion in a fluid of infinite extent (otherwise the walls and the bottom of the container will affect the resisting force). In steady motion or flow, the velocity of the fluid at any point is constant in time. Now let us consider a sphere falling vertically under its weight in a viscous fluid as shown in Figure 7.13. There are three forces acting on the sphere:

1. its weight,acting downwards
2. the upthrust, (upward buoyant force due to the weight of fluid displaced by the sphere) acting upwards and
3. the viscous drag (derived from Stokes' law).

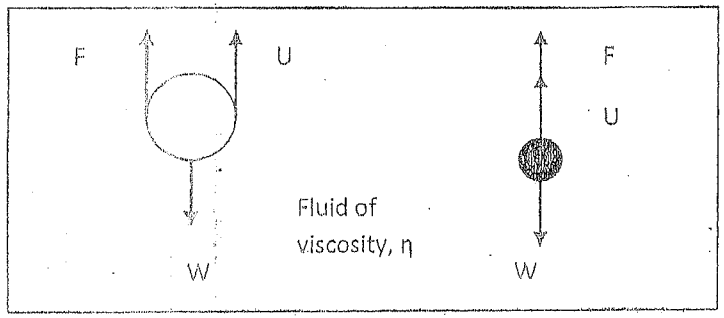


Fig. 7.13: A sphere falling/rising in a fluid of viscosity, η.

$$\Sigma F = W - U - F = ma \qquad 7.27$$

The resultant downward force equals *W-U-F* and causes the sphere to accelerate until its velocity and viscous drag reach certain values such that when σ is its density, the mass is

$$m = \sigma \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3\sigma. \qquad 7.28$$

and

$$\Sigma F = W - U - F = 0 \qquad 7.29$$

so that the sphere then begins to fall with a constant velocity known as the terminal velocity v_t. Now we find expressions for W, U and F to substitute in Equation 7.29. The weight W is given as

$$W = mass(m) \times acceleration(g) \qquad 7.30$$

But *mass of sphere = density of sphere* × *volume of sphere*.

$$\therefore W = mg = \frac{4}{3}\pi r^3\sigma \times g = \frac{4}{3}\pi r^3\sigma g \qquad 7.31$$

where σ is the density of the sphere. By Archimedes principle, the upthrust U is given by

$$U = (mass\ m\ of\ fluid\ displaced\ by\ sphere) \times (acceleration\ g) \qquad 7.32$$

But

mass of fluid displaced by sphere = density of fluid × volume of fluid displaced by sphere

$$\therefore m = \rho \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3\rho \qquad 7.33$$

Then the upthrust,

$$U = \frac{4}{3}\pi r^3 \rho g \qquad 7.34$$

where ρ is the density of the fluid. To find the drag force F, we apply Stokes' Law by assuming steady fluid flow when the sphere falls with the terminal velocity v_t

$$F = 6\pi\eta r v_t \qquad 7.35$$

Hence

$$\frac{4}{3}\pi r^3\sigma g - \frac{4}{3}\pi r^3\rho g - 6\pi\eta r v_t = 0 \qquad 7.36$$

$$\therefore v_t = \frac{2r^2(\sigma - \rho)g}{9\eta}$$

Experiment to Find the Viscosity η of a Fluid

In a laboratory experiment you are given metals of the same material but of different radii $r_1, r_2, r_3, \ldots$ etc and you are asked to drop them into a long graduated cylindrical tube containing a viscous fluid (e.g. glycerine) and then take measurements of the times $t_1, t_2, t_3, \ldots$ etc respectively of the descent of the balls from a marked level on the cylinder. We then calculate for each radii $r_1, r_2, r_3, \ldots r_n$ the corresponding terminal velocities $v_1 = \frac{x}{t_1}, v_2 = \frac{x}{t_2}, v_3 = \frac{x}{t_3} \ldots, v_n = \frac{x}{t_n}$, we then plot v against r^2. The slope of the graph is given by

$$Slope = \frac{\Delta v}{\Delta r^2} = \frac{2(\sigma - \rho)g}{9\eta} \qquad 7.38$$

Making η (viscosity) the subject we have

$$\eta = \frac{2(\sigma - \rho)g}{9 \times slope} \qquad 7.39$$

With the density of the sphere, ó and density of the liquid ñ given (or calculated) and g known we can then calculate the viscosity ç of a given fluid. If the velocity of the sphere in the liquid reaches a critical velocity v_c, the flow breaks up and form eddies as shown in Figure 7.14.

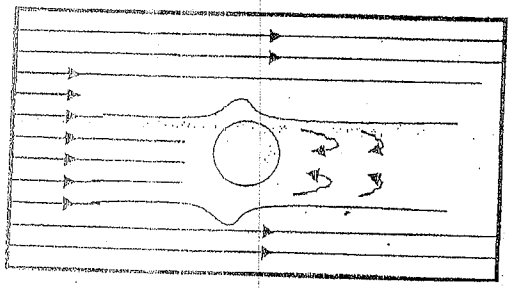


Fig. 7.14: If v (velocity of the sphere) ≥ v_c (critical velocity), the flow breaks up and form eddies

At velocities greater than the critical velocity the drag increases rapidly and then becomes approximately proportional to the square of the velocity which is given by

$$Drag = \frac{C\rho A v^2}{2} \qquad 7.40$$

where A is the cross-sectional area of the body normal to its velocity v, ρ is the fluid density and the drag coefficient C has a numerical value between 0 and 1.

From Stokes' law we gather that the resistance is proportional to the velocity of the sphere moving through the fluid. *Note* that for turbulent flow the resistance is dependent on the density of the fluid but not the viscosity of the fluid. To reduce drag and turbulence, bodies are streamlined by changing their shape. The drag and resistance are then reduced at a particular speed because steady flow replaces the turbulent flow. This modification to moving bodies in fluid is called *streamlining*. Streamlining is particularly important in the design of high-speed aircraft.

Poiseuilles Law

Viscosity makes analyzing fluid flow difficult. For example, when a fluid flows through a pipe, there is frictional drag between the liquid and the walls of the pipe and the fluid velocity is greater toward the centre of the pipe. This effect makes a difference in the average flow rate Q (which we will derive later) of the fluid which is given by

$$Q = \frac{\pi r^4 \Delta p}{8\eta L} \qquad 7.41$$

where $Q = A\bar{v} = \Delta V / \Delta t$ is the volume flow rate, Δp is the pressure difference between the ends of the pipe, r and L are the inner radius and length of the pipe respectively, and η is the viscosity of the fluid. Equation 7.41 is the Poiseuille's law. The SI unit of flow rate is cubic metres per second (m^3/s).

Derivation of Poiseuille's law

Let us consider the variation of velocity of a fluid with radius for a cylindrical pipe or tube with inner radius R. We can also consider the flow of a cylindrical element of fluid coaxial with the pipe or tube of radius r and length L as shown in Figure 7.15. The force on the left end of the inner tube is $p_1 \pi r^2$ (Force = Pressure × Area) and that on the right end is $p_2 \pi r^2$. The resultant force is thus

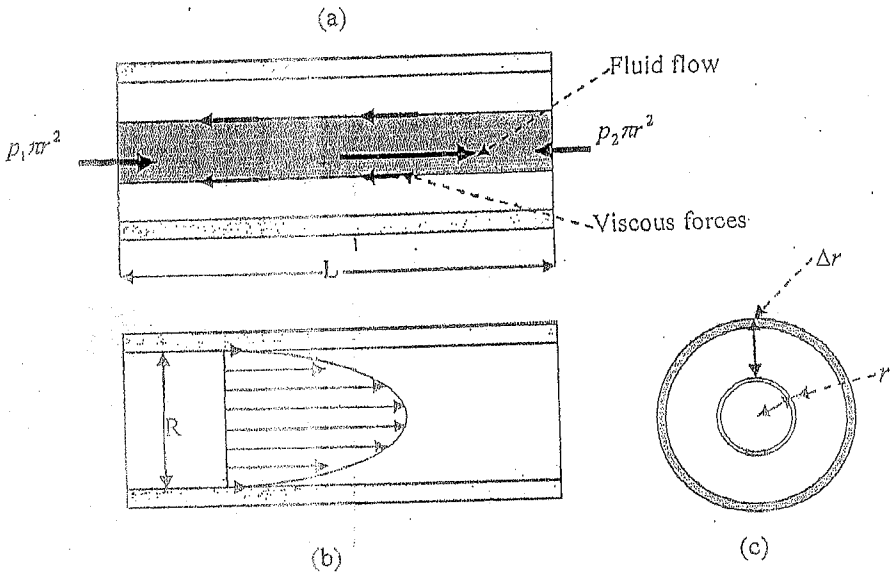


Fig. 7.15: Derivation of Poiseuille's law

$$F = p_1\pi r^2 - p_2\pi r^2 = (p_1 - p_2)\pi r^2 \qquad 7.42$$

Since the elemental volume does not accelerate, this force must balance the viscous retarding force at the surface of the volume element. This means that Equation 7.26 must be equal to Equation 7.42 but since the velocity does not vary uniformly with distance from the centre, we must replace v/L in this expression by dv/dr, where dv is the small change in velocity when we go from a distance of r to $r + dr$ from the axis. The area over which the viscous force acts is $A = 2\pi rL$. Thus the viscous force is

$$F = \eta 2\pi rL\frac{dv}{dr} \qquad 7.43$$

Equating Equation 7.42 to Equation 7.43, the resultant force due to the pressure on the ends is

$$F = \eta 2\pi rL\frac{dv}{dr} = (p_1 - p_2)\pi r^2 \qquad 7.44$$

Rearranging we have

$$\frac{dv}{dr} = -\frac{(p_1 - p_2)r}{2\eta L} \qquad 7.45$$

The negative sign indicates that the viscous force acts in the direction opposite to that of the pressure gradient force which is driving the fluid motion. This translates that as r increases v decreases. Equation 7.45 also shows that velocity changes rapidly as we go away from the centre (or the axis where $r = 0$) to the inner pipe or tube wall (where $r = R$). Rearranging again we have

$$-dv = \frac{(p_1 - p_2)}{2\eta L}dr \qquad 7.46$$

Integrating Equation 7.46 gives

$$-\int_v^0 dv = \frac{(p_1 - p_2)}{2\eta L}\int_r^R r dr \qquad 7.47$$

and

$$v = \frac{(p_1 - p_2)}{4\eta L}(R^2 - r^2) \qquad 7.48$$

This means that the velocity decreases from a maximum value of $(p_1 - p_2)R^2/4\eta L$ at the center or axis to zero at the wall. Thus the maximum velocity is proportional to the square of the pipe or the tube radius and also proportional to the pressure change per unit length $(p_1 - p_2)/L$, so that the total flow rate must also be proportional to this quantity. Let us consider the thin-walled element in Figure 7.15. The elemental volume of the fluid dV crossing the ends of the element in a time dt is $vdAdt$ where v is the velocity of the fluid at the radius r and dA the shaded area which equals $2\pi rdr$. From Equation 7.48 we have

$$dV = \frac{(p_1 - p_2)}{4\eta l}(R^2 - r^2)2\pi r dr dt \qquad 7.49$$

The volume flowing across the entire cross-section is obtained by integrating over all elements between and

$$dV = \frac{\pi(p_1 - p_2)}{2\eta L}\int_0^R (R^2 - r^2)dr dt \qquad 7.50$$

$$\Rightarrow dV = \frac{\pi}{8}\frac{R^4}{\eta}\left(\frac{p_1 - p_2}{L}\right)dt \qquad 7.51$$

The total volume flow per unit time is given by

$$Q = \frac{dV}{dt} = \frac{\pi R^4 \Delta p}{8\eta L} \qquad 7.52$$

where Δp is the pressure difference $(p_1 - p_2)$ from Equation 7.51. The volume flow rate is inversely proportional to viscosity (η), it is proportional to the pressure gradient $(\Delta p/L)$ along the pipe and to the fourth power of the radius R. If the radius is say halved, the flow rate is reduced by a factor of 16.

Example 7.12

A large artery in a goat has an inner diameter of $8\times10^{-3}\,m$. Blood flows through the artery at the rate of of $1cm^3s^{-1}$. Find (a) the average velocity of the blood (b) the pressure drop in a 0.1m long segment of the artery [$\eta = 2.084\times10^{-3}$ Pa s].

Solution

Given: $R = diameter/2 = 8\times10^{-3}\,m/2 = 4\times10^{-3}\,m$

(a) The average velocity is

$$\bar{v} = \frac{Q}{A} = \frac{Q}{\pi R^2} = \frac{10^{-6}\,m^3s^{-1}}{\pi(4\times10^{-3}\,m)} = 1.99\times10^{-2}\,ms^{-1}$$

(b) Given: $\eta = 2.084\times10^{-3}$ Pa s. The pressure drop is found from Equation 7.41 and is given as

$$\Delta p = \frac{8\mu LQ}{\pi R^4} = \frac{8\mu L\bar{v}}{R^2} = \frac{8(2.084\times10^{-3}\,\text{Pas})(0.1\text{m})(1.99\times10^{-2}\,\text{ms}^{-1})}{(4\times10^{-3}\text{m})^2} = 2.07\text{Pa}$$

Reynolds Number, Re

When the velocity of a fluid flowing in a tube exceeds a certain critical value (which depends on the properties of the fluid and the diameter of the tube), the nature of the flow becomes extremely complicated. Within a very thin layer adjacent to the tube walls (which we call the boundary layer) the flow is still laminar. The velocity of the fluid flow in the boundary layer is zero at the tube walls and increases throughout the layer. The properties of this boundary layer are necessary for understanding the transfer of heat to or from the moving fluid and in determining the resistance to the flow of the fluid.

Beyond the boundary layer the flow or the motion of the fluid parcels are highly irregular. The random local circular currents are called vortices and they develop within the fluid, with large increase in the resistance to flow. The fluid flow of this type is called turbulent. Experiments have shown that a combination of four factors determines whether a flow through a pipe or a tube is laminar or turbulent. This combination of factors is known as Reynolds number, Re and is defined as

$$\text{Re} = \frac{\rho v D}{\eta} \qquad 7.53$$

where ρ is the density, v is the average forward velocity, η the viscosity of the fluid and D is the diameter of the tube. The average velocity is defined as the uniform velocity over the entire cross-section of the tube which would result in the same volume rate of flow. The Reynolds number $\rho vD/\eta$ is a dimensionless quantity and it has the same numerical value in any constituent system of units.

The Reynolds number of a system forms the basis of the study of the behaviour of real systems through the use of small scale models. A common example is the wind tunnel, in which one measures the aerodynamic forces on a scale model of an aircraft wing. The forces on a full size wing are then deduced from the measurements. Two systems are said to be dynamically similar if their Reynolds numbers are the same. The letter D may refer, in general, to any dimension of a system, such as the span or chord of an aircraft wing. Thus the flow of fluid of given density ñ and viscosity ç, about half-scale model, is dynamically similar to that around the full-size object if the velocity is twice as great.

In tubes, the D in Equation 7.53 is $D = 2R$, where R is the radius of the tube. *Re* is found experimentally that if

$\text{Re} < 2000 \Rightarrow$ flow is laminar
$\text{Re} \geq 3000 \Rightarrow$ flow is turbulent
$2000 < \text{Re} < 3000 \Rightarrow$ flow is unstable (may change from laminar to turbulent and vice versa)

Example 7.13

In Example 7.3, the diameter of the artery is $8 \times 10^{-3}\,m$, the average velocity is

$\bar{v} = 1.99 \times 10^{-2}\, ms^{-1}$ and the viscosity. Given that the density of blood is $\rho_b = 1.0595 \times 10^3\, \text{kgm}^{-3}$, find the Reynolds number and determine whether the flow is laminar.

Solution

$$\text{Re} = \frac{2\rho\bar{v}R}{\eta} = \frac{2(1.0595\times10^3)(1.99\times10^{-2})(4\times10^{-3})}{2.084\times10^{-3}} = 80.9$$

This is much less than 2000, so the flow is laminar.

Fluid Flow Types

Fluid in motion leads to Poiseuille's Law and is exemplified in the types of fluid flow. This leads to the treatment of turbulent effects of fluid and to the study of Non-Newtonian fluids. In the study of fluid dynamics it is simple to classify fluid flow in terms of the following properties of the fluid:

1. steady or unsteady flow
2. rotational or irrotational flow
3. viscous or non-viscous flow and
4. compressible or incompressible flow. Fluid flow can also be described as laminar, turbulent or a mixture of both.

Laminar Flow

This is a flow pattern where all the particles in the fluid follow the same line of flow. These lines of flow can be visualized as "sheets" and are known as streamlines. In the case of a tube these streamlines are a set of concentric tubes, the particle velocity of which increases the closer to the centre one measures.

Turbulent Flow

When the fluid velocity at a given point changes, the flow is unsteady. Turbulence is an extreme example of unsteady flow. In turbulent flow, swirling vortices or little whirlpools of fluid appear. The vortices are not stationary but they move with the fluid. The flow velocity at any point changes erratically and prediction of the direction or speed of fluid flow under turbulent conditions is difficult. Turbulent fluid flow can be classified accordingly as:

1. Compressible flow and
2. incompressible flow.

A fluid flow is called compressible if the pressure variations in the flow are large enough to effect substantial changes in the density of the fluid. Phase changes in liquids with pressure variations much smaller than those required to cause a phase change (cavitation), or flow of gases involving speeds much lower than the isentropic sound speed are termed incompressible.

Steady and Uniform Flow

A flow is described as *uniform* if the velocity at a given instant is the same in magnitude and direction at every point in the fluid. If at any instant, the velocity changes from point to point, the flow is described as non-uniform. In practice, when a fluid flows past a solid boundary there will be variations of velocity in the region close to the boundary. However, if the size and shape of the cross-section of the stream of fluid is constant, the flow is considered to be uniform.

A *steady* flow is one in which the velocity, pressure and cross-section of the stream may vary from point to point but do not change with time. If, at any given point, conditions do change with time, the flow is described as unsteady. In practice there will always be slight variation of velocity and pressure, but, if the average values are constant, the flow is considered to be steady.

Newtonian Fluids

Fluids for which the equation $F = 6\pi\eta vr$ (Equation 7.26) holds are called Newtonian fluids. These Newtonian fluids are modeled by the coefficient of viscosity, which depends on the specific fluid. However, some of the other materials, such as suspensions (or dispersions), emulsions and some visco-elastic materials have more complicated non-Newtonian stress-strain behaviours. These materials include sticky liquids such as latex, honey, and lubricants.

Equation of Continuity

If there are no losses during the flow of a fluid within a uniform tube, the mass flowing into the tube at a given time Δt must be equal to the mass flowing out of the tube in the same time (this is, in other words, the conservation of mass). For example in Figure 7.16, the mass (Δm) entering the tube during a short time (Δt) is given by

$$\textit{mass} = \textit{density} \times \textit{volume}\,.$$

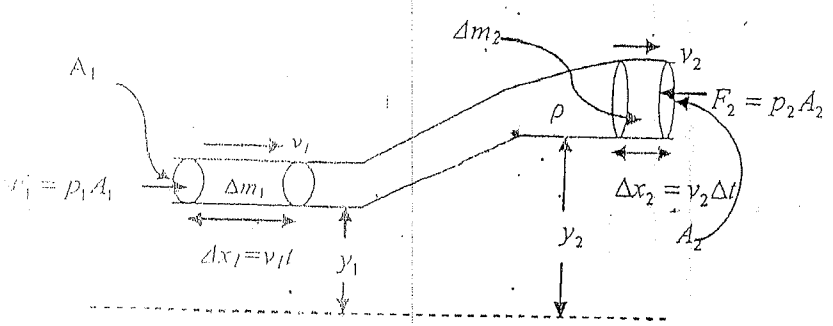


Fig. 7.16: An incompressible fluid flowing horizontally through a non-uniform tube.

$$\Delta m_1 = \rho_1 \Delta V_1 = \rho_1 (A_1 v_1 \Delta t) \qquad 7.54$$

where A_1 is the cross-sectional area of the tube at the entrance and, in a time Δt, a fluid particle moves a distance equal to $\Delta x_1 = v_1 \Delta t$. Thus the volume of mass Δm_1 entering the tube is $\Delta V_1 = A_1 \Delta x_1 = A_1 v_1 \Delta t$

$$\Rightarrow \Delta m_1 = \rho_1 \Delta V_1 = \rho_1 (A_1 v_1 \Delta t) \qquad 7.55$$

Similarly the mass leaving the tube in the same time interval is

$$\Delta m_2 = \rho_2 \Delta V_2 = \rho_2 (A_2 v_2 \Delta t) \qquad 7.56$$

Since mass is conserved i.e. $\Delta m_1 = \Delta m_2$, equating Equations 7.54 and 7.56 gives

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

and

$$\rho A v = const. \qquad 7.57$$

For an incompressible fluid, the density ñ is constant, so we have

$$A_2 v_2 \; A_1 v_1 = A_2 v_2 \text{ and}$$
$$\Rightarrow Av = const. \qquad 7.58$$

This is sometimes referred to as the flow rate equation. *Note* that the flow rate equation shows that the fluid velocity is greater where the cross-sectional area of the tube is smaller. That is $v_2 = (A_1/A_2)v_1$ and v_2 is greater than v_1 if A_2 is less than A_1. This effect is evident in the common experience that the speed of water is greater from a hose fitted with a nozzle than the same hose without a nozzle (see Figure 7.17). In Figure 7.17(a) the hose has no nozzle so water does not gush out unlike in figure 7.17(b) where a nozzle (having a smaller area than the hose) is fitted and water gushes out with a higher velocity.

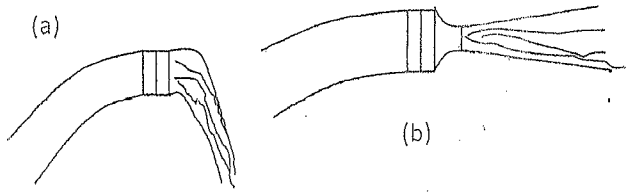


Fig. 7.17: Fluid velocity is greater in (b) where cross-sectional area of tube is smaller.

The flow rate equation can be applied to the flow of blood in your body. Blood flows from the heart into the aorta. It then makes a circuit through the circulatory system passing through the arteries, arterioles (small arteries), capillaries, venules (small veins) and back to the heart through the veins. High cholesterol in the blood can cause the fatty deposits called plaque to deposit on the inside of the arteries effectively causing the radius of the arteries to reduce. This is results is the medical condition commonly called *high blood pressure*.

Having dealt with flow patterns and factors affecting flow, we can now consider other effects that flow produces. Using the continuity equation we will look at the Bernoulli principle and how this gives rise to the Venturi and Coanda effects.

Example 7.14

Water flows out of a pipe at the rate of 3.0cms^{-1}. Find the velocity of the water at a point in the pipe where its diameter is 0.50cm.

Solution

From the equation of continuity, $\rho_1 A_1 v_1 = \rho_2 A_2 v_2$, we have (since ρ is constant) $\rho_1 = \rho_2$

$\pi(0.25)^2 v = 3.0$, so $v = 15.3 ms^{-1}$

Bernoulli's Principle

Looking again at the ideal fluid flowing through a tube and considering the work done by the forces F_1 and F_2, F_1 does positive work i.e. the direction of the force is the same as that of the fluid motion and F_2 does negative work because the direction of F_2 is opposite to that of the fluid motion. The net work done on the system by these forces is

$$W = F_1\Delta x_1 - F_2\Delta x_2 = (p_1A_1)(v_1\Delta t) - (p_2A_2)(v_2\Delta t) \qquad 7.59$$

The flow rate equation requires that ($A_1v_1 = A_2v_2$), so we write the work equation as

$$W = A_1v_1\Delta t(p_1 - p_2) \qquad 7.60$$

Recall from the previous derivation of the equation of continuity that $\Delta m_1 = \rho_1\Delta V_1 = \rho_1(A_1v_1\Delta t)$ and $\Delta m_1 = \Delta m_2$, so we write in general that

$$W = \frac{\Delta m}{\rho}(p_1 - p_2) \qquad 7.61$$

The total work done on the system by the external force F_1 and F_2 must be equal to the change in the total mechanical energy. That is

$$W = \Delta E = \Delta K + \Delta U \qquad 7.62$$

Considering kinetic energy changes, ΔK, of the elemental mass Δm we have

$$\Delta K = \frac{1}{2}\Delta m(v_2^2 - v_1^2) \qquad 7.623$$

The corresponding change in gravitational potential energy is given by

$$\Delta U = \Delta mg\,(y_2 - y_1) \qquad 7.64$$

Thus,

$$W = \frac{1}{2}\Delta m(v_2^2 - v_1^2) + \Delta mg(y_2 - y_1) \qquad 7.65$$

Then from Equation 7.62

$$\frac{\Delta m}{\rho}(p_1 - p_2) = \frac{1}{2}\Delta m(v_2^2 - v_1^2) + \Delta mg(y_2 - y_1) \qquad 7.66$$

Canceling Dm and rearranging gives the common form of Bernoulli's equation.

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho gy_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho gy_2 \qquad 7.67$$

or

$$p + \frac{1}{2}\rho v + \rho gy = cons\tan t \qquad 7.68$$

Bernoulli's equation, Equation 7.67, can be summed up as a mathematical representation for the universal law for the conservation of energy for a non-compressible, non-viscous fluid undergoing laminar flow. The sum of the pressure, kinetic and potential energies per unit volume, remains a constant at all points along the line of flow.

Note that, the terms in Bernoulli's equation are work or energy per unit volume (J/m^3), so that $W = F\Delta x = (pA)\Delta x = p\Delta V$ and $p = \frac{W}{\Delta V} = \left(\frac{work}{volume}\right)$.

Similarly we have $\rho = m/V$, then $\frac{1}{2}\rho v^2 = \frac{\frac{1}{2}mv^2}{V} = \left(\frac{energy}{volume}\right)$ and $\rho gy = \frac{mgy}{V} = \left(\frac{energy}{volume}\right)$.

From Equations 7.67 and 7.68, if we alter the energy of one portion of the system, it has an effect on the rest of the system. So if the kinetic energy rises, the potential energy and pressure must fall. When we apply the Bernoulli principle in practice, we can ignore the portion due to gravity to make the analysis a little simpler.

Bernoulli's Effects

The equations of hydrostatics are special cases of Bernoulli's equations in which velocity is held at zero. There are a number of phenomena and applications which concern Bernoulli's equation and this includes

1. Venturi effect/metre
2. The Torricelli effect

3. the Coanda effect
4. the mechanisms involved in the use of the Pitot and Prandtl tubes
5. the lift of an aircraft wing, etc. We will discuss only the Venturi and the Coanda effects in this book.

The Venturi Effect/Metre

Consider the tube with a narrowing at point B in Figure 7.18. If there are no leaks, the volume of fluid at point A is the same as at point C. Consequently the narrowing at point B means that the fluid has to speed up in order to fulfill this continuity. As an analogy, imagine you are sitting in a boat and it starts to gain water. You have a cup and your boating partner has a bucket. In order to remove the same quantity of water you have to move quicker, which is analogous to you being the flow at point B. In so doing you have to work harder and so expend more energy.

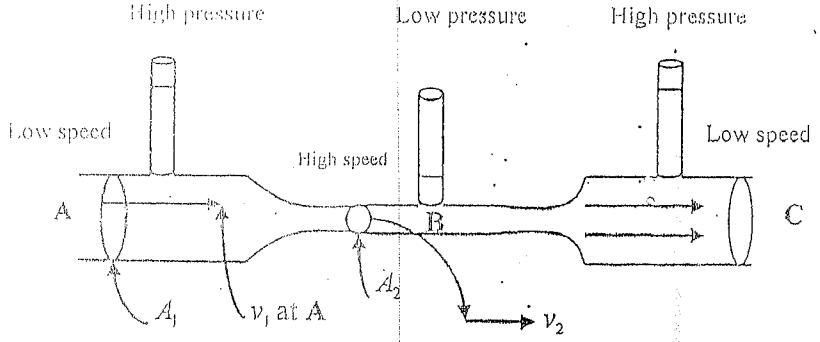


Fig. 7.18: The Venturi Metre

We take a look again at Equation 7.68. This means that point B has an increase in velocity compared with point A or C, so the pressure falls at point B. This is a consequence of Bernoulli's principle. The pressure drop induced by the increase in velocity of a fluid passing through a narrow column is used by chimneys and smoke stacks to draw exhaust from fire places and mud ovens.

The Coanda Effect

This effect was named after a Romanian aircraft designer, Henri Coanda, after an aircraft he designed went up in flames as a consequence of this effect. Essentially any fluid coming into contact with a curved surface will cling to this surface and alter its direction of flow. You can illustrate this yourself by running a thin stream of water

from a tap, and bringing the curved surface of a spoon to touch it. The water follows the surface of the spoon. It does so because the solid stationary surface of the spoon slows the layer in immediate contact. This has a drag effect on the other layers, in effect pulling them into the line of the curved surface.

Surface Tension

Numerous effects of liquids reveal that liquid surfaces behave like a stretched elastic skin i.e. it is in a state of tension. For example, a steel needle which has a density far higher than that of water when placed gently on water will float. The question is what kind of force acts counter to the weight of the needle to help it float. This effect is known as surface tension. Small droplets are nearly spherical as seen when water drops from the tap; a sphere has the minimum surface area for a given volume. This tendency of surfaces of liquids to shrink and form a drop of minimum area can be defined by the concept of surface tension.

Effects of Surface Tension

Surface tension results in the following phenomenon or physical effects.

Liquid Surfaces

The surface of a liquid must always be at right angles to the resultant forces acting to the surface, otherwise there will be a component of this force that acts parallel to the liquid surface and will cause motion. Normally a liquid surface is horizontal, that means the surface will be at right angles to the force of gravity, but where it comes into contact with the container or a solid it is usually curved.

Capillary Action

When liquids are constrained in vessels whose dimensions are small, compared to the relevant length scales, surface tension effects become important leading to the formation of a *meniscus* through capillary action.

When a glass capillary tube of a small bore is dipped in water (as shown in Figure 7.19a), the water rises in the tube, but if the tube is immersed in mercury (as shown in Figure 7.19b), the mercury is depressed

Drops

Without surface tension, drops would not be able to form. The dimensions and

stability of drops are determined by surface tension. A liquid emerging from the tip of an eye drop is not continuous but a succession of drops.

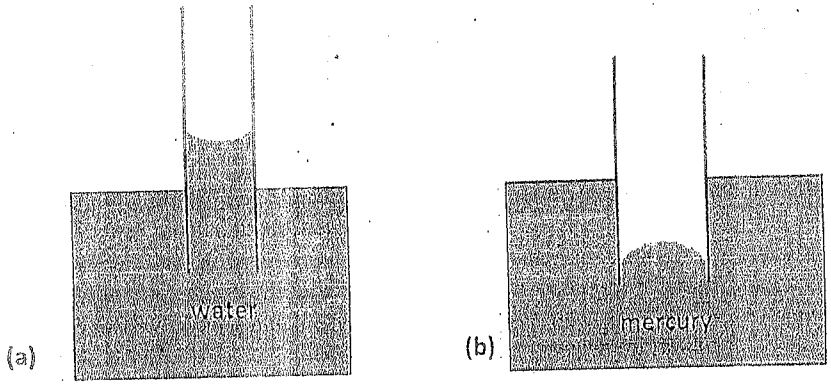


Fig. 7.19: (a) Water rises (b) Mercury depresses in capillary tube

Bubbles

In an underwater air bubble, the surface tension of the water surface tries to contract the bubble while the pressure of the enclosed air pushes outward on the surface. In equilibrium, the air pressure inside the bubble must be larger than the water pressure outside so that the net outward force due to pressure balances the inward force due to surface tension.

Needle Floating on Water

A sewing needle, if placed carefully on a water surface rests there without sinking even though the material used to make the needle (steel) has density about ten times that of water

The above observed phenomena ((a) to (e)) due to surface tension indicates that the surface of a liquid can be considered to be in a state of tension or stress such that for any line lying in or bounding the surface, the material on either side of the line exerts a pull on the other side. This pull is planar and is perpendicular to the line. Figure 7.19 shows a flexible wire ring which has a loop of thread attached to it. When the wire loop is dipped in a soap solution and removed, the loop is surrounded by and is bounding a film of soap solution. The film of soap solution in the middle of the wire loop is then punctured.

After puncturing the film of soap solution in the loop of thread, the thread spreads out into a circular shape as shown in Figure 7.18b, as if the surfaces of the liquid were pulling radially outward on it (as shown by the direction arrows in Figure 7.18b).

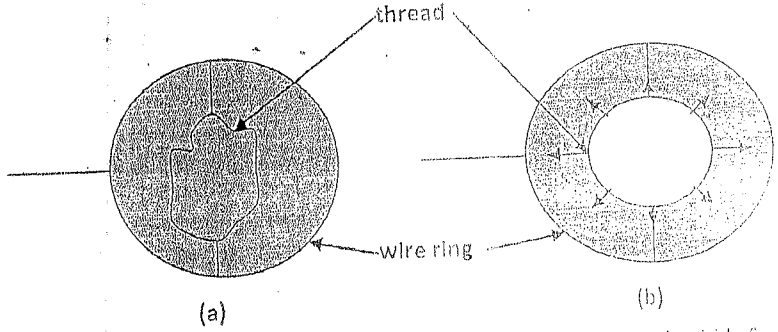


Fig. 7.20: Wire ring with a flexible loop of thread, dipped in a soap solution (a) before (b) after puncturing the film inside the loop.

We can presume the same forces were present and acting before the film was punctured, but since there was a film of soap solution on both sides of the thread we say that the resultant force exerted by the film on every portion of the thread was zero.

Experiment Demonstrating Effects of Surface Tension

The apparatus for this experiment is made up of a piece of wire which is bent into a U-shape and a second piece of wire used as a slider (as shown in Figure 7.21a)

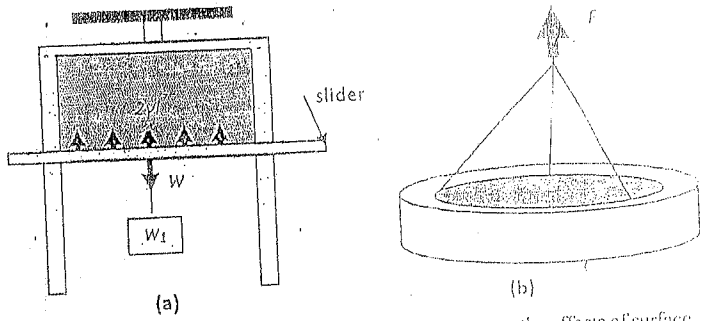


Fig. 7.21: Shows a simple apparatus which demonstrates the effects of surface tension (a) wire with a slider (b) a converted form of (a)

When the whole apparatus is dipped into a soap solution and removed, the slider (if its weight w_1 is not too great) is quickly pulled to the top of the U-shape. It may be held stationary or in equilibrium by adding a second weight w_2.

Amazingly the same net force $F = w_1 + w_2$ will hold the slider stationary at any position, regardless of the surface area of the liquid film, provided the liquid film remains at a constant temperature. In other words, the force does not increase as the surface is stretched further. This is very different from the behaviour of a sheet of rubber (or elastic materials in general), for which the net force would be greater as the sheet is stretched. Even though a soap film is very thin, its thickness is still very large compared with the sizes of the molecules making up the soap solution. Thus we consider the soap film to be made mainly of bulk liquid, bounded by two surface layers a few molecules thick. When the slider in Figure 7.21a is pulled down and the surface area increased, molecules formerly in the main body of the liquid move into the surface layers. That is, these layers are stretched as a rubber sheet would be stretched; this is by virtue of the fact that more of the surface is created by molecules moving from the bulk liquid. Let l be the length of the wire slider. Since the film has two surfaces, the total length along which the surface force acts on the slider is $2l$. The surface tension γ in the film is defined as the ratio of the surface force (which is perpendicular to the length) to the length along which the force acts. Thus with our apparatus we define surface tension mathematically as:

$$\gamma = \frac{F}{2l} \qquad 7.69$$

The SI units of surface tension are Newton per metre (Nm^{-1}). The cgs unit is the dyne per unit centimetre ($dyn\ cm^{-1}$) and the conversion is done to SI units by the following.

$$1N \cdot m^{-1} = 1000 dyn \cdot cm^{-1} \qquad 7.70$$

A converted form of this apparatus (shown in Figure 7.21b) which consists of a circular wire of circumference l is lifted from a body of the liquid of which we want to determine its surface tension. There is an additional force which is needed to balance the surface forces $2\gamma l$ due to the two surface films on each side. This force can be measured by either a stretch of a delicate spring or by the twist of a torsion wire. The surface tension is then calculated using Equation 7.69. Surface tension usually decreases as temperature increases.

Exercises 7

7.1 The area of a piston of a force pump is $8cm^2$. What force must be applied to the piston to raise oil (ρ= 0.78g/cm3) to a height of 6.0m? Assume the upper end of the oil is open to the atmosphere. (Ans: 37N)

7.2 A metal cube, 2.0cm on each side, has a density of $6600kgm^{-3}$. Find its apparent mass when it is totally submerged in water. (Ans: 44.8g)

7.3 A solid wooden cube, 30cm on each edge can be totally submerged in water if it is pushed downward with a force of 54N. What is the density of the wood? (Ans: $800kgm^{-3}$)

7.4 Downward forces of 45N and 15N respectively are required to keep a plastic block totally immersed in water and in oil. If the volume of the block is $8000cm^3$, find the density of oil. (Ans: $620kgm^{-3}$)

7.5 A cube of wood floating in water supports a 200g mass resting on the centre of its top face. When the mass is removed, the cube rises 2cm. Determine the volume of the cube. (Ans: $1000cm^3$)

7.6 Oil flows through a 4cm (internal diameter) pipe at an average velocity of $2.5ms^{-1}$. Find the flow in m^3s^{-1} and cm^3s^{-1}. (Ans: $3.14\times10^{-3}\ m^3s^{-1} = 3140\ cm^3s^{-1}$)

7.7 How much work does the piston in a hydraulic system do during one 2cm stroke if the end area of the piston is $0.75cm^2$ and the pressure in the hydraulic fluid is 50kPa? (Ans: 7.5mJ)

7.8 Water flows steadily through a horizontal pipe of varying cross section. At one place the pressure is 130kPa and the speed is $0.60ms^{-1}$. Determine the pressure at another place in the same pipe where the speed is $9.0ms^{-1}$. (Ans: 90kPa)

7.9 Fuel oil of density $820kgm^{-3}$ flows through a venturi meter having a throat diameter of 4cm and an entrance diameter of 8cm. The pressure drop between entrance and throat is 16cm of mercury. Find the flow. The density of mercury is $13600Kgm^{-3}$ (Ans:$9.310^{-3}\ m^3s^{-1}$)

7.10 Find the maximum amount of water that can flow through a 3cm (internal diameter) pipe per minute without turbulence. Take the Reynolds number for non-turbulent flow to be 2000. For water, η=1.010^{-3} Pa s (Ans: $0.00283m^3$)

Chapter Eight

KINETIC THEORY AND THERMODYNAMICS

Atoms and Molecules

A great number of the ancient Greek philosophers perceived that all substances were composed of tiny particles or atoms and it was not until the nineteenth century that this idea developed into a useful theory for explaining some of the chemical and physical properties of matter. In 1808, the English chemist, John Dalton, produced experimental evidence to demonstrate that chemical compounds consist of molecules which are groups of atoms of various elements united in the same simple numerical proportion. A substance which cannot be split into simpler substances is an element while an atom is the smallest portion of an element which can take part in a chemical change.

Atomic Mass Unit

The unified atomic mass unit (u), or Dalton (Da), is a small unit of mass used to express atomic and molecular masses. It is defined to be one twelfth of the mass of an unbound atom of the carbon-12 nuclide, at rest and in its ground state. Now the number of molecules in 1 mole of carbon is 6.02×10^{23}, the Avogadro constant, and since carbon is monatomic, there are atoms of carbon. These have a mass of 12g.

Therefore, mass of 1 atom of carbon

$$= \frac{12}{6.02\times10^{23}}\,g = \frac{12}{6.02\times10^{26}}\,kg = 12u$$

$$\therefore 1u = \frac{12}{12\times6.02\times10^{26}}\,kg$$

$$= 1.66\times10^{-27}\,kg$$

The symbol amu for atomic mass unit can sometimes still be found, particularly in older works. Atomic masses are often written without any unit and then the atomic mass unit is implied. In biochemistry and molecular biology literature (particularly in reference to proteins), the term "dalton" is used, with the symbol Da.

Because proteins are large molecules, they are typically referred to in kilodaltons, or "kDa", with one kilodalton being equal to 1000 daltons. The unified atomic mass unit, or dalton, is not an SI unit of mass, although it is accepted for use with SI under either name.

The unit is convenient because one hydrogen atom has a mass of approximately $1u$, and more generally an atom or molecule that contains n protons and neutrons will have a mass approximately equal to nu. (The reason is that a carbon-12 atom contains 6 protons, 6 neutrons and 6 electrons, with the protons and neutrons having about the same mass and the electron mass being negligible in comparison.) This is an approximation, since it does not account for the mass contained in the binding energy of an atom's nucleus; this binding energy mass is not a fixed fraction of an atom's total mass. The differences which result from nuclear binding are generally less than $0.01u$, however. Chemical element masses, as expressed in u, would, therefore, all be close to whole number values (within 2% and usually within 1%) were it not for the fact that atomic weights of chemical elements are averaged values of the various stable isotope masses in the abundances with which they naturally occur. [For example, chlorine has an atomic weight of $35.45u$ because it is composed of 76% Cl-35 ($34.96u$) and 24% Cl-37 ($36.97u$)].

Another reason the unit is used is that it is experimentally much easier and more precise to compare masses of atoms and molecules (determine relative masses) than to measure their absolute masses. Masses are compared with a mass spectrometer.

Avogadro's number (N_A) and the mole are defined so that one mole of a substance with atomic or molecular mass $1u$ will have a mass of precisely 1 gram. For example, the molecular mass of water is $18.01508u$, and this means that one mole of water has a mass of 18.01508 grams, or conversely that 1 gram of water contains $N_A/18.01508 \approx 3.3428\times10^{22}$ molecules.

History

The chemist, John Dalton, was the first to suggest the mass of one atom of hydrogen as the atomic mass unit. Francis Aston, inventor of the mass spectrometer, later used $1/16$ of the mass of one atom of oxygen-16 as his unit.

Before 1961, the *physical atomic mass unit* was defined as $1/16$ of the mass of one atom of oxygen-16, while the *chemical atomic mass unit* was defined as $1/16$ of the *average* mass of an oxygen atom (taking the natural abundance of the

different oxygen isotopes into account). Both units are slightly smaller than the unified atomic mass unit, which was adopted by the International Union of Pure and Applied Physics in 1960 and by the International Union of Pure and Applied Chemistry in 1961.

Mass spectrometry looking at isotope ratios is most often used by archaeologists to determine the ages of old man-made objects based upon the isotopic proportions of carbon. Having the atomic mass unit based upon oxygen added needless error; now that it is based upon an isotope of carbon this is obviated. (Mass spectrometry is used in many other fields, such as forensic science, but in these fields the goal is usually identification of specific molecules, not comparison of isotope ratios)

Avogadro's Number

Avogadro's number, also called **Avogadro's constant** (N_A), named after Amedeo Avogadro, in German scientific literature also known as **Loschmidt number**, is formally defined to be the number of carbon-12 atoms in 12 grams (0.012 kg) of unbound carbon-12 in its rest-energy electronic state. The current best estimate of this number is $(6.0221415 \pm 0.0000010) \times 10^{23}$.

A mole is defined in the SI as Avogadro's number of particles of any kind of substance (atoms, ions, molecules, or formula units). In the SI, this unit is abbreviated mol. The mole is widely used as a macroscopic unit of amount of substance.

History

Avogadro's number is named after the early nineteenth century Italian scientist Amedeo Avogadro. It appears that Jean Baptiste Perrin was the first to name it. Perrin called it "Avogadro's constant" and it is still sometimes known by that name. The numerical value was first estimated by Johann Josef Loschmidt in 1865 using the kinetic gas theory. In German-speaking countries, the number may still be referred to as Loschmidt's number. Unfortunately, in a few cases (mainly in the older literature) Loschmidt's number refers to the number of atoms (or molecules) in a cubic centimetre, a usage now ridiculed.

In the nineteenth century, physicists measured the mass of one atom of hydrogen to be about $1/6.023 \times 10^{23}$ grams; they were trying to evaluate how many molecules of an ideal gas would fit in 1 cubic centimetre at standard conditions for temperature and pressure (STP), which is related to Avogadro's number via the ideal gas law. Before 1960, there were conflicting definitions based on 16 grams of oxygen: physicists generally used oxygen-16 while chemists generally used the "naturally occurring" isotope ratio. Switching to 12 grams of carbon-12 ended this dispute.

Application

Avogadro's number can be applied to any substance. It corresponds to the number of atoms or molecules needed to make up a mass equal to the substance's atomic or molecular mass, in grams. For example, the atomic mass of iron is $55.847u$, so Avogadro's number of iron atoms (i.e. one mole of iron atoms) has a mass of 55.847g. Conversely, 55.847g of iron contains Avogadro's number of iron atoms. Thus Avogadro's number N_A corresponds to the conversion factor between grams (g) and atomic mass units:

$$1\ \text{g} = N_A \text{u}.$$

Chemical Significance of Avogadro's Number

The definition of Avogadro's number depends on the definition of the kilogram. The latter is based on arbitrary convention, namely the mass of a particular "standard" cylinder of metal in France. This means that the particular value of Avogadro's number is the result of convention; there is no physical reason for it. For this reason, Avogadro's number is not considered a fundamental constant in the strictest sense. However, for practical purposes, Avogadro's number is regarded as a chemical constant.

Avogadro's number can be regarded as a conversion factor between the microscopic mass system (atomic mass units or Daltons) and the kilogram system. The microscopic mass system is based on the mass of carbon-12, while the kilogram system is currently based on the mass of a particular "standard" cylinder of metal in France. So naturally there's no simple conversion factor between the two. However, if a method were developed to count atoms, it would be possible to redefine the kilogram in a way that did not depend on an arbitrary cylinder of metal. The number of atoms picked would presumably be equal or close to the latest accepted value of Avogadro's number. In that case, the kilogram would be redefined as 83.33 × the mass of Avogadro's number of carbon-12 atoms (i.e. $83.33 \times 12\ \text{g} = 1\ \text{kg}$).

Additional Physical Relations

Because of its role as a scaling factor, Avogadro's number provides the link between a number of useful physical constants when moving between an atomic mass scale and a kilogram scale. For example, it provides the relationship between:

1. the universal gas constant R and the Boltzmann constant k_B : $R = k_B N_A$
2. the Faraday constant F and the elementary charge e : $F = eN_A$

In the 19th century physicists measured the mass of one atom of hydrogen to be about $6.02214199 \times 10^{-23}$ grams. The gram was originally defined to be the mass of a cubic centimetre of pure water at standard temperature and pressure. As experiments became more accurate, it was found that water was contaminated with variable amounts of heavy water, which made it undesirable to maintain a standard with hydrogen having one atomic mass unit. Carbon was found to have a more constant isotopic composition, and it was also possible to separate pure carbon-12. Therefore, the atomic mass unit was changed to $\frac{1}{12}$ the mass of an atom of carbon-12. Hence 12 grams of carbon-12 has about 6.0221415×10^{23} atoms.

Numerical Value

At present it is not technologically feasible to count the exact number of atoms in 12 g of carbon-12, so the precise value of Avogadro's number is unknown. The 2002 CODATA (Committee on Data for Science and Technology) recommended value for Avogadro's number is

$$N_A = 6.0221415 \times 10^{23} \pm 0.0000010 \times 10^{23}\ \text{mol}^{-1}.$$

A number of methods can be used to measure Avogadro's number. One modern method is to calculate Avogadro's number from the density of a crystal, the relative atomic mass, and the unit cell length determined from x-ray crystallography. Very accurate values of these quantities for silicon have been measured at the National Institute of Standards and Technology (NIST) and used to obtain the value of Avogadro's number.

Connection to Masses of Protons and Neutrons

A carbon-12 atom consists of 6 protons and 6 neutrons (which have approximately the same mass) and 6 electrons (whose mass is negligible as a first approximation, being about $\frac{1}{1840}$ of the mass of the proton). One could therefore think that N_A is the number of protons or neutrons that have a mass of 1 gram. While this is approximately correct, the mass of a free proton is 1.00727 *amu*, so a mole of protons would actually have a mass of 1.00727 g. Similarly, a mole of neutrons has a mass of 1.00866 g. Clearly, 6 moles of protons combined with six moles of neutrons would have a mass greater than 12 g. So it would appear that one mole of carbon-12 atoms, which should consist of 6 moles each of protons, neutrons, and electrons would have a mass greater than 12 g. The discrepancy or mass defect (as it is known) is related to the equivalence of matter and energy discovered by Albert Einstein as part of the theory of special relativity. When an atom is formed, the protons and neutrons in the nucleus are bound together by the strong nuclear force. This binding results in the formation of a low energy state and is accompanied by a large release of energy. Since energy is equivalent to mass (which means that all energy *has* mass), the released energy has mass and carries away the loss in the mass of the nucleus relative to that of the separated protons and neutrons (note that mass is conserved in this process just as energy is). Thus, protons and neutrons in the nucleus have masses that are less (about 0.7 percent less) than those of free protons and neutrons. The precise amount of mass loss is related to the binding energy of the nucleus and varies depending on the type of atom.

One may, therefore, say that N_A is approximately the number of *nuclear* neutrons or protons (nucleons) that have a mass of 1 gram. This is approximate because the precise mass of a nuclear proton or neutron depends on the composition of the nucleus, as explained above. For example, iron nucleons will have a significantly lower mass than those in hydrogen or plutonium.

Avogadro's Number in Life

Avogadro's number may also yield practical reasonings in real life. For example, the fact that a known number of atoms are in a given amount of a substance is a reason for scientific criticism of homeopathy, in which medicinal substances are often diluted to the extent that a single molecule appears in only one dose amongst the hundreds or thousands prepared, as a simple calculation involving Avogadro's number will reveal.

Another common sense application shows that without determining the actual weight of a substance, a good rule of thumb to use is that a cubic centimetre of solid matter contains about 10^{24} atoms.

Molecular Mass

The *molecular mass* (abbreviated Mr) of a substance, formerly also called *molecular weight* and abbreviated as MW, is the mass of one molecule of that substance, relative to the unified atomic mass unit u (equal to $\frac{1}{12}$ the mass of one atom of carbon-12). Due to this relativity, the molecular mass of a substance is commonly referred to as the *relative molecular mass*, and abbreviated to Mr.

Conversion factor of molecular mass is to molar mass is *molar mass* = *molecular mass* × (6.022×10^{23}).

The molecular mass can be calculated as the sum of the atomic masses of all the atoms of any one molecule. The molecular mass can also be measured directly

using mass spectrometry. In mass spectrometry, the molecular mass of a small molecule is usually reported as the *monoisotopic mass*, that is, the mass of the most common isotope of each element. The masses used to compute the monoisotopic molecular mass are found on a table of isotopic masses and are not the same as found on a typical periodic table. The *average* molecular mass is often used for larger molecules since molecules with many atoms are unlikely to be composed exclusively of the most abundant isotope of each element. This *average* mass can be calculated using the elemental mass numbers on a typical periodic table, since there is likely to be a statistical distribution of atoms representing the isotopes throughout the molecule.

The molar mass of a substance is the mass of 1 mol (the SI unit for the basis SI quantity *amount of substance*, having the symbol n). This has numerical value of the molecular mass multiplied by Avogadro's constant, and the SI unit is $^{kg}/_{mol}$, but usually also found as stated in $^{g}/_{mol}$. For example, the atomic mass of hydrogen is $1.00784u$ and that of oxygen is $15.9994u$; therefore, the molecular mass of water with formula H_2O is $(2 \times 1.00784u) + 15.9994u = 18.01508u$. Thus, one mole of water has a mass of 18.0151 grams. However, the exact mass of hydrogen-1 is 1.00783, and the exact mass of oxygen-16 is 15.9949, so the mass of the most common single molecule of water is $18.0105u$. Both hydrogen -1 and oxygen -16 are the most common isotope. Molecular mass or molar mass are used in stoichiometry calculations.

Since molecules are created by *chemical reactions*, not *nuclear reactions*, a molecule's molecular mass exactly equals the sum of the atomic masses of its constituent atoms.

The *gram-molecular weight* is the molecular weight of a chemical compound expressed in grams. Thus the molecular weight of calcium carbonate is 100, therefore, the gram-molecular weight is 100 g. The gram molecular weight is frequently used, particularly for calculating the concentration of solutions. Thus a molar solution will contain the gram molecular weight of a chemical dissolved to produce one litre of solution.

Temperature Scales

The temperature of a substance is a number which expresses its degree of hotness on some chosen scale. Thus, the temperature of a bucketful of warm water is lower than that of a hot electric lamp filament; the water contains a much larger quantity of internal energy than the filament. Temperature is measured by means of a thermometer, of which a number of different types are available, but each makes use of a particular thermometric property of a particular thermometric substance.

Some depend on the expansion of a liquid when heated; others on the expansion of a compound strip of two metals. In order to establish a temperature scale it is necessary to make use of fixed points. A fixed point is the single temperature at which it can confidently be expected that a particular physical event (e.g. the melting of ice under specific conditions) always takes place. Three such points are defined below.

The upper fixed point or steam point is the temperature of steam from water boiling under standard atmospheric pressure of 760 *mmHg*.

The lower fixed point or ice point is the temperature of pure melting ice.

The triple point of water is that unique temperature at which pure ice, pure water and water vapor can exist together in equilibrium. The triple point is particularly useful, since there is only one pressure at which all three phases (solid, liquid and gas) can be in equilibrium with each other.

The difference in temperature between the two fixed points is called the fundamental interval Figure 8.1. This is divided into 100 equal degrees, the ice point being called 0°C and the steam point 100°C. This subdivision of the interval was suggested by a Swedish astronomer named Celsius, and is now called the *Celsius scale*.

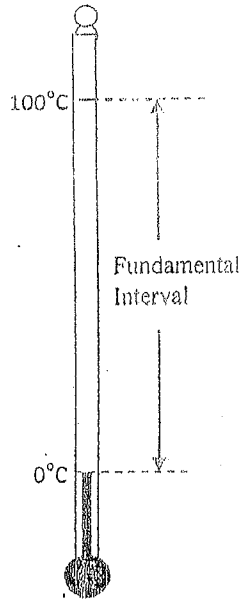


Fig. 8.1: A thermometre

The SI unit of temperature is the Kelvin (K). An interval of one Kelvin is defined as being $^{1}/_{273.16}$ of the temperature of the triple point of water as measured on the thermodynamic scale of temperature. The triple point of water is the fixed point of the scale and is assigned the value of 273.16 K. On this basis absolute zero is 0 K, the ice point is 273.15 K, and the steam point is 373.15 K.

To see the relationship between the Celsius scale and the Kelvin scale, define

$$\theta = T - 273.15 \qquad 8.1$$

where, θ = temperature in °C, and T = temperature in K.

It follows from the equation that a temperature change of 1K is exactly equal to a temperature change of 1°C.

Absolute Zero

Charles's law shows that, if we plot the volume of a given mass of any gas at constant pressure against its temperature, a straight line graph results as shown in Figure 8.2.

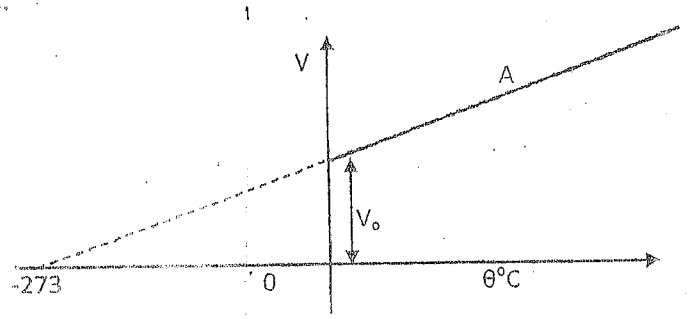


Fig. 8.2: Absolute zero.

If this line is produced backwards, it will meet the temperature axis at -273 °C. This temperature is called the *absolute temperature*. It is the lowest possible temperature where nothing could be colder and no heat energy (or kinetic energy) remains in a substance. Absolute zero is the point at which molecules stop and they have no movement, retaining only quantum mechanical, zero-point energy-induced particle motion.

By international agreement, absolute zero is defined as precisely

- 0 K on the Kelvin scale, which is a thermodynamic (absolute) temperature scale, and
- −273.15 °C on the Celsius scale.

Absolute zero is also precisely equivalent to

- 0 °R on the Rankine scale (also a thermodynamic temperature scale), and
- −459.67 °F on the Fahrenheit scale.

While scientists cannot fully achieve a state of "zero" heat energy in a substance, they have made great advancements in achieving temperatures ever closer to absolute zero (where matter exhibits odd quantum effects). In 1994, the National Institute of Standards and Technology (NIST) achieved a record cold temperature of 700 nK (billionths of a kelvin). In 2003, researchers at Massachusetts Institute of Technology (MIT) eclipsed this with a new record of 450 pK (0.45 nK).

Absolute Temperature Scales

As mentioned earlier, absolute or thermodynamic temperature is conventionally measured in Kelvins (Celsius-size degrees), and increasingly rarely in the Rankine scale (Fahrenheit-size degrees). Absolute temperature is uniquely determined up to a multiplicative constant which specifies the size of the "degree", so the *ratios* of two absolute temperatures, T_1/T_2, are the same in all scales. The most transparent definition comes from the classical Maxwell-Boltzmann distribution over energies, or from the quantum analogs: Fermi-Dirac statistics (particles of half-integer spin) and Bose-Einstein statistics (particles of integer spin), all of which give the relative numbers of particles as (decreasing) exponential functions of energy over kT.

Kinetic Theory

This is an attempt to explain the experimentally observed properties of gases by considering the motion of the molecules (atoms) of which they are composed. A number of assumptions are made.

1. The molecules of a particular gas are identical.
2. Collisions between the molecules and with the container are perfectly elastic.
3. The molecules exert no forces on each other except during impacts and the effect of gravity is ignored so that:
   (a) between collisions the molecules move in straight lines at constant speed, and
   (b) the motion is random.
4. The size of the molecules is negligible compared to their separation.
5. The laws of Newtonian mechanics apply.

Pressure Exerted by a Gas

Consider a cube-shaped box, each side of length L, filled with molecules of an ideal gas. A molecule of ideal gas is like a bouncy rubber ball; whenever it is involved in a collision with a wall of the box, it rebounds with the same kinetic energy it had before hitting the wall. Similarly, if ideal gas molecules collide, the collisions are elastic, so no kinetic energy is lost.

Now consider one of these ideal gas molecules in this box, with a mass m and velocity c at any instant and this will have components of u, v, w respectively in the direction of the three perpendicular axes Ox, Oy, Oz as shown in Figure 8.3. If this

molecule bounces off one of the walls perpendicular to the x-direction, the y and z components of the molecule's velocity are unaffected, and the x-component of velocity reverses. The molecule maintains the same speed, because the collsion is elastic. How much force, on average, does it exert on the wall?

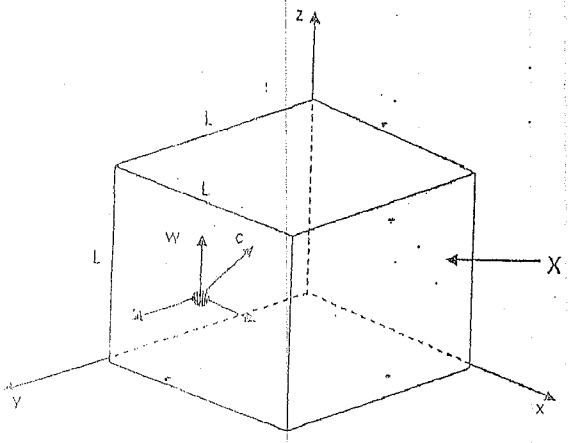


Fig. 8.3: Calculation of gas pressure

To answer this, we just have to think about momentum, and impulse. Momentum is a vector, so if the particle reverses its x-component of momentum there is a net change in momentum. Just before impact, the momentum of the molecule due to u is mu. After impact, the momentum is $-mu$, since the momentum reverses. Thus

$$\text{Momentum change on impact} = mu - (-mu) = 2mu$$

If we're dealing with the average force, the time interval is just the time between collisions, which is the time it takes for the molecule to make a round trip in the box and come back to hit the same wall. This time simply depends on the distance travelled in the x-direction $2L$, and the x-component of velocity; the time between collisions with that wall is $2L/u$. This is the time for one impact. The number of impacts per second t on a given face $= \dfrac{1}{2L/u} = \dfrac{u}{2L}$.

Therefore, momentum change per second $= t \times$ one momentum change

$$= \frac{u}{2L} \times 2mu = \frac{mu^2}{L}$$

$$\therefore \text{ force on X} = \frac{mu^2}{L} \qquad 8.2$$

This is the magnitude of the average force exerted by the wall on the molecule, as well as the magnitude of the average force exerted by the molecule on the wall; the forces are equal and opposite.

$$\therefore \text{ pressure on X} = \frac{force}{area} = \frac{mu^2}{L \times L^2} = \frac{mu^2}{L^3} \qquad 8.3$$

This is just the average pressure exerted on one wall by one molecule selected at random. To find the total pressure, we need to include all the other molecules, which travel at a wide range of speeds. Let us now take into account the N molecules in the cube. Each has a different velocity and hence a component of different magnitude in the direction Ox. Let these be represented by $u_1, u_2, u_3, \ldots, u_N$, it follows from Equation 8.3 that the total pressure on X, p, is given by

$$p = \frac{mu_1^2}{L^3} + \frac{mu_2^2}{L^3} + \ldots + \frac{mu_N^2}{L^3}$$

$$= \frac{m}{L^3}\left(u_1^2 + u_2^2 + \ldots + u_N^2\right) \qquad 8.4$$

Now, let $\overline{u}^2$ represent the average of all the squares of the components in the Ox direction, i.e.

$$\overline{u}^2 = \frac{u_1^2 + u_2^2 + \ldots + u_N^2}{N}$$

$$N\overline{u}^2 = u_1^2 + u_2^2 + \ldots + u_N^2 \qquad 8.5$$

From Equation 8.5, we have

$$p = \frac{Nm\overline{u}^2}{L^3} \qquad 8.6$$

Consider now how this average x-component velocity compares to the average velocity. For any molecule, the velocity can be found from its components using the Pythagorean Theorem in three dimensions:

$$c^2 = u^2 + v^2 + w^2 \qquad 8.7$$

so that the mean square $\overline{c}^2$ is given by

$$\overline{c}^2 = \overline{u}^2 + \overline{v}^2 + \overline{w}^2 . \qquad 8.8$$

With a large number of molecules of varying speed in random motion, the mean square of the component speed in any one of the three axes is the *same*.

$$\therefore \overline{c}^2 = \overline{v}^2 = \overline{w}^2 \qquad 8.9$$

So that

$$\overline{u}^2 = \frac{\overline{c}^2}{3} \qquad 8.10$$

Hence, from Equation 8.6,

$$p = \frac{Nm\overline{c}^2}{3L^3} \qquad 8.11$$

The number of molecules per unit volume, $n = N/L^3$. Thus

$$p = \frac{nm\overline{c}^2}{3} \qquad 8.12$$

nm is the product of the number of molecules per unit volume and the mass of one molecule and is, therefore, the total mass of the gas per unit volume which is its density ρ

$$\therefore p = \frac{\rho\overline{c}^2}{3} \qquad 8.13$$

Relationship between Molecular Kinetic Energy and Temperature

On the basis of the kinetic theory of gases and using Equation 8.13,

$$p = \frac{\rho\overline{c}^2}{3} \qquad 8.13$$

For any volume V of gas,

$$pV = \frac{\rho V\overline{c}^2}{3} \qquad 8.14$$

Therefore,

$$pV = \frac{1}{3}M\overline{c}^2 . \qquad 8.15$$

where M is the mass of volume V of gas.

Equation 8.15 may be rewritten as

$$pV = \frac{2}{3}N\left(\frac{1}{2}m\overline{c}^2\right) \qquad 8.16$$

where N = the total number of molecules in volume V, and m = the mass of one molecule.

The ideal gas equation for n moles of a gas of volume V and pressure p is

$$pV = nRT \qquad 8.17$$

where R = the universal molar gas constant, and T = the temperature in Kelvin.

Thus the predictions of the kinetic theory of gases are in agreement with idealized experimental observation if

$$\frac{2}{3}N\left(\frac{1}{2}m\overline{c}^2\right) = nRT \qquad 8.18$$

i.e. $\frac{1}{2}m\overline{c}^2 = \frac{3}{2}\frac{nR}{N}T$ and

$$\frac{1}{2}m\overline{c}^2 = \frac{3}{2}\frac{R}{N_A}T \qquad 8.19$$

since $\frac{N}{n}$ is the number of molecules per mole, i.e. N_A, the Avogadro constant.

Since R and N_A are universal constants, so also is $\frac{R}{N_A}$ and it is called the Boltzmann's constant, k ($= 1.38 \times 10^{-23}\ JK^{-1}$) and is the gas constant per molecule. The left-hand term of Equation 8.19 is the translational kinetic energy of a single molecule, and therefore, average translational kinetic energy of a molecule is given as

$$\frac{3}{2}kT = \frac{3}{2}\frac{R}{N_A}T \qquad 8.20$$

A further assumption to those already mentioned is that the average translational kinetic energy of a molecule is equal to $\frac{3}{2}kT$. This is a reasonable assumption, since putting heat into a gas increases its temperature and must also increase the kinetic energy of its molecules because there is no other way that the energy can be absorbed.

Example 8.1

A tank used for filling helium balloons has a volume of 0.300 m^3 and contains 2.00 mol of helium gas at 20.0°C. Assuming that the helium behaves like an ideal gas, (a) what is the total translational kinetic energy of the molecules of the gas? (b) What is the average kinetic energy per molecule?

Solution

(a) $E_{trans} = \frac{3}{2}nRT = \frac{3}{2}(2.00mol)(8.37J/mol.K)(293K)$

$E_{trans} = 7.30\times10^{3}J$

(b) $\frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_nT = \frac{3}{2}(1.38\times10^{-23}J/K)(293K)$

$= 6.07\times10^{-21}J$

Example 8.2

Helium gas occupies a volume of 0.04 m^3 at a pressure of 2 × 10^5 Pa (Nm^{-2}) and temperature 300 K. Calculate (a) the mass of helium, (b) the r.m.s speed of its molecules, (c) the r.m.s speed at 432 K when the gas is heated at constant pressure to this temperature, (d) the r.m.s speed of hydrogen molecules at 432 K. (Relative molecular mass of helium and hydrogen = 4 and 2 respectively, R = 8.3 $Jmol^{-1}K^{-1}$)

Solution

(a) *Mass* $PV = nRT$

For n mols, $n = \frac{PV}{RT} = \frac{2\times10^5\times0.04}{8.3\times300} = 3.2$

$\left(\frac{3P}{\rho}\right)^{\frac{1}{2}}$

$n = \frac{m}{M}$

3.2

Hence mass of helium = 3.2 × 4g = 12.8g

(b) *r.m.s speed*

Pressure P = 2 x 10^5 Pa

Density $\rho = \frac{mass}{volume} = \frac{12.8\times10^{-3}kg}{0.04m^3} = 0.32kgm^{-3}$

So rms speed $= \left(\frac{3P}{\rho}\right)^{\frac{1}{2}} = \left(\frac{3\times2\times10^5}{0.32}\right)^{\frac{1}{2}} = 1369ms^{-1}$

(c) *Temperature 432 K*

Since rms speed is proportional to $\sqrt{T}$, the new value c_r at 432 K is given by

$\frac{c_r}{1369} = \left(\frac{432}{300}\right)^{\frac{1}{2}} = 1.2$

So $c_r = 1.2\times1369 = 1643ms^{-1}$

(d) *Hydrogen*

One mole of hydrogen has a mass of 2 g and one mole of helium has a mass of 4 g. So ration of molar masses = 2 : 4 = 1 : 2.

But rms speed at a given temperature is proportional to $\frac{1}{\sqrt{M}}$, where M is the molar mass. So at 432 K, rms speed of hydrogen molecules = $\sqrt{2}\times1643$ m/s.

Mean Free Path

In physics and kinetic theory, the **mean free path** of a particle, such as a molecule, is the average distance the particle travels between collisions with other particles. The mean free path or average distance between collisions for a gas molecule may be estimated from kinetic theory. If the molecules have diameter *d*, then the effective cross-section for collision can be modeled by using a circle of diameter 2*d* to represent a molecule's effective collision area while treating the "target" molecules as point masses. In time t, the circle would sweep out the volume shown in Figure 8.4 and

the number of collisions can be estimated from the number of gas molecules that were in that volume.

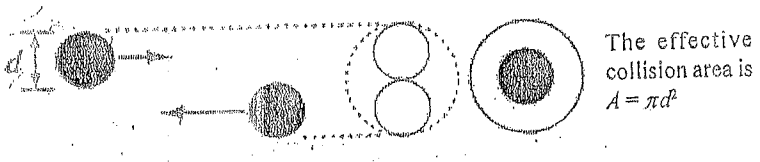


Fig. 8.4: Effective cross-section for collision

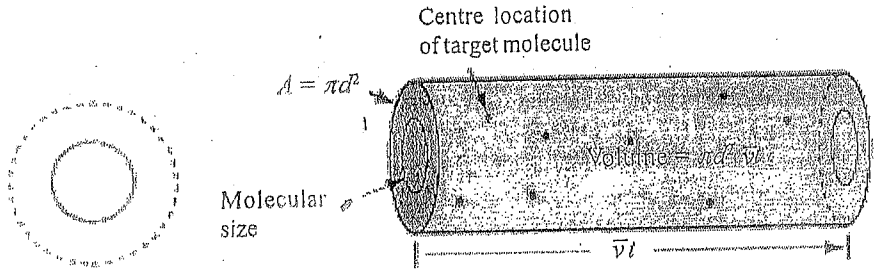
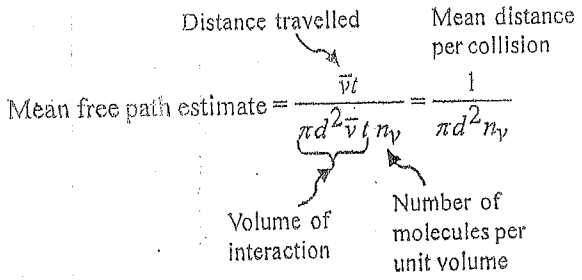


Fig. 8.5: Estimating the number of gas molecules in a volume

The mean free path could then be taken as the length of the path divided by the number of collisions.



The problem with this expression is that the average molecular velocity $\bar{v}$ is used, but the target molecules are also moving. The frequency of collisions depends upon the average relative velocity of the randomly moving molecules.

Refinement of Mean Free Path

The intuitive development of the mean free path expression suffers from a significant flaw — it assumes that the "target" molecules are at rest when in fact they have a high average velocity. What is needed is the average relative velocity, and the calculation of that velocity from the molecular speed distribution yields the result

Average relative velocity $\bar{v}_{rel} = \sqrt{2}\,\bar{v}$

which revises the expression for the effective volume swept out in time t

Effective volume of targets swept $\pi d^2 \sqrt{2}\,\bar{v}\,t$ The number of collisions is $\sqrt{2}$ times the number with stationary targets

The resulting mean free path is

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n_v} \qquad 8.21$$

The number of molecules per unit volume can be determined from Avogadro's number and the ideal gas law, leading to

$$n_v = \frac{nN_A}{V} = \frac{nN_A}{\frac{nRT}{P}} = \frac{N_A P}{RT}$$

Mean free path

$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P} \qquad 8.22$$

Application

A classic application of mean free path is to estimate the size of atoms or molecules. Another important application is in estimating the resistivity of a material from the mean free path of its electrons. For example, for sound waves in an enclosure, the mean free path is the average distance the wave travels between reflections off the enclosure's walls.

Heat Energy

In everyday life we sometimes loosely refer to the "heat energy in a body", instead of using the term "internal energy". In physics, heat is defined as the energy which

is transferred from one place to another owing to a temperature difference between them. When you heat a substance, you are transferring energy into it by placing it in contact with surroundings that have a higher temperature. This is the case, when a bucket of cold water is put on a stove burner — the burner is at a higher temperature than the water, and so the water gains energy.

Thermal Conductivity

A substance is in a steady state when the temperatures at all points in it do not change with time. Consider a thin disc of some material of cross sectional area A and thickness δx as shown in Figure 8.6. Suppose that the hotter face is maintained at a temperature θ and that the other face is maintained at $\theta - \delta\theta$. Suppose also that the rate of heat flow from the hotter face to the colder face is $\delta Q/\delta t$.

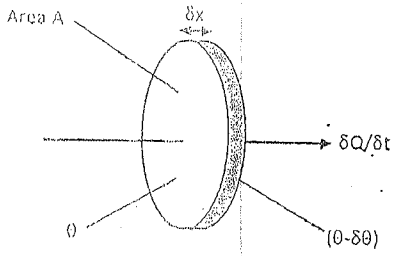


Fig. 8.6: Thermal conductivity

Experiment shows that if there are no heat losses from the sides and steady state conditions prevail, then

$$\frac{\delta Q}{\delta t} \propto A\frac{\delta\theta}{\delta x}$$

With the introduction of a constant of proportionality k this can be written as

$$\frac{\delta Q}{\delta t} = -\kappa A\frac{\delta\theta}{\delta x} \qquad 8.23$$

which in the limit as $\delta x \to 0$ becomes

$$\frac{dQ}{dt} = -\kappa A\frac{d\theta}{dx} \qquad 8.24$$

where dQ/dt is the rate of flow of heat from the hotter face to the colder face and is at right angles to the faces (unit = W).

$d\theta/dx$ is called the temperature gradient across the section concerned (unit = Km^{-1})

κ is a constant whose value depends on the material of the disc. It is called the coefficient of thermal conductivity of the material (unit = $Wm^{-1}K^{-1}$).

Notes

1. When heat is flowing in the positive direction of x as in Figure 8.6 the temperature gradient is negative and, therefore, the presence of the minus sign in Equation 8.24 makes κ a positive constant.
2. It is the existence of the temperature gradient which causes the heat to flow. If it were not for the fact that the two faces are being maintained at their respective temperatures, the effect of the heat flow would be to destroy the temperature gradient by warming the cooler regions.
3. Equation 8.24 is used to define κ. Thus:

   The coefficient of thermal conductivity, κ, of a material is the rate of flow of heat per unit area per unit temperature gradient when the heat flow is at right angles to the faces of a parallel-sided slab of the material under steady state conditions.

Suppose that a long, uniform rod of length L is thermally insulated so that energy cannot escape by heat from its surface except at the ends. One end is in thermal contact with an energy reservoir at temperature T_1, and the other end is in thermal contact with a reservoir at temperature $T_2 > T_1$. When a steady state has been reached, the temperature at each point along the rod is constant in time. In this case if we assume that k is not a function of temperature, the temperature gradient is the same everywhere along the rod and is

$$\frac{d\theta}{dt} = \frac{T_2 - T_1}{L} \qquad 8.25$$

Thus the rate of energy transfer through the rod is

$$\frac{dQ}{dt} = \kappa A\frac{(T_2 - T_1)}{L} \qquad 8.26$$

Substances that are good thermal conductors have large thermal conductivity values, whereas good thermal insulators have low conductivity values.

Example 8.3

Calculate the quantity of heat conducted through 2 m^2 of a brick wall 12 cm thick in 1 hour if the temperature on one side is 8°C and on the other side is 28°C. (Thermal conductivity of brick = 0.13 $Wm^{-1}K^{-1}$).

Solution

Temperature gradient = $\frac{28-8}{12\times10^{-2}} Km^{-1}$ and t = 3600 s

$\therefore Q = kAt \times$ temperature gradient

$$= 0.13\times2\times3600\times\frac{28-8}{12\times10^{-2}} J$$

$$= 156000 J$$

Heat Capacity and Specific Heat

When energy is added to a substance and no work is done, the temperature of the substance usually rises except when the substance undergoes a change of state. The quantity of energy required to raise the temperature of a given mass of a substance by some amount varies from one substance to another.

As an example, the quantity of energy needed to raise the temperature of one 1 kg of water by 1°C is 4186 J, but the quantity of energy needed to raise the temperature of 1 kg of copper by 1°C is only 387 J. The heat capacity C of an object is the proportionality constant between an amount of heat and the change in temperature that this heat produces, i.e., it is defined as the amount of energy needed to raise the temperature of that object by 1°C. Thus

$$Q = C\left(T_f - T_i\right) = C\Delta T \qquad 8.27$$

In which T_f and T_i are the final and initial temperatures of the object and ÄT is the temperature change.

Two objects made of the same material will have heat capacities proportional to their masses. We therefore define a "heat capacity per unit mass" or specific heat c that refers to a unit mass of the material of which the object is made. Then

$$Q = cm\Delta T \qquad 8.28$$

Specific heat is essentially a measure of how thermally insensitive a substance is to the addition of energy. The greater a material's specific heat, the more energy must be added to a given mass of the material to cause a particular temperature change. The energy required to raise the temperature of 0.5 kg of water by 3°C is $(0.5kg)(4186J/kg.\ °C)\ (3°C) = 6.28 \times 103J$. When the temperature increases, Q and ΔT are both taken to be positive and energy flows into the system. When the temperature decreases, Q and ΔT are both negative, and energy flows out of the system.

Adiabatic Processes

An adiabatic process is one that occurs so rapidly or occurs in a system that is so well insulated that no transfer of heat occurs between the system and the environment. If a gas is compressed (or expanded) very rapidly, very little energy is transferred out of (or into) the system by heat, and so the process is nearly adiabatic. The very slow expansion of a gas that is thermally insulated from its surroundings is also an example of an adiabatic process.

Consider an ideal gas undergoing an adiabatic expansion. At any time during the process, assume that the gas is in an equilibrium state, so that the equation of state $PV = nRT$ is valid.

The pressure and volume at any time during an adiabatic process are related by the expression

$$PV^{\gamma} = k \qquad 8.29$$

which is a constant, where $\gamma = C_p/C_v$ is assumed to be constant during the process. Thus all three variables in the ideal gas law — P, V, and T — change during an adiabatic process. C_p and C_v are the specific heat capacities at constant pressure and constant volume respectively.

If we wish to introduce the temperature, T, into an adiabatic change, then for one mole of gas,

$$PV = RT \qquad 8.30$$

and

$$PV^{\gamma} = k \qquad 8.31$$

Now $P = RT/V$, then

$$\frac{RT}{V} V^{\gamma} = k \qquad 8.32$$

So that $RTV^{\gamma-1} = k$. But R is a constant.

$$\therefore T.V^{\gamma-1} = \text{constant} \qquad 8.33$$

Example 8.4

An ideal gas at 17°C has a pressure of 1.0×10^5 Pa, and is compressed (a) isothermally, (b) adiabatically until its volume is halved, in each case reversibly. Calculate in each case the final pressure and temperature of the gas, assuming $\gamma = 1.4$.

Solution

(a) Isothermally, PV = constant

$$\therefore P \times \frac{V}{2} = 1.0 \times 10^5 \times V$$

$$\therefore P = 2.0 \times 10^5 Pa$$

The temperature is constant at 17°C.

(b) Adiabatically, PV^{γ} = constant, and $\gamma = 1.4$.

$$\therefore P \times \left(\frac{V}{2}\right)^{1.4} = 1.0 \times 10^5 \times V^{1.4}$$

$$\therefore P = 1.0 \times 10^5 \times 2^{1.4} = 2.6 \times 10^5 Pa$$

But since $TV^{\gamma-1}$ = constant,

$$\therefore T \times \left(\frac{V}{2}\right)^{0.4} = (273+17) \times V^{0.4}$$

$$\therefore T = 290 \times 2^{0.4} = 383K$$

∴ Temperature is 110 °C.

Example 8.5

A quantity of oxygen is compressed isothermally until its pressure is doubled. It is then allowed to expand adiabatically until its original volume is restored. Find the final pressure in terms of the initial pressure assuming for oxygen $\gamma = 1.4$.

Solution

Let P_0, V_0 = the original pressure and volume of the oxygen.

Since PV = constant for an isothermal change,

∴ new volume = $\frac{V_0}{2}$ when new pressure is $2P_0$

If the gas expands adiabatically to its volume V_0, when the pressure is P, then

$$P \times V_0^{1.4} = 2P_0 \times \left(\frac{V_0}{2}\right)^{1.4}$$

$$\therefore P = 2P_0 \times \left(\frac{1}{2}\right)^{1.4} = 0.8P_0$$

Example 8.6

Air at 20.0 °C in the cylinder of a diesel engine is compressed from an initial pressure of 1.00 atm and volume of 800.0 cm³ to a volume of 60.0 cm³. Assume that air behaves as an ideal gas with $\gamma = 1.4$ and that the compression is adiabatic. Find the final pressure and temperature of the air.

Solution

We know that

$$P_f = P_i \left(\frac{V_i}{V_f}\right)^{\gamma} = (1.00atm)\left(\frac{800.0cm^3}{60.0cm^3}\right)^{1.40}$$

$$P_f = 37.6atm$$

But PV = RT is valid during any process and since no gas escapes from the cylinder

$$\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}$$

$$T_f = \frac{P_f V_f}{P_i V_i} T_i = \frac{(37.6atm)(60.0cm^3)}{(1.00atm)(800.0cm^3)}(293K)$$

$$T = 826^0C = 553^0C$$

Example 8.7

A tank used for filling helium balloons has a volume of 0.300 m³ and contains 2.00 mol of helium gas at 20.0 °C. Assuming that the helium behaves like an ideal gas, (a) what is the total translational kinetic energy of the molecules of the gas? (b) What is the average kinetic energy per molecule?

Solution

(a)
$$E_{trans} = \frac{3}{2}nRT = \frac{3}{2}(2.00mol)(8.37J/mol.K)(293K)$$

$$E_{trans} = 7.30\times10^3 J$$

(b) $$\frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_B T = \frac{3}{2}(1.38\times10^{-23}J/K)(293K) = 6.07\times10^{-21}J$$

Thermodynamics

The First law of Thermodynamics

The first law of thermodynamics is a generalization of the law of conservation of energy that encompasses changes in internal energy. It is a universally valid law that can be applied to many processes and provides a connection between the microscopic and macroscopic worlds. Suppose that a system undergoes a change from an initial state to a final state, both the work W done and the heat Q exchanged depend on the nature of the process. Experimentally, however, we find that *the quantity $Q - W$ is the same for all processes.* This quantity depends only on the initial and final states and does not depend at all on how the system gets from one state to the other.

All other combinations of Q and W, including or alone, , and , are path dependent; only the quantity is not.

The quantity must represent a change in some intrinsic property of the system. We call this property the internal energy and write

$$\Delta E_{int} = E_{int,f} - E_{int,i}$$

8.34

$$\Delta E_{int} = Q - W$$

Equation 8.34 is the **first law of thermodynamics**. If the thermodynamic system undergoes only a differential change, we can write the first law as

8.35

$$dE_{int} = \delta Q - \delta W$$

The internal energy E_{int} of a system tends to increase if energy is added via heat Q and tends to decrease if energy is lost via work W done by the system.

In the statement of the first law of thermodynamics above, it is assumed that there are no changes in the kinetic energy or the potential energy of the system as a whole; that is $\Delta K = \Delta U = 0$. The first law equation is an energy equation specifying that the only type of energy that changes in the system is the internal energy E_{int}. We consider some special cases.

First, consider an isolated system — that is, one that does not interact with its surroundings. In this case no energy transfer by heat takes place and the value of the work done by the system is zero and hence the internal energy remains constant. That is, because, $Q = W = 0$, it follows that $\Delta E_{int} = 0$, and thus thus $E_{int,f} = E_{int,i}$. It is concluded that the internal energy E_{int} of an isolated system remains constant.

Second, consider the case of a system (one not isolated from its surroundings) that is taken through a cyclic process — that is, a process that starts and ends at the same state. In this case the change in the internal energy must again be zero and, therefore, the energy added to the system must equal the work done by the system during the cycle. That is, in a cyclic process,

8.36

$$\Delta E_{int} = 0$$

and

8.37

$$Q = W$$

If the value of the work done by the system during some process is zero, then the change in internal energy E_{int}, equals the energy transfer Q into or out of the system;

$$\Delta E_{int} = Q \qquad 8.38$$

The work done on a system is always the negative of the work done by a system. If energy enters the system, then Q is positive and the internal energy increases. For a gas, we can associate this increase in internal energy with an increase in the kinetic energy of the molecules. Conversely, if no energy transfer occurs during some process but work is done by the system, then the change in internal energy equals the negative value of the work done by the system:

$$\Delta E_{int} = W \qquad 8.39$$

As an example, if a gas is compressed by a moving piston in an insulated cylinder, no energy is transferred by heat and the work done by the gas is negative; thus the internal energy increases because kinetic energy is transferred from the moving piston to the gas molecules.

Work Done by a System

Consider some gas, at a pressure p, in a cylinder fitted with a piston as shown in Figure 8.7. If the piston has an area, the force on it is

$$F = pA \qquad 8.40$$

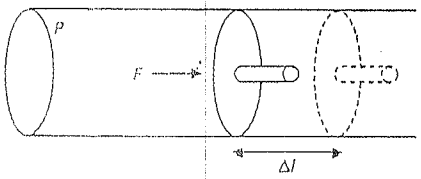


Fig. 8.7: Gas in a cylinder fitted with a piston.

If the piston is allowed to move outwards a distance Δl, the gas will expand, and its pressure will fall. But by making the distance very short, we can make the fall in pressure so small that we may consider the pressure constant. The force F is then constant, and the work done is

$$\Delta W = F\Delta l = pA\Delta l \qquad 8.41$$

The product $A\Delta l$ is the increase in volume, ΔV, of the gas, so that

$$\Delta W = p\Delta V \qquad 8.42$$

The product of pressure and volume, in general, therefore represents work. If the pressure p is in Nm^{-2}, and the area A is in m^2, the force F is in Newton. Also if the movement Δl is in *metre*, the work $F\Delta l$ is in Newton x metre or joule (J). The increase of volume, , is in m^3. Thus the product of pressure in Nm^{-2}, and volume in m^3, represents work in joule. Therefore,

Work done = pressure x volume change 8.43

So if the volume of a gas at constant pressure of 10^5 Nm^{-2} expands by 0.01 m^3, then work done = $10^5 \times 0.01 = 1000$ J.

Exercise 8

8.1 The pressure of helium gas in a tube is 0.200mmHg. If the temperature of the gas is 20°C, what is the density of the gas? (use $M_{HC} = 4$kg/kmol) **(Ans: 4.410^{-5} Pa)**

8.2 At what temperature will the molecules of an ideal gas have twice the rms velocity they have at 20 °C? **(Ans: 1170K)**

8.3 In certain region of outer space there are an average of only five molecules per cm^3. The temperature there is about 3K. What is the average pressure of this very dilute gas? **(Ans:210^{-16} Pa)**

8.4 The rms velocity of nitrogen molecules in the air at S.T.P is about 490ms^{-1}. Find their mean free path and the average time between collisions. The radius of a nitrogen molecule can be taken to be 210^{-10}m.
(Ans: 5.210^{-8}m ; 1.110^{-10}s)

8.5 The mean free path of a gas molecule (radius =2.510^{-10}m) in an ideal gas at 500°C when the pressure is 710^{-6}mmHg? **(Ans: 10m)**

8.6 A 2.0Kg metal block (c=0.137cal/g°C) is heated from 15°C to 90 °C. By how much does its internal energy change? **(Ans: 86kJ)**

8.7 If a certain mass of water falls a distance of 854m and all the is effective in heating the water, what will be the temperature rise of the water? (Ans: 2.00°C)

8.8 How many joules of heat per hour are produced in a motor that is 75 percent efficient and requires 0.25hp to run it? (Ans: 168kJ)

8.9 How much external work is done by an ideal gas in expanding from a volume of 3 litres to a volume of 30 litres against a constant pressure of 2atm? (Ans: 5.45kJ)

8.10 As an ideal gas is compressed isothermally, the compressing agent does 36J of work. How much heat flows from the gas during the compression process? (Ans: 8.6cal)

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mole of P = $\frac{12.09}{30.97}$

= 0.387

mole of Cl_2 = $\frac{35.0}{70.97}$ = 0.4931

2 = 2

0.387 = x

x = 0.387

3 2

0.4931 = x

x = 0.328

$k = \frac{1}{2}mv \times x$

$\left(\frac{1}{2}kA^2\right)C$